

NetJets Transportes Aéreos, S.A.

Phenom 300 Handbook

NetJets Transportes Aéreos S.A.
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CHAPTER 1 - INTRODUCTION

1.1 GENERAL

Disclaimer: This document does not replace any NTA manuals or OEM documentation, and it should not be used as a primary source document. The handbook is not an ANAC approved document and is intended for crew reference only.

This is a document updated using the best information and advice available at the time of writing. The Fleet Handbook is created and maintained by the Phenom team on the fleet. It is not intended to duplicate other official documents but to have one place where useful information can be found or easily amended without the need for a prolonged process. It is intended that this document will continuously evolve with information being added or removed periodically. This process will be accomplished via the standard NetJets amendment procedure.

1.2 CREW INPUT

The content of the handbook will primarily be driven by the wishes of crew members. Crews are encouraged to submit any information they would like to see published in the handbook to lis-phenom@netjets.com for assessment and possible inclusion.

1.3 BEST PRACTICE

Items within the handbook are taken from many sources and it is hoped that the handbook will evolve to represent best practice under operational conditions. Crews are welcome to submit practical advice on all aspects of our operation.

Many of the best practices we have on the fleet come from Embraer FOLs. It is not the intention to repeat all of the FOLs in this manual, however, because of their nature, it will be normal for some of them to appear in here as we emphasise certain aspects of our operation.

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CHAPTER 2 - PRE-FLIGHT

2.1 EXTERNAL WALKROUND GUIDE

Note: REFER TO POH FOR LATEST REVISION

This walkround guide is designed to add some pictures as well as best practices to help and remind while checking the outside. We follow the External Inspection Checklist from the Embraer AFM to carry out the walkround but this guide is here to remind you of some of the smaller details.

2.1.1 EXTERNAL LIGHTS CHECK

An External Lights Check must be carried out on the first flight of the day. Check the following lights:

Navigation lights red, green, and 2 x white facing rearwards near the wing tips.



Red Beacon light on the tail that comes on automatically when either Engine Start / Stop knob is set to Run.

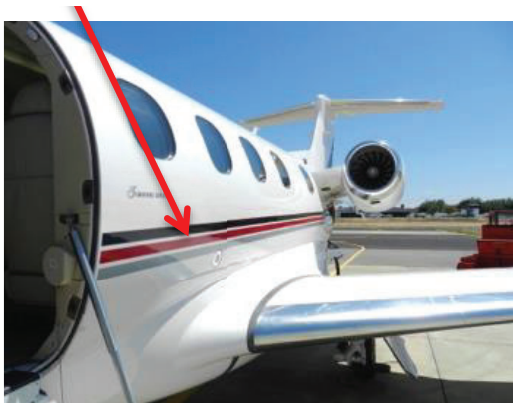


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Taxi and Landing lights - 4 under belly lights. All 4 come on in Dim Mode for taxi, when Landing Lights are selected, the middle two lights change to fully bright.



Wing Inspection Light, left side only and is the only halogen bulb, all of the other lights are LED.



Strobe/Anti Collision lights are located on both upper and lower surface of the fuselage.



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2.1.2 EMERGENCY LIGHTING CHECK

Make sure the Emergency Lights are tested with the batteries OFF, to check they work off their own power supply.

Main Aisle Lights
 (Red lights in front of exits)



Overhead Lights and EXIT Signs
 Illuminated



Main Door EXIT Sign



Emergency EXIT Light



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Main Stair Lights



Emergency Exit Egress Lights



Ensure that all lights are turned off following the check especially if using battery power.

Proceed to carry out the External Inspection check starting to the left of the main door and moving towards the nose.

Emergency Lights Troubleshooting

If you have flickering EMER lights or lights ON when in ARM:

- This is a weak or low Emergency Light Battery.
- Charge the battery with a GPU or in flight for a minimum of 45 minutes and then check the battery again.

This is a known 'fault' and Embraer are looking at making changes to the batteries or the system logic. Most faults have been diagnosed as either not having enough charge getting into the battery or a logic issue with the charge memory that requires a reset. Please refrain from the practice of using the Emer Lights after the Batts are OFF to do the paperwork when closing the jet down.

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2.1.3 WALKROUND CHECKS

During the external check the overall condition of the aircraft for any signs of damage to any of the surfaces and the general condition.

IASP.....Condition, No Obstructions
Remove cover and check for damage or obstructions. Cover should be placed into box inside left side front baggage bay.



Oxygen Blow Out Disc..... Green Disk in place
If required, the oxygen pressure can be checked using the gauge inside the left side nose baggage bay.



Ice detector..... Condition



Antennas Condition
Check the Antennas underneath the front of the aircraft for damage.

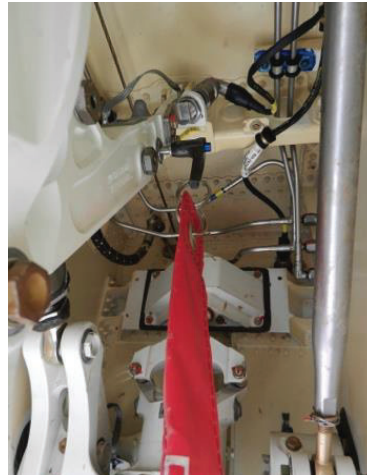
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Nose Landing Gear and Bay..... Condition

Check the general condition of the nose wheel and compartment. Check the strut is clean and the extension looks normal for aircraft weight. Examine the tyre for excessive wear or cuts.

Ensure the nose landing gear safety pin is removed.

NOTE: this can be difficult to see and the flag can be blown up inside the nose bay.



Check condition of the nose wheel steering torque link connection. If not required to be towed ensure it is connected and secured, and the additional pin is in place and locked (using the extra clip).

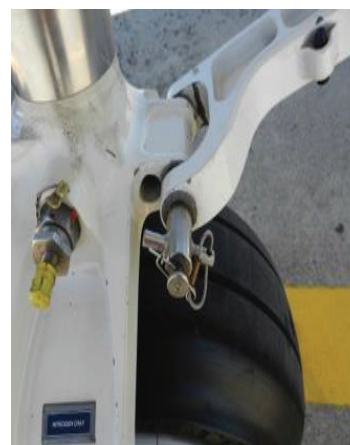
NOTE: the aircraft towing arm adapter can only be fitted with the steering disconnected.



Connected



Disconnected



Safety Pin Secured

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Forward LH Baggage Compartment door..... Locked
(if not required or when loading is complete)



ALL DOORS AND PANELS SHOULD BE LEFT EITHER OPEN OR FULLY LOCKED TO ENSURE THAT THEIR CONDITION IS KNOWN AND OBVIOUS TO SEE.

Radome Condition
Check condition of the radome.



Forward RH Baggage Compartment Door Locked

IASP..... Condition, No Obstruction



Standby Pitot Tube Condition

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Refuelling Panel..... Closed



Ldg/Taxi Lights..... Condition

Fuselage Air Intake Condition

Engine Fan..... Condition



Starter Generator Air Inlets Condition

Fuel Drains..... Drain and check for contamination
(We are not required to carry out this check because the aircraft are flown so often).

Fuel Drains and Dump Valves No Leaks

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MLG Doors Wheels Brakes and Tyre Condition
 Each wheel has two brake wear pins. The brakes need to be changed when the pin is flush with the main housing.



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MLG Safety Pin..... Removed



Right Wing Leading Edge Condition



Fuel Cap Closed and Locked



Fuel Tank Air Inlet No Obstruction

Navigation Strobe Lights..... Condition

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Right Winglet Condition



Right Aileron Condition



Static Dischargers.....Number and Condition

Check condition and number of static discharge wicks. See the MEL CDL page 23 G-CDL-5 for the minimum number of Static Dischargers required for dispatch. The aeroplane is allowed to be dispatched with 1 static discharger per control surface, provided the total number installed on the aeroplane is equal or greater than 8.

Right Spoilers Condition



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Right Flap..... Condition



Battery Access Door Secured



Toilet Servicing Door Secure



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Oil Level Check



Cowlings Latched



Drain Masts Condition
 Engine Exhausts Clear



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Heat Exchanger Air Exhaust Clear



Pylon Condition



Vertical stabilizer Condition



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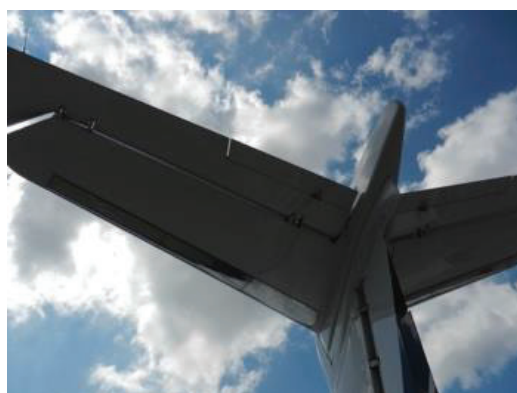
Rudder Condition
Rudder Static Dischargers Check number and Condition



Yaw Trim tab Condition
Horizontal Stabilizer Condition



Elevator Condition
Pitch Trim Tab Condition



Elevator Static Dischargers Number and Condition
Antennas Condition

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Ventral Rudder Condition



Air Exhausts No Obstructions



Aft Baggage Bay Secure



Pylon Condition

Cowlings Latched

Exhausts Clear

Drain Masts Condition

Oil Level Check

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DC Power Receptacle Check



Hydraulic Reservoir Access Door Closed

The upper door allows access to check the hydraulic lines and the red chip detectors. Check the pipes, connections, and chip detectors, and then ensure that both doors are closed.



The lower door allows access to see the hydraulic reservoir and contents. Make sure the blue diagram of the fluid indication is within normal range, within the green mark. The green range corresponds to the dispatch ability range.

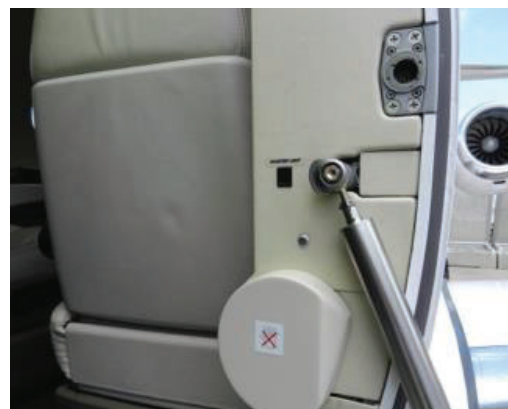


If there is any doubt of the content of the hydraulic system carry out a full check (POH 5-30). The parking brake will need to be de-pressurised by pumping the handle until the pressure is zero (if the contents are low or close to low, releasing all of the parking brake pressure will add fluid to the system to check the 'correct' fluid level). It is vital that the aircraft is chocked and that the chocks are NOT removed until after the first engine start.

Left Flap	Condition
Left Spoilers	Condition
Left Aileron Static Dischargers	Number and Condition
Left Aileron	Check Free
Roll Trim Tab	Condition
Navigation / Strobe Lights	Condition
Left Winglet	Condition
Left Winglet Leading Edge	Condition
Fuel Cap	Closed and Locked
Fuel Tank Air Inlet	No Obstruction
MLG Doors, Wheels, Brakes, and Tyres	Condition
MLG Safety Pin	Removed
LDG / Taxi Lights	Condition
Wing Inspection Light	Condition
Fuselage Air Inlet	No Obstruction
Engine Fan	Condition
Engine Air Inlet	No Obstruction
Starter / Generator Air Inlets	Clear

2.1.4 MAIN DOOR

Check for condition, ease of movement, lights.



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2.2 ADDITIONAL CHECKS AND GUIDANCE

2.2.1 GENERATORS

The generators on the aircraft were all changed in 2017 for a new manufacturer to try to reduce or eliminate the number of generator failures we have suffered in the past. The new generator also came with a new GCU and whilst there have been no failures of the new generators we have had a few issues with the GCU. Embraer are aware of the problems that show themselves as GEN OFF BUS CAS, the GPU not connecting or no engine rotation on start-up. The GCUs will all be replaced again, planned to be completed by the end of 2018. **FOL 006/13 has been updated to Rev. 3 (Jun 2018) and gives advice on all the issues stated above (the advice in the FOL has been proven to work on numerous occasions).**



Previously soot underneath the engine as shown below was a sign of impending failure. Now, however, this soot is normal because the new generators have a different internal seal system so any soot is now blown outside.

2.2.2 BATTERIES

The aircraft batteries have been the subject of many briefs and advice since we have had the aircraft. During this time we have had support from Embraer in various ways including a new No.2 Battery with more capacity.

Basic operation means that we limit the amount of time we are on just battery power to a minimum and we try to get a GPU for revenue flights where available or for all flights in cold conditions.

During 2017 we changed several batteries and sent them all for checking- none failed and all were cleared to continue in service.

The batteries themselves are 24VDC are we cannot expect them to stay up and around 24.6 V for any great amount of time, however, they will probably drop to near 24.1 and hold the charge there. In the cold they will drop in charge much quicker towards 24.0.

The AFM limit for starting engines on the battery is 24.0V and that is the limit still in the OM-B, however, Embraer have said that they are content for us to use 23.5 as the limit for start in line with the EMER check after start.

The voltage before start and the EMER check after start are two different checks. A battery at 23.6V to start an engine does not mean it will fail the 45 mins check if required in flight. The 23.5 V check is to ensure that we will get the minimum time guaranteed if we found ourselves on just battery power.

By adapting our checklists and learning to use the batteries carefully we have got through the winter with hardly any failures.

If the battery is used to start and was low on volts it will require around 90 mins on either a GPU or an engine generator to fully recharge. If the GPU is used make sure the volts are above 27V and ideally above 28.3V because below 27 it will not charge and below 28.3 it will not charge fully in the 90 mins.

2.2.3 WHEELS AND TYRES

The tyres on the Phenom have two grooves which are approximately 7 mm deep when the tyre is new. The tyre is normally replaced when either groove is not visible. This is before Embraer require the tyre to be changed and even if the groove is not seen the tyre still has at least 10 cycles remaining. As soon as the first groove starts to reach the limit let maintenance know and they will start the process of getting the tyres changed. A picture of the tyre is extremely helpful and allows us to make a judgement call.



New Tyre



Inner tread still just visible

There are criteria that require the tyre to be replaced sooner and these are shown in **FOL 007/14 Rev 04 (Jun 2018)**, and also shown in the Aircraft Maintenance Manual an extract of which is shown below.

EXTRACT FROM AIRCRAFT MAINTENANCE MANUAL 32-45-00

Wheels, Tyres, and Brakes - Inspection/Check

General

(1) This task gives the procedure to do an inspection of the tyres of the NLG and MLG. This inspection is a visual examination of the tyres for wear, damage, and general condition.

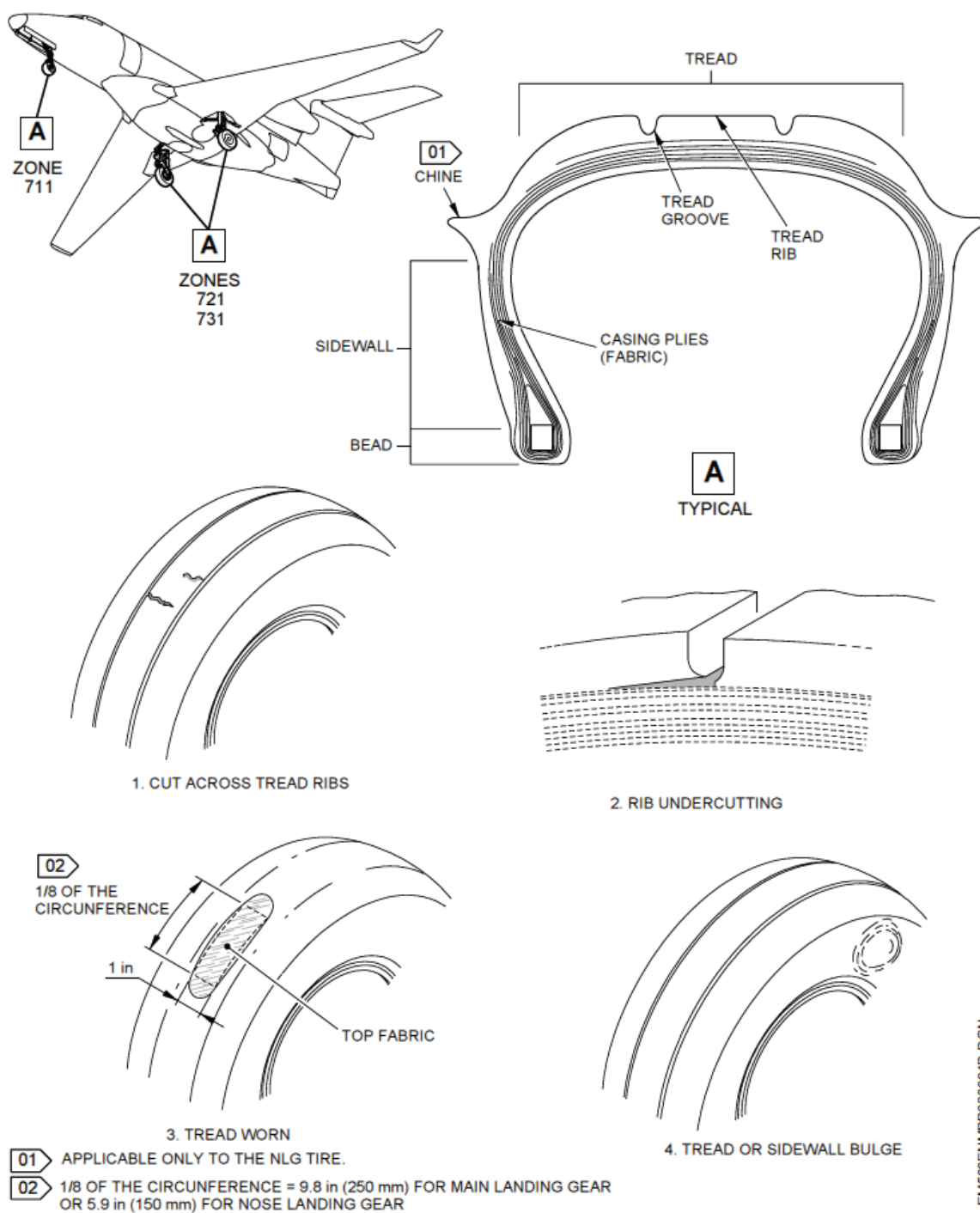
WARNING: Do not go near hot tyres installed on the aircraft. If it is necessary to measure the tyre temperature, go near the tyre from its front, not from its side. High temperature can cause tyre explosion.

(1) Examine the tyres of the NLG and MLG. Replace the tyre if you find on a tyre one or more of the conditions that follow:

- a. Cut into the casing plies.
- b. Cut into the tread groove. ÀREA DE NÃO INTERFERÊNCIA.
- c. NLG tyres, damage, cut, or wear on the chine.
- d. Cut across the tread rib to the fabric (1).
- e. Rib undercutting (2).
- f. Tread worn (3), with top fabric worn through more than 1/8 of the circumference of the tyre or more than 1 in (25 mm) wide.

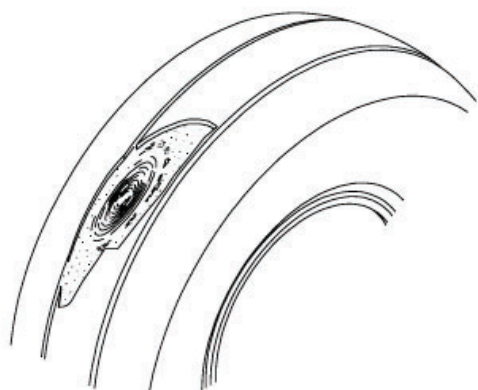
NOTE: Tyres worn beyond this limit may be unsafe and must be replaced before next flight. Schedule the tyre replacement when any point of the tyre tread is worn to the bottom of the groove.

- g. Tread or side wall bulge (4).
- h. Peeled rib (5).
- i. Thrown tread (6).
- j. Open tread splice (7).
- k. Heat damage in bead (8).
- l. Impact break (9).
- m. One or more of the conditions that follow with exposed fabric:
 - 1. Groove cracking (10).
 - 2. Dry braking flat spot (11).
 - 3. Tread chunking (12).
 - 4. Chevron cutting (13).
 - 5. Sidewall crack (14).
 - 6. Sidewall cut or snag (15).
 - 7. Radial or circumferential crack (16).
- n. Balance affected because of wet braking flat spot (17) or dry braking flat spot

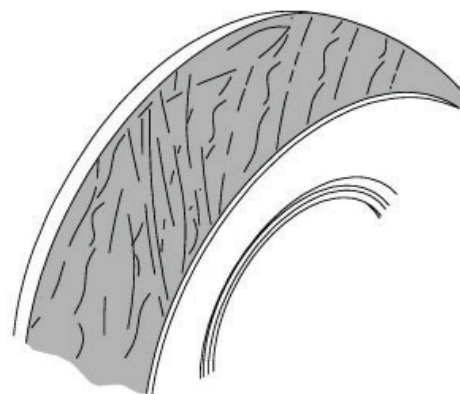


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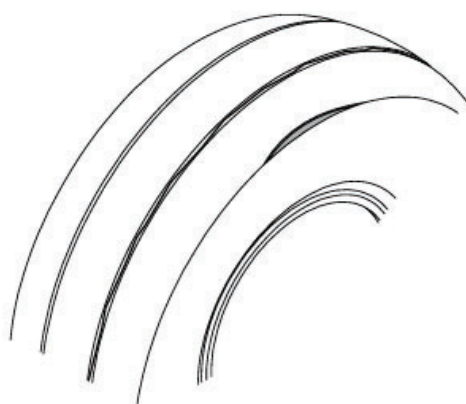
EM500ENMPP320081D.DGN



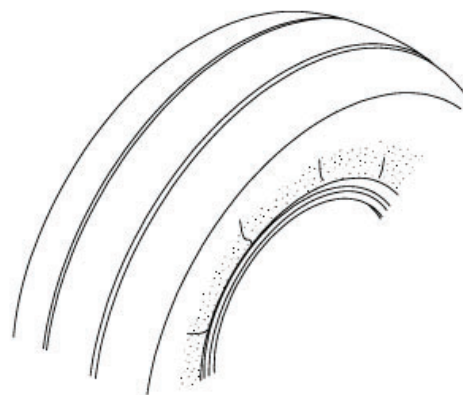
5. PEELED RIB



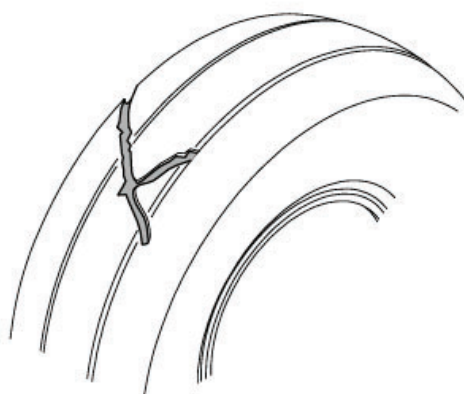
6. THROWN TREAD



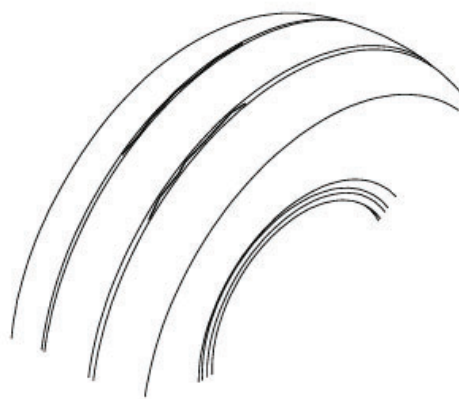
7. OPEN TREAD SPLICE



8. HEAT DAMAGE

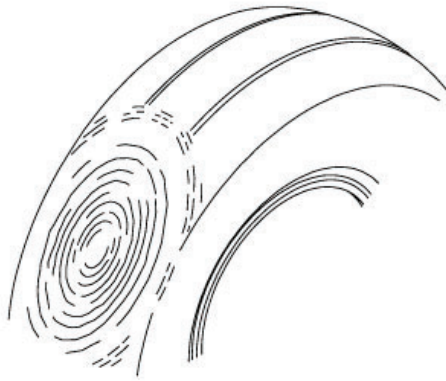


9. IMPACT BREAK

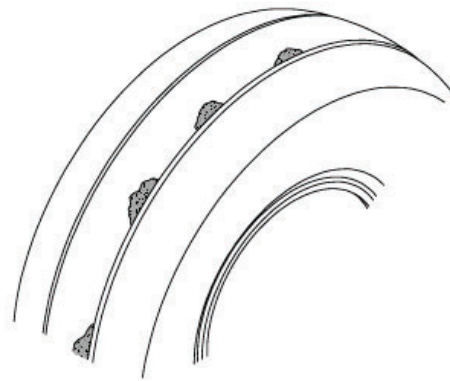


10. GROOVE CRACKING

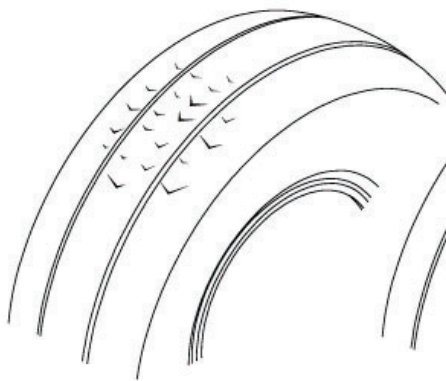
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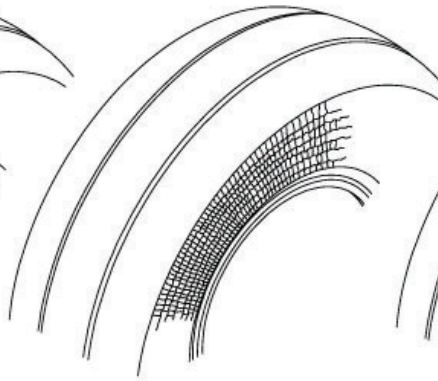
11. DRY-BRAKING FLAT-SPOT



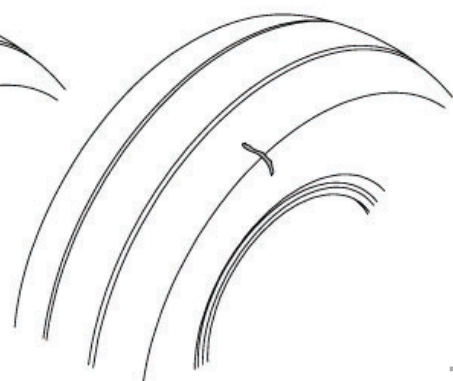
12. TREAD CHUNKING



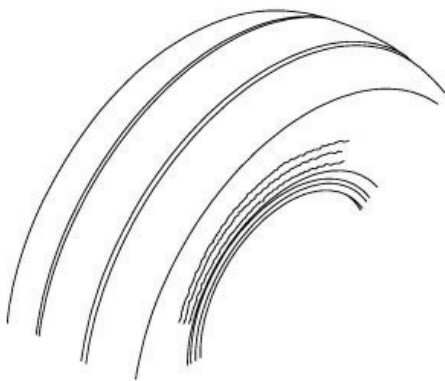
13. CHEVRON CUTTING



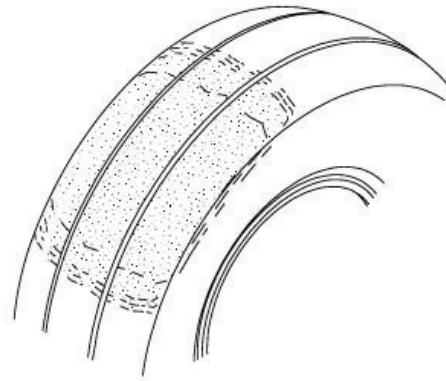
14. SIDEWALL CRACKS



15. CUT OR SNAG



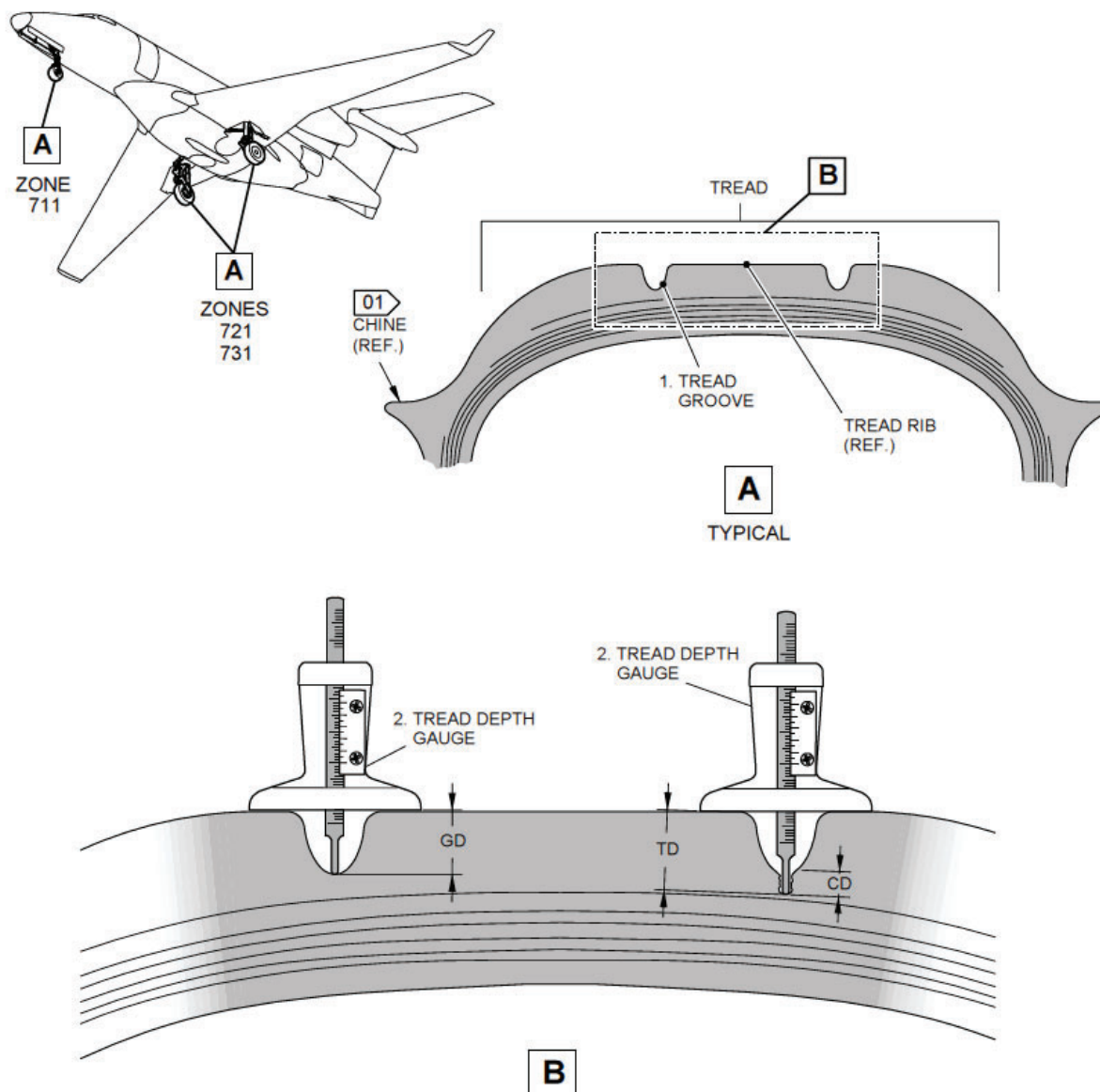
16. RADIAL OR CIRCUMFERENTIAL CRACK



17. WET BRAKING FLAT-SPOT

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EM500ENMPP320004B.DGN



$$CD = TD - GD$$

LEGEND:

CD = CUT DEPTH
TD = TOTAL DEPTH
GD = GROOVE DEPTH

01 APPLICABLE ONLY TO THE NLG TIRE.

EM500ENMPP320319B.DGN

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2.2.4 BRAKES, RIVETS, AND SURROUNDS

All aircraft have now been updated with the BCU 10. This update has been carried out to solve some of the braking issues that were seen during 2016.

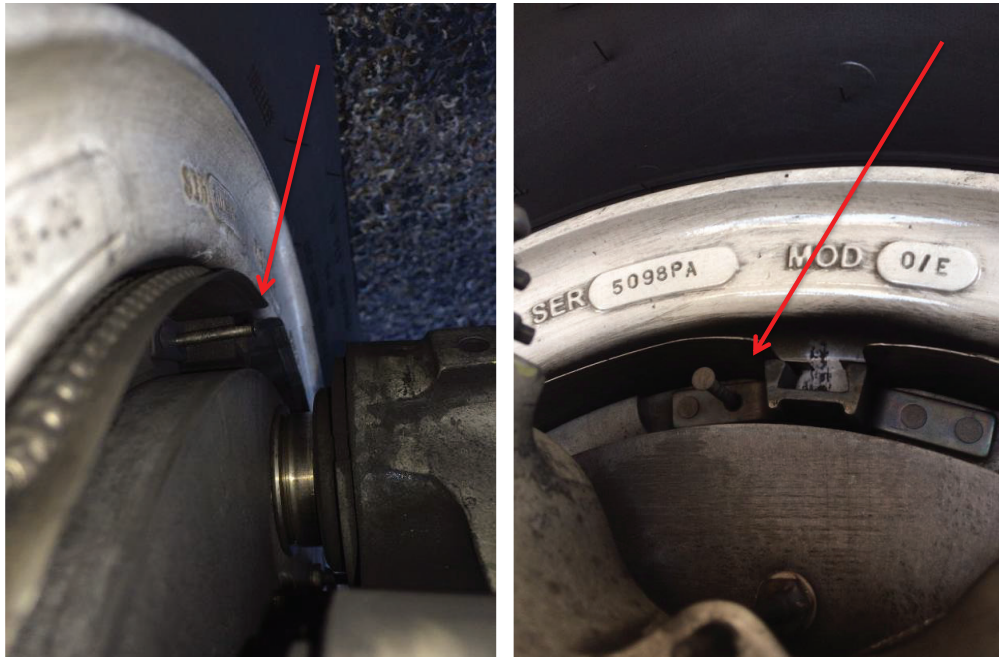
The new Brake Control Unit should solve the following:

1. Smooth out and remove the reports of brake grab, lurch, and asymmetric braking.
2. Remove the chance of a BRK FAIL CAS during Start and Shut-down.
3. Prevent BRK FAIL CAS on touchdown.

The aircraft will now transmit a smooth braking curve to control the aircraft when the pilot chooses to apply the brakes gradually. On some landings pilots may feel that the brakes are not as responsive as before but this is the BCU smoothing out the pressures to prevent the brake 'grab' we sometimes felt on the longer runways when trying to stop gradually.

MAX BRAKING PERFORMANCE IS NOT AFFECTED BY THIS UPDATE.

The previous brake rivets (shown below) have been replaced by solid ones that should not come out. Crew must still check the brake pads and surrounds closely but Embraer guidelines are that each 'pair' of rivets may have one missing as long as the other is fully intact.



If the Tyres or Brakes are close to limits please make sure you send an email to Maintenance before the limit is reached. This will give them a chance to plan the replacement of the worn parts rather than suddenly being told the aircraft is grounded in a remote location. The record for a tyre change is 45 mins in Paris because we could plan ahead and have the tyre and team ready for us. This was simply because of you sending in emails warning of the units getting close to limits and giving maintenance the ability to plan rather than just react.

2.2.5 NOSE BAGGAGE BAY

The nose bay is not a specific part of the pre-flight check other than to make sure it is secured. However, it is strongly recommended to check the bay (this is also where the gear pins and pitot covers are stowed) especially after maintenance procedures. There has been one case where the EMER BUS BAR was accessed through the screwed down panel inside the nose bay and it was left open with a c/b pulled and tagged. Maintenance procedures have been improved to include this check, but it is a good idea to check both sides of the baggage bay before the first flight of the tour and certainly after any maintenance action.



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2.2.6 NOSE STEERING AND TOWING ARM

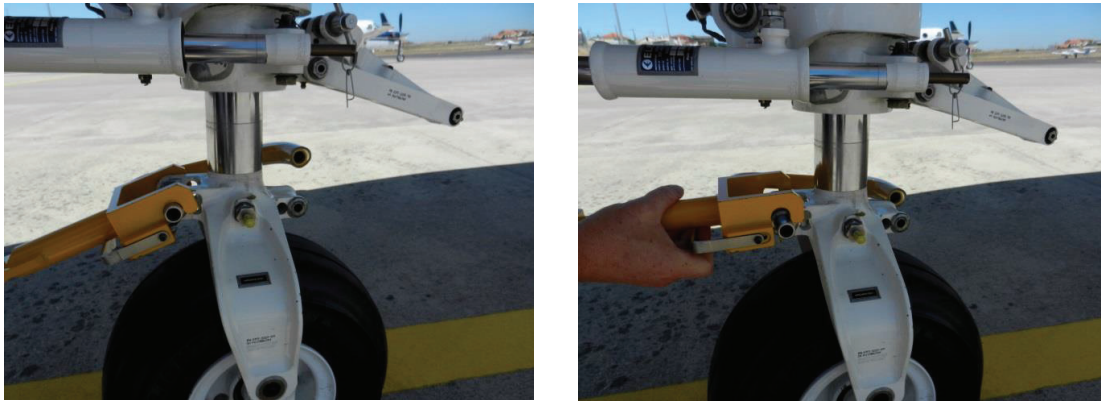
The nose steering is connected to the pedals by the scissor style torque link bracket and secured with a pin.



Inside the nose baggage bay is a towing arm attachment which can be used to attach the aircraft to a tow bar arm. The tow bar head attachment includes a shear pin and is connected to the aircraft with the nose wheel steering disconnected. To prevent damage to the steering, the towing bar attachment can only be attached correctly with the nose wheel steering disconnected.



Steering still connected - attachment will not connect



Steering disconnected - Arm attached correctly

The Phenom is authorised to be moved by a towbarless vehicle and normally the aircraft is secured to this vehicle using a strap that goes around the nose gear.

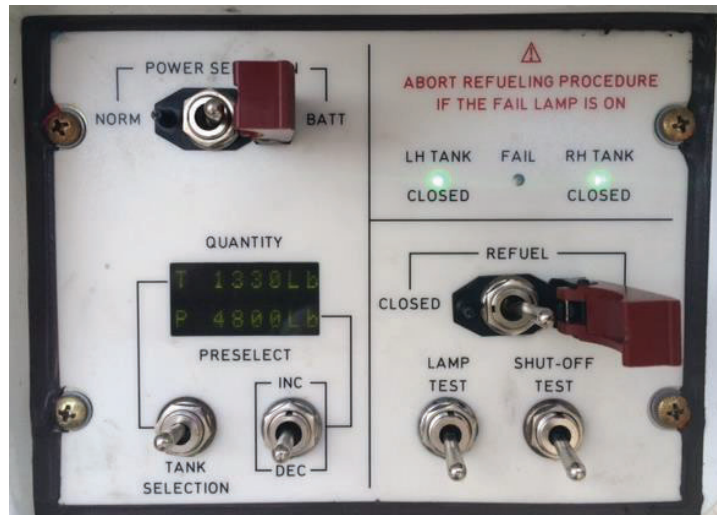
2.2.7 REFUELLING PANEL

The aircraft has Pressure and Over-Wing gravity refuelling capability. Pressure refuelling is normally used but there are a couple of techniques that need to be highlighted.

We have had some fuel spillages during refuelling and these have been linked to the over flow valve in the wing not closing correctly when the fuel is being filled to near maximum or a miscalibrated fuel sensor.

Crew must monitor the refuelling especially during the later stages and be prepared to manually stop the refuel at the first sign of any spillage from the wing tips.

In addition, crew should always select the power supply to battery even with a GPU connected. There is a chance that if the GPU has a power drop then the refuel panel selection could default to FULL. This would increase the chance of a spill but probably more importantly could over fill the jet with regards to the flight.



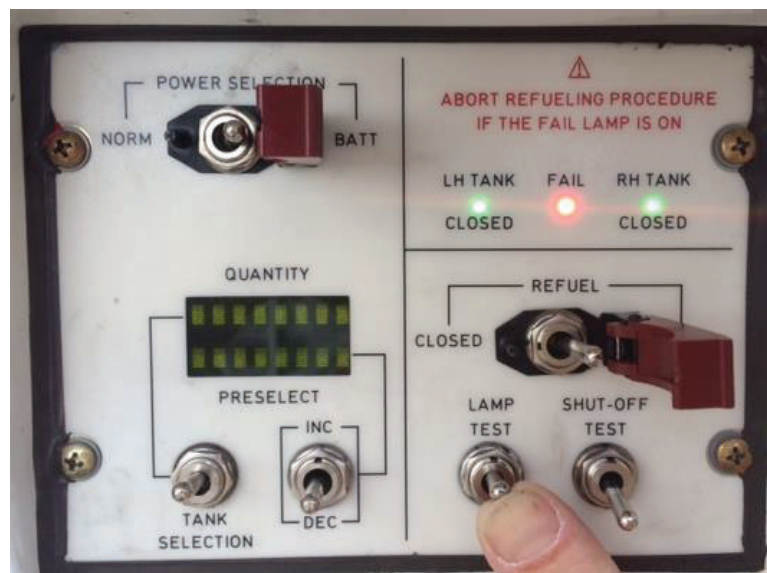
EXTRACT FROM POH 5-35 SEE POH FOR LATEST REVISION

REFUEL SwitchCLOSED

Make sure that the REFUEL switch is in the CLOSED position.

Lamp TestPERFORM

Observe that the LH TANK and RH TANK indicator lights keep lighted on, the FAIL indicator light lights on, and the display shows all the segments that form the characters lighted on.



Fuel Panel with Battery ON and Lamp Test being carried out

Fuel Quantity.....SELECT

Actuating IND/DEC switch, select the desired fuel quantity. If the quantity is not selected the refuel will stop at the maximum capacity of the tank or when the operator closes manually the refuel valve.

Pressure Filler Adapter OPEN
Open the pressure filler adapter and introduce the fuelling nozzle into the pressure filler adapter. Pressure 10 to 50 psig (0.7 to 3.4 bar).

Fuel Nose Valve Handle OPEN
Check if pressure is between 10 to 50 psig (0.7 to 3.4 bar).
Observe that the LH TANK and RH TANK indicator lights are on and, there is no fuel flow.

Fail Light OBSERVE
Make sure that the FAIL indicator light is off. If the FAIL indicator light is on, stop the refuelling procedure immediately.

REFUEL Switch OPEN
Observe that the LH TANK and RH TANK indicator lights go off.

Shut-off Test..... PERFORM
Observe that right and left tanks closed lamps are illuminated, and the FAIL lamp is off.

The fuel system can also overfill compared to the requested amount. Monitor the refuel especially towards the end of the uplift and try to ensure the requested fuel amount is taken and not more. One technique that seems to work to help reduce over-filling is to use the Shut Off/Test switch with about 100 lbs to go to stop the fuel flow. You can then hear the fuel settling in the tank and when the fuel settles release the switch to allow the last 100 lbs into the tank. This technique has been used successfully especially on jets where there is a tendency for the automatics to stop the fuel but the quantity continues to rise.

Fuel Cold Wx Ops

Last winter there were a few occasions in colder temperatures (-5°C) where the refuel panel did not work and refuelling had to be completed using the over-wing gravity feed. Normally this showed itself by the refuel valves staying shut when fuel pressure was applied or the fail light staying ON.

If the valves do not open make sure the REFUEL switch is set to CLOSED and then ensure fuel pressure from the truck, then OPEN the switch and see if the pressure can overcome the cold valves.

If the FAIL light remains ON or the valve lights stay green you are basically stuck and the solution is to wait until the system warms up or carry out over-wing refuelling.

On all occasions reported so far the fuel system has worked normally again once the aircraft was flown to a warmer place.

2.3 INTERNAL CHECKS

2.3.1 CABIN CHECKS

Make sure to check the cabin properly and do not assume that others have done the full checks before you. Make sure the documents are all present and correct and especially check the things that could get used or moved such as fuel cards and copies of the AOC.

When checking and cleaning the cabin make sure each one of the seats can move around and can fully recline. When in the recline position check the leather between the seat and the back of the chair which can collect food and crumbs and dirt. Also make sure you lift and clean under the headrest which is also a magnet for dirt if not used for a few trips.



The magazine rack is popular with passengers and it seems to work well on seat 6 as shown below. If you do have 6 pax then you can put the holder in the roller cabinet after either offering magazines or putting a couple in the seat pockets for that trip.



2.3.2 Wi-Fi

The NJE Phenom 300 has an internet system based on SwiftBroadBand (SBB) Technology using Cobham Aviator 200 system. It provides internet to the UCS-5000 multimedia server, router, and WiFi access point.

The Wi-Fi is switched on and off using the IFE switch. Make sure this is checked / recycled first if there is any problem with the connection.

The Wi-Fi connection will be seen when on-board the aircraft as 'Your Aircrafts Wi-Fi'. There is no password and all normal style passenger devices (iPhone, Android etc) should connect easily to the aircraft through the usual settings application.

Due to the size of the aerial the Wi-Fi is very slow and max is about 16mb/sec. You may wish to remind the pax that it can be used for checking mail and sending texts but not much more. The Wi-Fi used for the Go-Go streaming comes from the Go-Go box itself in the roller cabinet and so can give the impression of fast data streaming.

On the ground the aircraft is also connected via the Wi-Fi aerial but the actual data comes from a 3G modem so the connection on the ground may seem to be faster than in the air.

If you are testing or troubleshooting the Wi-Fi system make sure that you do it while airborne to avoid the possibility of a broken connection appearing to work when on ground.

During revenue flights, crew are to ensure that their devices are in Flight Safe and the Wi-Fi is deselected. Should crew need to contact the office you can email, SMS or call via the GTC using the Iridium connection.

During ferry flights, use of the Wi-Fi is allowed but should be limited to light data usage apps, for example Whatsapp.

A briefing card should be on all aircraft for the passengers.

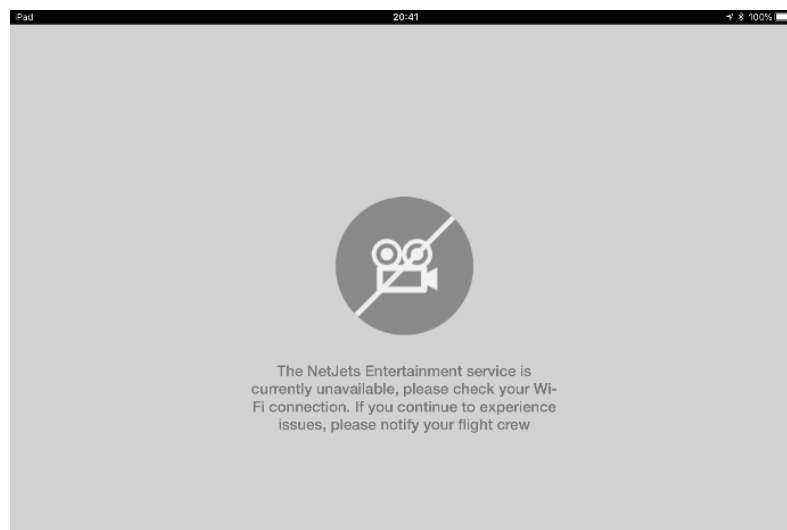
The Wi-Fi and Go-Go entertainment system is in the space underneath the roller cabinet and the additional c/b's are on the panel behind the left-hand pilots seat. There is no need for the crew to remove the access panel however it might be required to check the c/b's if there is a problem.



Wi-Fi Troubleshooting

The Go-Go has 3 main issues that we have seen and continue to see on a regular basis:

1. NetJets Entertainment App displays the following with a Wi-Fi error message:



Read what the message says (just in case it has connected to a ramp Wi-Fi instead of the aircraft one) and either connect to 'Your Aircrafts WiFi' or try connecting your iPhone to the Wi-Fi.

If 'Your Aircrafts WiFi' is not showing as available then recycle the IFE switch.

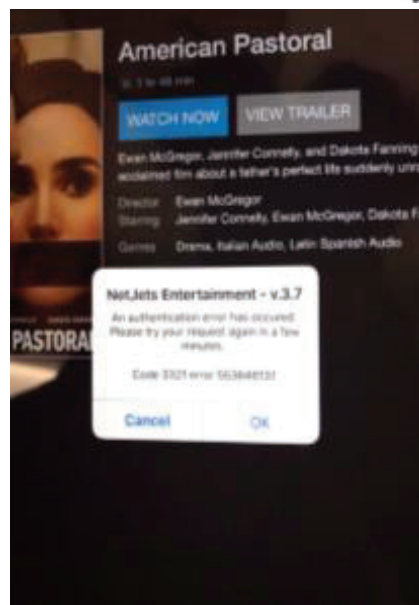
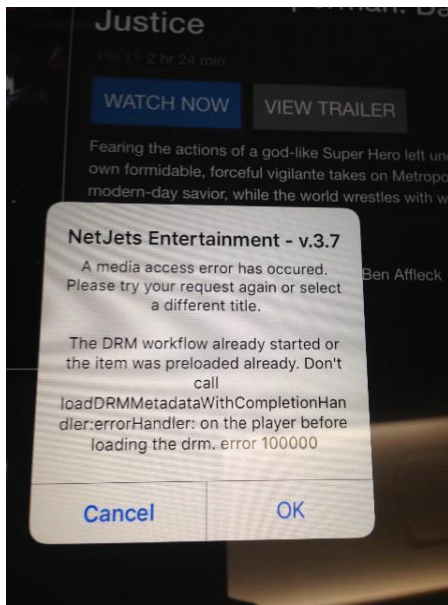
Confirm aircraft cabin power is on for at least 10 minutes after power on or reset (the system needs time to boot) before trying to reconnect to any device.

Any Wi-Fi failure can be reported to nje-wireless@netjets.com.

2. DRM Errors

DRM (Digital Rights Management) errors may occur for several reasons and show different error messages. In general, they are caused by a lack of communication between the UCS-5000 or the iPad app and the internet.

There are several methods that can be tried to solve it. Some different DRM error examples are shown below.



- Power up the aircraft and allow approximately 5-10 minutes for UCS to completely boot up (or do this in flight).
- Power on the iPad.
- Double tap the home button, close all running apps.
- Open settings -> wifi - connect to Your Aircraft's WIFI.
- Open the browser. Type this URL in the address bar: splash.gogo.aero.
- Wait for the page to completely load so you know that you have an internet connection.
- Close all browser tabs.
- Double tap the home button, close all running apps.
- Open the NetJets IFE app, tap the movie tile, tap a movie, tap 'watch now'.

If the above does not work connect the iPad to the internet away from the aircraft (it may be any Wi-Fi) and call nje-wireless@netjets.com to unlock the device, remove the app, and reinstall the app.

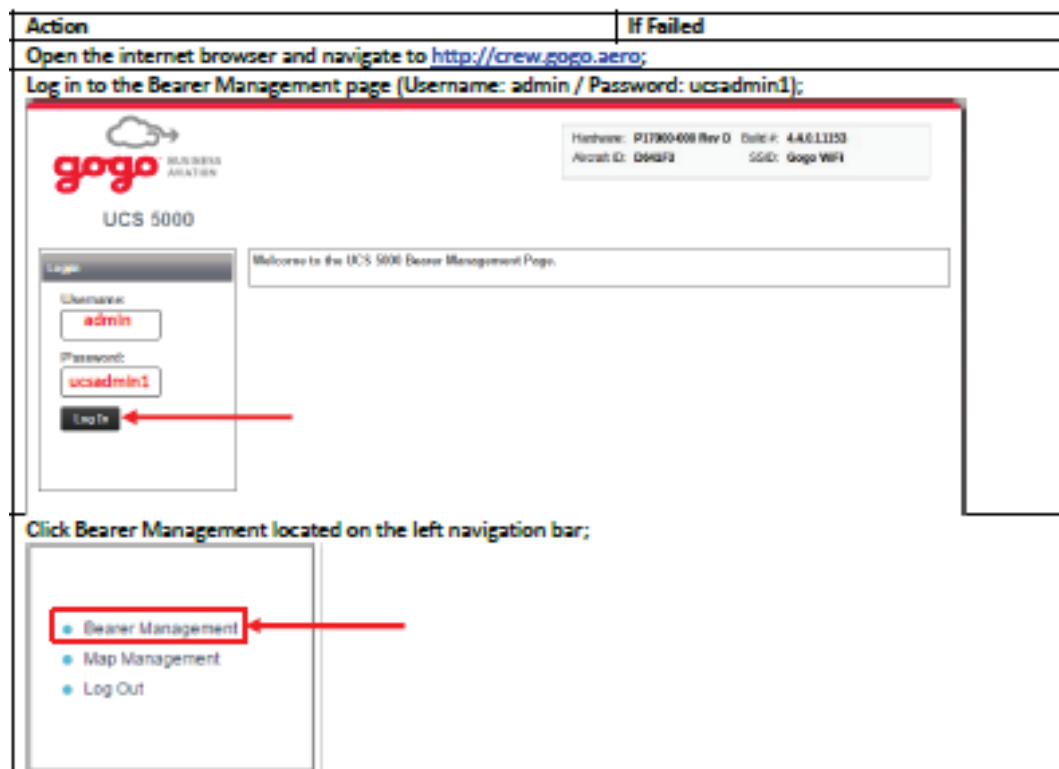
3. No Media available / App blank screen / no storefront available

Notes:

1. Occurs randomly on several occasions, most notably after contents update with poor 3G coverage.
2. Internet through the UCS-5000 can be provided by several means, but usually:
 - On ground internet is provided through the UCS-5000 internal 3G modem.
 - If no 3G network is available, then SBB is used.
 - If SBB is also not available, the 'Connection unavailable' below is shown.
 - In flight, the 3G modem is disabled, so only SBB is used.

Occurs if the UCS-5000 is not able to connect to the internet through any means. To check for SBB internet on ground, the 3G TM must be manually disabled.

On the iPad or aircraft laptop carry out the instructions below:



Disable Internet access via the Terrestrial Modem for connected devices to the UCS. To do this, uncheck the Internet Access Enabled checkbox;

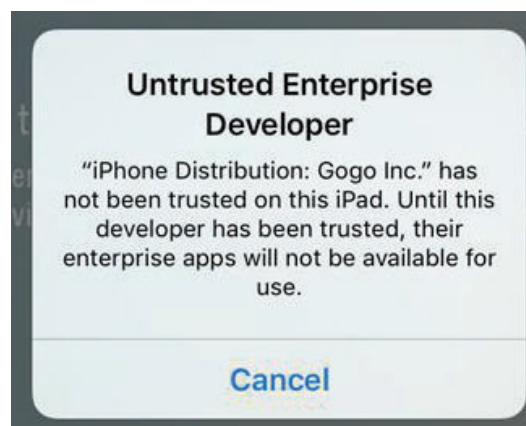
DATA BEARERS				
INTERFACE	STATUS	AVAILABILITY	BEARER ENABLED	INTERNET ACCESS ENABLED
Swift BroadBand	Data ENABLED	Data NOT CONNECTED	☒	
Terrestrial Modem	Data ENABLED	Data CONNECTED		☐

Note: This setting reverts to normal on the next power cycle.

Using the internet browser and navigate to <http://splash.gogo.aero/>:

Check access to internet	Go to 8.g
Set "IFE/ Refreshment Center" switch to OFF, wait 2 minutes and set it back to ON. Recheck Internet.	Contact Maintenance Control and fill in the ATL for maintenance follow-up.

4. This error is not very frequent, but has happened before. It is thought to be related to a lack of communication of the iPad with NetJets Device Management server. Several procedures have been used successfully on different events, a short list is presented below.



- Close all the apps;
- Ensure the iPad is connected to the internet (any Wi-Fi);
- Run the NetJets Entertainment app again;
- Hard-reset the iPad (press and hold home button and power button until device reboots with Apple logo);
- Ensure the iPad is connected to the internet (any Wi-Fi);
- Run the app again.

The above are hopefully solutions to some of the more common Go-Go faults that we can solve. The full brief for these issues can be found in the Maintenance Memo NTA 12/2017 Rev 1.

A total grey screen with no error messages is usually known as the 'Grey Screen of Death' and this normally is an indication that the Go-Go box is corrupted which sometimes occurs after an update to content has been tried. New solutions are being trialled at the time of publishing that will include remote access with the crew assisting using the laptop but at the moment the only way to correct this is access to the Go-Go remotely from the company through a maintenance base. The jet goes to Maintenance and Go-Go then access the box and reboot the content.

2.3.3 NESPRESSO COFFEE MACHINE

The coffee machine has an instruction book that should be kept alongside the machine. We have had some issues with no power to the machine and this has been caused by the circuit breakers above the baggage bay tripping. These are inaccessible to the crew and are on a separate inverter so if they pop it is a maintenance action to fix them.

The socket for the coffee machine must only be used for the coffee machine. It is on a separate circuit to the sockets in the cabin and the one in the cockpit.

THE COFFEE MACHINE WORKS OFF 120V DC. THIS IS AN AMERICAN VOLTAGE AND NOT THE EUROPEAN 240V ELECTRICAL STANDARD. DO NOT, UNDER ANY CIRCUMSTANCES PLUG IT INTO ANY POWER SUPPLY OTHER THAN THE AIRCRAFT SOCKET DESIGNED FOR THE MACHINE.

The coffee machine has a hot water spout that can be found in the side of the machine and also a milk container which is kept in the small cabinet underneath the machine.



Both of these fit into the slot on the front of the machine and are used from the settings panel on the top.

The coffee machine instruction book has guidance for using the milk container should any passengers ask however we do not generally advertise it to owners because it can be difficult to use and clean.

Once again the instruction book is your friend but the milk container fits in the front where we normally fire the hot water spout and then you are using the settings on the right side of the display. If you have the time, try the milk out on a ferry or hotspare but make sure the container is properly cleaned and dry before hiding it back in the cupboard.



2.3.3.1 COFFEE MACHINE TROUBLESHOOTING

No Power

If there is a problem with the machine firstly make sure the door is fully open, because there is a secondary power switch in the door which turns the machine off when the door is closed. If you need to carry out diagnostics try the coffee machine in another aircraft socket to see if it is the machine or the socket that has gone. The coffee machine socket has a fairly low power output tolerance and so plugging anything in that requires a lot of amps such as the vacuum could kill the socket. To test the socket only use an iPhone charger to see if it works.

**ONLY PLUG THE COFFEE MACHINE INTO THE AIRCRAFT DO NOT
PLUG INTO AN FBO OR HOTEL SOCKET**

(if you do, there will be a pop and a burning smell)

Empty Internal Tank

Another event that can seem a failure is when the power button is pressed the lights flashes quickly and then goes out. This can be a lack of water in the internal tank. The valve at the base of the water tank can stick and not allow a refill of the machine after a coffee is made. Remove the water container and exercise the valve at the base to make sure water flows out. Then reseal the water container firmly and check the machine again.

Cold Wx Ops

If the jet is staying for a long time in the cold, remove the machine with the rest of the liquid stock. In winter 2017 we lost 3 coffee machines due split pipes and water leaks. This was due to the aircraft remaining on the ground in cold temperatures for a couple of days with the water still inside the pipes. The machine froze and split the pipe which was then an expensive fix. The instruction manual shows the settings (page 7) and there is one called EMPTY FLUID SYSTEM which when pressed will empty the internal workings of the machine. In winter if there is ANY chance of the jet staying out in the cold you must as a minimum carry out this procedure. However, it is best to put the machine into the FBO.

Limescale

The Evian water we have on board the aircraft has a very high level of minerals that cause limescale inside the coffee machine. We have had several machines damaged by the limescale even though we descale the machines a minimum of once a year normally in November when we remind you to do it after the busy season (see instructions below).

If you start to have issues with the coffee not coming out of the spout properly then do not wait to be asked to carry out a descale please do one as soon as you can. The kit should be on the jet normally in the nose but are available from all re stock locations.

To try and help with the limescale damage we have placed large bottles of Volvic water on board which is just to be used for the coffee machine. Whilst it can be slightly trickier using the large bottle to fill the container please do use the Volvic water for the machine to try and prevent us losing machines to limescale.

We have spare coffee machines kept in Lisbon (used for Latitude and Global fleets as well). Maintenance have a company near the office who fixes our machines so the spares are kept here and then shipped to where they can be fitted and the broken unit returned to here. This means that the turnaround time for a broken machine should not be too long after the office has been informed.

2.3.3.2 COFFEE MACHINE DESCALING

The coffee machine also has the ability to be descaled and cleaned. The descaling pouches are at stock locations and each box contains two pouches. Only one pouch is used at a time so the other can be stored in the aircraft nose bay or returned at a stock location.



The kits are used by us and the Global and Challenger.

Nespresso have said that the Evian water we use has a very high concentration of mineral content and that we should descale the machines on a regular basis. We suggest carrying out this operation at least every 12 months or sooner if any passenger comments are received about taste. It is intended that we will send out a reminder in each year to carry out a fleet wide descaling, for example during November, after the summer rush.

The instructions for descaling the machine can be found on page 16 and 17 of the instruction manual kept next to the machine, however, the basic instructions are shown below.

The process is very simple but requires a 1 litre container to collect the water as it comes out of the coffee and hot water spout in turn. The stock locations should all have a flask that can be used for this or, if not available, you will need to borrow a suitable container. It also cleans the drip tray and water comes into the tray through the slot on the left hand side. It is difficult to collect the water from here so the suggestion is to use kitchen roll to mop up the water as it filled the drip tray and put them into the bucket.

Instructions:

- Add the 100 ml sachet of descaling fluid to 500 ml of water in the water tank.
- Make sure there is no coffee capsule in the mechanism and empty the drip tray of water and any collected coffee capsules.
- Make sure the hot water spout is attached to the front of the machine.
- Enter the menu option on the top of the machine and select descaling. It will then guide you through the process.
- Place the collecting container underneath the coffee spout.
- Press the confirm button on the top of the machine.
- Hot water with descaling fluid will now be passed alternately through the normal coffee spout, then the hot water spout, and finally the drip tray. This is repeated 3 or 4 times.

The machine will tell you when the descaling is finished and it will prompt you to fill the water tank with clean water and it will then carry out a flushing process in the same sequence as the descaling process. Once again you will need to collect the water.

The whole thing will take around 20 minutes to complete.

Please carry out a descaling process as soon as you get the chance and send an email to lis-phenom@netjets.com when completed.

Note: REFER TO COFFEE MACHINE BOOKLET FOR LATEST REVISION

2.3.4 PASSENGER BAGGAGE

Every year we have several Case Management issues where pax baggage is the subject due to either missing bags or additional bags carried on the flight.

It can be impossible to be 100% certain about the bags but if there is any doubt please check with the passengers to make sure you have their luggage.

Some of the events we have had are mentioned below:

- Pax bag left in FBO vehicle - passengers 'helped' with unloading luggage, main cases went into the baggage bay but a smaller bag was left in the car because it was believed the passenger had carried it aboard himself. Car was not checked by FBO staff after unloading.
- Extra bag brought from hotel by the hotel staff. All luggage in hotel courtesy car was unloaded in the car park after the pax has headed inside. The car was emptied and the luggage went straight through to the aircraft. The extra bag was not seen until landing when the pax said it was not theirs.
- Extra bag was carried after security checks. The security person had left his bag in the security scanner area and it was collected by the FBO staff along with the passenger luggage after it had all been screened. Bags went to vehicle and pax got onto aircraft.
- Busy FBO delivered 2 sets of bags to 2 NJE aircraft but put all of the bags onto the one jet (that had 6 pax) and none on the other aircraft (with 2 pax). The crew of the jet with no bags questioned the owners but by this time the first jet had taken-off.
- Crew arrived on a recovery and as they went to meet the 6 pax the FBO staff pointed to baggage on a trolley and said that was all of their pax bags. Bags were loaded onto the aircraft but on landing it was found that 2 bags belonged to another crew and not the passengers.

Some locations naturally make us nervous, a busy Mikonos in summer can be a complete nightmare but even at the more usual safer places make sure you are vigilant about the bags. If you meet the owner, see the car and bags unloaded then your job is easier but how many times have we been both at the aircraft when the bags are brought out by the FBO.

The above are events that have actually happened and are shown purely to give examples of the different scenarios that can occur that are almost totally outside our control.

The bottom line is that with our operation we will not be able to be 100% sure every single time but if you have any doubt then try and check with the passengers.

Also remember to brief your colleague about any unusual or specific instructions he may need if unloading the bags at destination. Simple things like a reminder that 'I put 4 bags in the cabin and 3 in the hold' can assist both of you to make sure all the bags leave the aircraft.

Photographs can sometimes help – some crew take pictures of the bags before they get taken off to the jet by the FBO staff. You do not necessarily need to email the pictures to your colleague but just let them know the number and type of bags and then have a quick check yourself while the pax get on board.

It is worth taking an extra few minutes if you have any doubt at all about the bags.

2.3.5 AIRPORT SLOTS AND A-CDM GUIDANCE

From EuroControl - Airport CDM Implementation Manual Ver 5.0 Mar 2017:

Airport Collaborative Decision Making is the concept which aims at improving Air Traffic Flow and Capacity Management (ATFCM) at airports by reducing delays, improving the predictability of events and optimising the utilisation of resources.

Implementation of Airport CDM allows each Airport CDM Partner to optimize their decisions in collaboration with other Airport CDM Partners, knowing their preferences and constraints and the actual and predicted situation.

The decision making by the Airport CDM Partners is facilitated by the sharing of accurate and timely information and by adapted procedure, mechanisms and tools. The decision making by the Airport CDM Partners is facilitated by the sharing of accurate and timely information and by adapted procedure, mechanisms and tools.

Note: EXTRACT FROM AIRPORT SLOT BRIEFING DATED AUG 2015 (C365)

Please find below some guidance and some background information in terms of SLOT adherence

1. Crews should always plan to comply with the OFP times (Off-Block times).
2. The OFP times consider the EOBT and airport slot.
3. All deviations have to be coordinated with FltDispatch.
4. CTOT are issued by ATC (Slot tolerances are -5+10 and need to consider taxi times) and override Airport Slots.
5. Ready early: Coordinate new ETD with FltDispatch.

If in doubt: Call FltDispatch.

BACKGROUND INFORMATION

We, as a company, have been having some issues with airport slots that have resulted in delays or heavy fines because FCMs may be unaware of the procedure at Slot coordinated airports.

At these airports, and other airports where airport slots are required, you need to be aware of and comply with, your OFP time - this is your Estimated Off Block Time (**EOBT**).

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PLAN 3634 NJE938N EGWII TO EGLC C5XS LRC/F IFR 08/19/15  
NONSTOP COMPUTED 1317Z FOR ETD 1410Z PROGS 1906UK CSDXV LBS
```

Always assess that your EOBT is connected to a Slot. This will be reflected in your briefing. It will typically be shown with your call sign and tail number and should be the same as the **EOBT**. Sometimes the briefing shows “**Needs Review**”; this means the slot has been requested, but is not confirmed yet.

DEFINITIONS

The **Airport CDM (A-CDM)** project integrates processes and systems aiming at improving the overall efficiency of operations at European airports, particularly focusing on the aircraft turn-round and pre-departure sequencing process. This in turn allows the ATM Network to run more fluently.

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A-CDM is about partners - airport operators, aircraft operators, ground handlers, air traffic control and the Network Manager - working together more efficiently and transparently in how they work and share data. It allows better decision making, based on more accurate and timely information, with all airport partners having the same operational picture.

The benefits are visible at a network level, with more accurate take-off information feeding into the air traffic flow and capacity management system run by EUROCONTROL's Network Management. The network will be able to use the available capacity more efficiently. More effective use of slots results in reduced delays, improved predictability of events during a flight and optimised use of resources at airports.

During daily operations, more specific information can be found in the Jeppesen airport briefing pages (10-1 section 3 Departures), details printed on flight folders or leaflets available at the handlers/FBOs.

EOBT. Estimated Off Block Time - This is designed to give all parties involved an idea at what time you will be finished with all flight preparations including boarding of the passengers and if applicable, De-icing on stand.

TOBT: Target Off-Block Time – the time at which the operator or handler estimates that a flight will be ready, doors closed and ready to start up. You may be asked by the FBO when you will be ready, so that the FBO can pass on the TOBT.

TSAT: Target Start-up Approval Time – the time, provided by ATC, at which a flight may expect start-up clearance in order to achieve an optimised departure sequence at the takeoff runway. TSAT takes into account the TOBT and any EUROCONTROL slot (**CTOT**).

TTOT: Target Take-Off Time – the expected take-off time, taking into account the TOBT/TSAT, as well as taxi time, wake turbulence and SID separation for the active runway. Although different airports may apply small variations in their local procedures, the principle remains the same: at participating airports, you should keep the FBO/handling agent informed of the progress of your turnaround.

When your **EOBT** is put into the A-CDM process, it becomes a Target off block time (**TOBT**): – the time at which the FBO estimates that a flight will be ready, doors closed and ready to start up. You may be asked by the FBO when you will be ready, so that the FBO can pass that on as the **TOBT**.

Target Start-Up Approval Time (**TSAT**) is the target time for start-up approval according to the A-CDM procedure. It also takes into account a possible **CTOT** (tolerance -5+10 mins). As a rule, **TSAT** and any changes to the **TSAT** are transmitted to the person responsible for the **TOBT** (in our case the FBO) who then forwards it to the flight crew.

You are only allowed to call for start-up and clearance not earlier than five minutes before **TSAT** (**TOBT**).

After clearance to start has been received, you are expected to call for taxi +/- 5 minutes of your **TSAT**.

If you want to depart earlier than your Flight planned **EOBT** (on passenger request or other reason) you need to request a new **EOBT** through Flight Dispatch.

Any delays need to be coordinated through Flight Dispatch and the FBO.

Again: If in any doubt at all, call Flight Dispatch.

Note: Further background reading can be found in IATA 'Airport CDM Implementation' Manual.

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CHAPTER 3 - IN-FLIGHT

3.1 SOCKETS AND PED-BELTS SWITCH

The aircraft is cleared for normal use of electronic equipment in accordance with the OM-A. The PED should be in flight-safe mode but may still be connected to the aircraft internet and Bluetooth speaker.

Embraer Technical Disposition ETD-2015-P300-00217735:

The outlets on the Embraer aircraft model EMB-505 were tested using devices representing PED, without known interference on aircraft systems. Based on this fact, Embraer informs that PEDs can be plugged into the AC outlets at or below 10,000 feet provided that they are in AIRCRAFT MODE with only Wi-Fi enabled.

3.2 OVER-WEIGHT LANDINGS

There is no buffer above the max landing weight of the Phenom (17042 lbs) and if that mass is exceeded by any amount at all, a full overweight inspection must be carried out.

There have been 3 landings that were assessed to be overweight and all were found after the event. This meant that the aircraft had to go to maintenance after the event at great additional costs for the overweight checks to be carried out.

Please ensure that you are well within the max landing weight limit. If you have a short flight, there is a large taxi bias you did not use or you get a large ATC shortcut then make sure you calculate the landing weight accurately. Do not try and land just under the max weight because the financial penalty is severe. Make sure there is a good buffer of at least a couple of hundred pounds.

Make sure that you put the correct ZFW and pax weight into the GTC. Use the figures accurately from the OFP to make sure your weight is correct and then the jet will tell you the fuel and final landing mass. The overweight landings we have had were less than 100 lbs out but still required an inspection.

The Ops Alert on Comply365 explains that Dispatch will not ask you to tanker max fuel and will aim to give a buffer of at least 700 lbs purely on fuel loaded. Therefore, with an accurate weight of pax you should not be 'surprised' by the FDM data after the flight.

3.3 WINDSHEAR DETECTION SYSTEM

The aircraft has had a windshear detection system for over 12 months now and we are still analysing the first year's results in terms of how sensitive or accurate it is.

The windshear system is based on the same calculations as the stall warning system and uses multiple inputs to provide the warnings we see and hear. The windshear system should be used alongside other recognition tools including the windshear conditions shown in the OM-B to create a full picture of the environmental conditions.

Just like the stall warning system, if the warning is an AOA trigger, then the warning will show for the required 3 seconds and then disappear. We have seen full download traces of the stall warning from Embraer where the AOA gauge goes from 0 to 10 and back down again in a second but the stall warning is still, correctly triggered. However, in these situations, crew are able to use their experience and other parameters to know they are not stalling.

So just like the stall warning system, the windshear detection system should be used as an aid to detection and not necessarily a black and white go, no go system.

But exactly the same as the stall warning system, safety is always paramount and all reactions to any warnings should always be safe and positive.

CHAPTER 4 - POST-FLIGHT

4.1 PASSENGER BELONGINGS CHECK

Make sure you check the back of the cabin for any items that may have been left by the passengers. Sometimes difficult to see please make sure to co-ordinate with your colleague but try not to let the passengers leave until you have checked the cabin.

Some items however may not be seen so easily:



Remember to check down the side of each seat. There is a recess which hides things very well.

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4.2 POST FLIGHT EXTERNAL INSPECTION

The aircraft should be checked after flight as soon as possible after the pax have disembarked to check for any in-flight damage or wear and tear. Crew should check the aircraft for any possible bird strike or other damage as well as the tyres and brakes.



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4.3 ENGINE OIL

The Oil level check should be carried out between 10 and 20 minutes after engine shutdown. If it is carried out when the engine is cold the oil level will indicate lower than actual due to migration of the fluid to the bottom of the engine after cooling. Check the oil level after shutdown and if required add any extra oil when convenient, but do not add more than required. Aim not to fill to the MAX line, the Pratt and Whitney representative recommends that the engine is kept close to the MIN.

Any oil above the MIN will be 'expressed' by the engine, so if you have added oil, do not assume the engine is burning unusual amounts if you check it and see that it has fallen to the MIN or below.

The most accurate check is achieved with the step ladder avoiding any parallax.

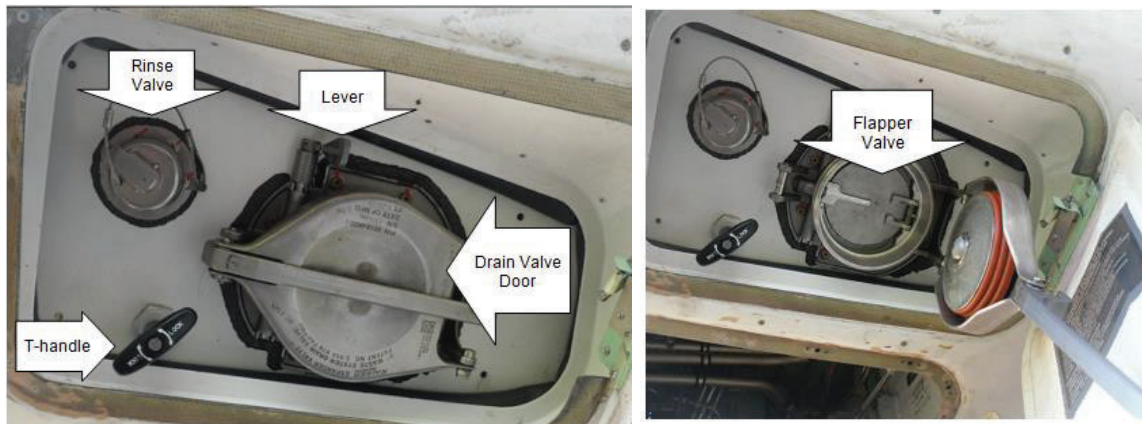


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4.4 TOILET SERVICING

The toilet is relatively simple but we have had a few issues that can cause problems for both us and passengers.

FOL 008/14 gives guidance about prevention of leaks in cold weather and also gives advice about blockages.



This guide is added to show how the servicing that should be carried out by ground staff. Only 4 litres of fluid should be added to complete the procedure. Because some passengers put tissues and hand wipes down the toilet it can be worthwhile to ask the toilet servicing staff to carry out a flush through a couple of times with the rinse water before adding the 4 litres and closing up. Make sure you monitor the toilet service if you are there. On more than one occasion excess fluid has been put into the toilet and this can prevent the system working. If the toilet is overfilled the flush is inhibited.



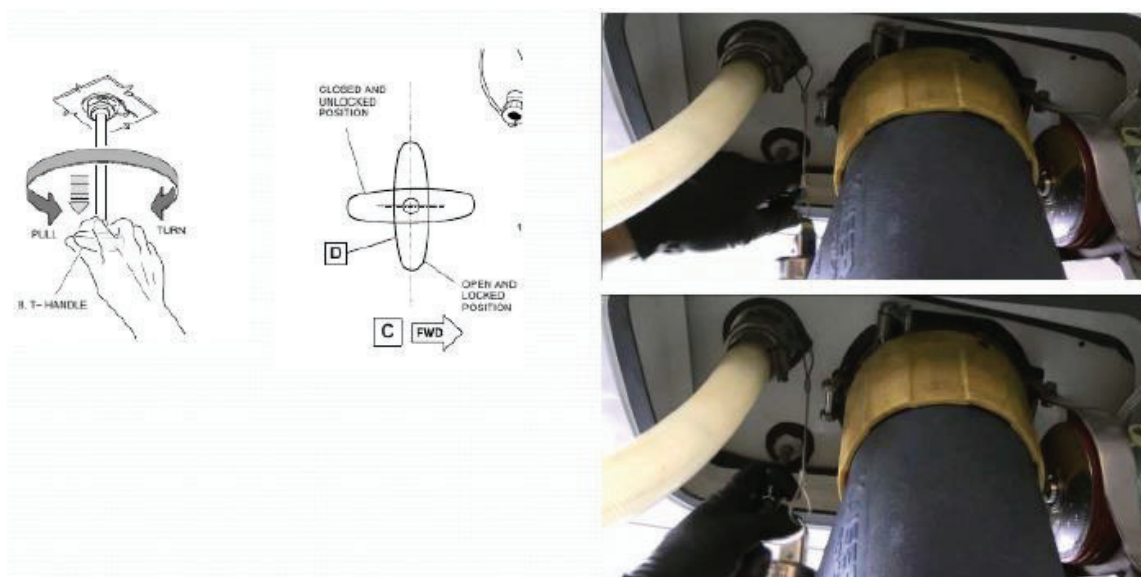
The panel is prepared for the service by opening the inlet rinse valve and the drain valve door. No fluid should be leaking from the internal door. It is very important that the internal flapper is not opened until the waste pipe is in place. Even though no waste should come out until the T handle is pulled, it has been seen that the internal flap can be held open or jammed open after a previous service. It is important that the pipe is clear of fluid after the service to ensure that freezing does not occur and the toilet will not be able to be serviced.

The waste pipe and the rinse pipe are attached as shown below:



Pull the T handle out, turn it to lock, and open the RTS waste drain valve inside the toilet itself. Continue the waste tank drainage until the waste flow stops.

NOTE: The T handle must not be turned many times in the same direction because it may cause damage.



Flush the waste tank with water through the inlet rinse valve for about 3 minutes to clean the toilet fully. Then close the T handle and add 4L of water which will remain in the toilet system for the flush mechanism.



Note: It has been known for the T handle to come unscrewed and 'disappear'. Twice it has been found in the space behind the hinges for the servicing door. The T handle can become loose after several servicing where the bolt holding the handle on becomes unscrewed without realising it.

Once the service is complete the waste pipe and rinse pipe are removed, and the doors closed. Make sure the inner door is secured before closing the outer door. The inner door will be closed by the action of closing the outer door but it is probably worth checking if needed.



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Closing the Service Panel drain valve Cap automatically closes the flapper valve.



NOTE: After every service make sure that the Rinse (Inlet Cap) is closed properly. On a few occasions this cap has not been connected properly and this has only become apparent after take-off when a vibration noise is heard and the toilet itself is vibrating as the cap bounces around the servicing area on its cable. In addition to this, the fluid from the toilet will be syphoned out and down the back of the aircraft fuselage.



The cap is not screwed on but the ball bearings shown above are locked in place by the cap lever. On occasions the ball bearings have become jammed and do not lock when the lever on the cap is closed. After servicing there is no need to remove the cap but just give a decent pull to make sure it is locked, if it comes off check inside and look at the condition of the 3 bearings and see if they can be freed.

Toilet Troubleshooting

Not Enough Water

Four litres of water should be enough to work the toilet without incident. If the toilet does not flush when you check it during pre-flight then it is possible (as long as the flush button is not red in colour) that there is not enough water in the bowl. Either check inside the toilet using a torch or just add another litre of water and try again. Because of the overflows and problems we have had and continue to experience, we have been very conservative with the required water amount for the toilet. The overflow capability is 16 litres (which includes the bowl itself) but we need to stay as far away from that as possible. However, adding another couple of litres to the toilet should not cause any problems.

Another reason to use just four (4) litres of fluid is because numerous of our floods have been because the servicing team did not actually empty the toilet when they should have done, therefore, if four (4) more litres are added then we should still be below the flood level even with double fluid.

It is important if you have any doubt to the ability of the servicing team that you monitor from the inside and check primarily that the toilet was emptied before any flush is done.

Too Much Water

Too much water normally shows itself with a flood, however, if the system works correctly, then the flush button should turn red at approximately 90% capacity which is about 10 litres (the tank holds 12). Therefore, if the flush does not work, check the button first to see if it is red. If it is, then do not use the toilet anymore and get a service when possible.

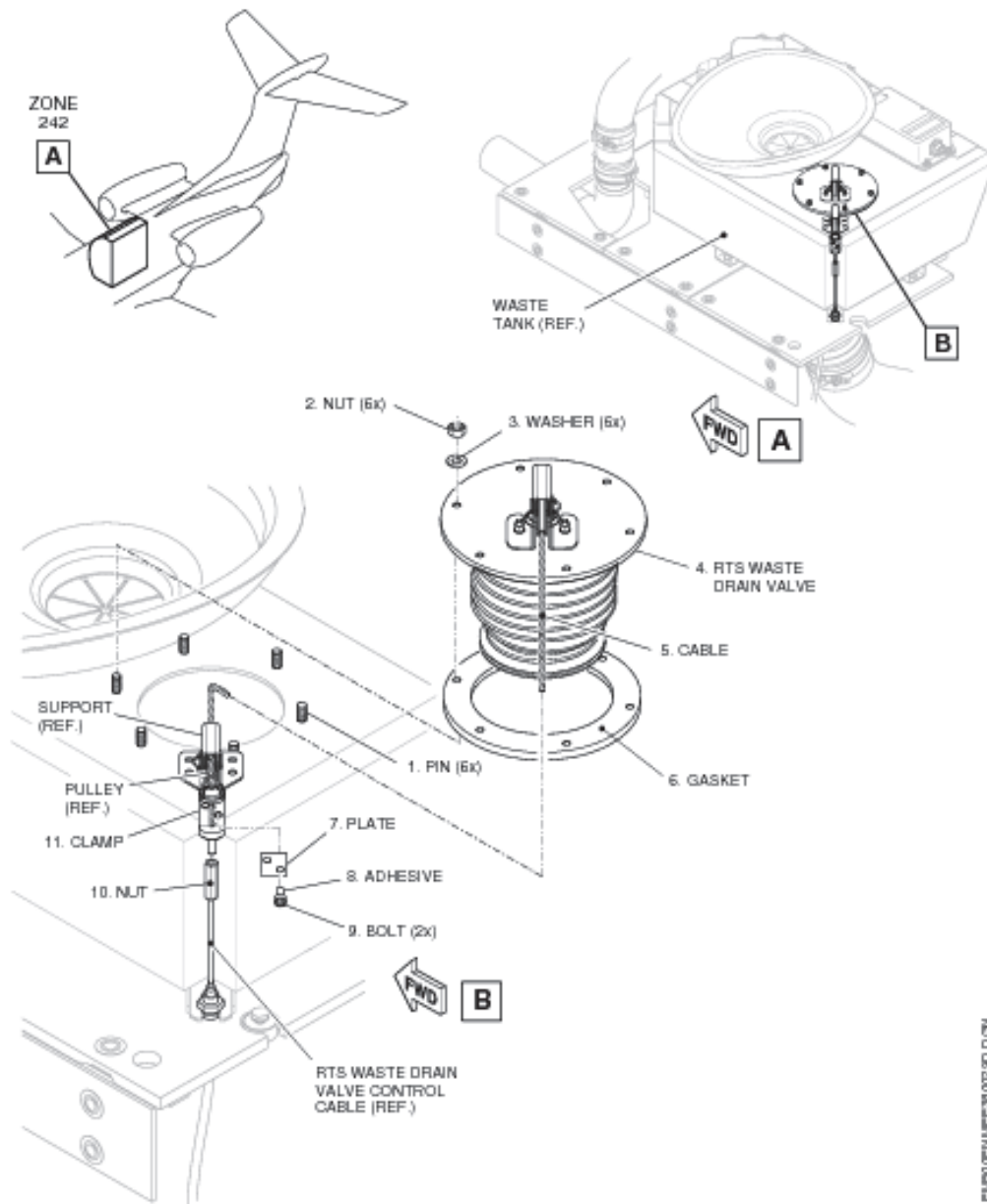
Dirty / Frozen Outflow Valve

The outflow valve can attract tissues and other stuff thrown into the toilet. This can show itself by the toilet developing a slight leak of the outflow valve not being able to be opened in the worst case.

If there has been some fluid loss and the toilet cannot be emptied, this normally means that the outflow valve has leaked into the outflow tube and then frozen there blocking the pipe. This is then the classic scenario for the servicing team to think the toilet is empty and pour a load more water in.

If icing is suspected, try and get the servicing done after the jet has been on the ground in warmer air. Make sure the servicing team put a flush rinse though the bowl so empty, fill, empty, fill.

Check the valve again and if it still is not clean, then you may need to resort to cleaning it yourself.



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CHAPTER 5 - AIRCRAFT SYSTEMS

5.1 MAIN BRAKE SYSTEM

Note: EXTRACT FROM 'RUNWAY OVERRUN PREVENTION MEMO' DATED FEB 2015

The main brake consists of a brake-by-wire system controlled by either the pilot or copilot via rudder pedals. Rudder pedals actuate the pedal transducers that send the brake inputs to the Brake Control Unit (BCU), which controls both LH and RH brakes independently. Then, the BCU, which is connected to the EMERGENCY BUS, receives all brake interface signals and controls the shut-off valve (SOV) and both brake control valves (BCVs) for braking capability. Each wheel brake is commanded by a dedicated, electro-hydraulic BCV. The BCU measures the output from the wheel speed transducer, pedal transducer, and pressure transducer and provides a commensurate electrical command to the associated BCV. Hydraulic pressure is available to the BCV through a brake SOV, electronically controlled by the BCU. This SOV provides pressure only when the pedals are pressed and the aeroplane is on the ground. It also provides pressure to the BCVs during the BCU built in tests. In case of failure or leakage in the main brake system, the SOV prevents this problem from affecting the other hydraulic consumers.

The brake system includes the following functions:

- Locked wheel protection;
- Anti-skid protection;
- Touchdown protection;
- Gear retract braking;
- Initiated Built In Test (IBIT).

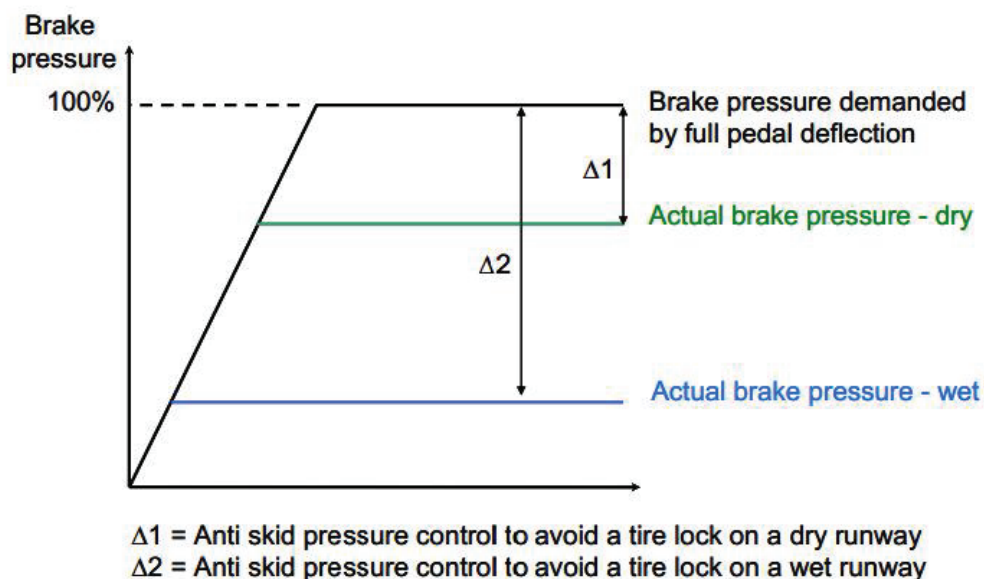
In case of hydraulic system failure, the emergency/parking brake is available and must be used carefully to stop the aeroplane.

Some aspects of the brake system design should be considered to better understand the expected behaviour of the aircraft during the landing:

- **Brake by wire:** Electrical transducers will transmit pedal deflection signals to the brake control unit (BCU). While no hydraulic components are installed on the pedal assembly, the only force feedback expected to the pilots is from the force spring. That means, regardless of the runway condition and pressure being applied to the brakes, the pedal resistance force, felt by the pilot, will be the same.
- **Anti-skid logic:** This feature provides optimum brake performance avoiding a wheel to lock or skid. It will deliver the best brake pressure possible to the runway conditions. The BCU receives inputs from the pedal transducers, understanding the amount of brake pressure demanded by the pilot, and wheel speed signal, to monitor the wheel deceleration and make the adjustments on the pressure as necessary.

BCU, due to the anti-skid function, will automatically calculate the maximum pressure that can be delivered to the brakes, based on the pavement conditions. As a result, pilots must be aware that for a same pedal deflection, in different pavement conditions, different amounts of hydraulic pressure will be applied to the brakes.

For example, on a full brakes application the pressure that is actually being applied to the brakes, by the BCU, will be higher at a dry runway and lower on a contaminated runway. As a result, pilots will notice lower deceleration on a contaminated runway, compared with a dry runway. The graph below shows a comparison between the braking pressure on a contaminated runway and on a dry runway.



5.1.1 EMERGENCY / PARKING BRAKE

Emergency / Parking brake system has two functions: first as the aeroplane parking brake and second as an emergency brake. It is operated through a T-handle located at the central pedestal of the cockpit. The T-handle is linked to the emergency/parking brake valve via a steel cable. The emergency/parking brake has a pressure accumulator isolated from the hydraulic system by a check valve and a shut-off valve, which is open during engine start, when the aircraft is on the ground and both Thrust Levers are in idle for more than 50 seconds. The emergency/parking brake is mechanically actuated and provides pressure to all brakes allowing the pilot to modulate brake pressure in emergency situations. The accumulator has sufficient pressure to provide six full-brake applications. The emergency/parking brake handle, when pulled up, causes hydraulic pressure to activate the brakes. When pressure is equal or higher than the brake contact pressure, a pressure switch turns on a white lamp at the cockpit front panel to alert pilot.

5.1.2 EMERGENCY BRAKING TECHNIQUE

The proper emergency brake utilisation consists in pulling the emergency/parking brake handle with care until the parking brake light illuminates. Initiate braking actuation using very little handle displacement. If it is necessary, in order to adjust the aeroplane deceleration, the handle must be carefully moved up as required. Steadily hold the emergency/parking brake handle at the desired position. Do not keep moving the handle up and down in order to minimise the possibility of tyre skidding.

NOTE:

- *Anti-skid protection is not available for emergency braking. Therefore rapid emergency / parking actuation can lead to tyre skidding.*
- *In case of tyre skidding, move the emergency / parking brake handle a little and maintain normal aeroplane directional control using the steering system.*

5.1.3 BRAKE FAIL CAS

We have had two incidents in the USA (NJA) where the crews received a BRK FAIL CAS at touchdown and the EMER BRK procedure had to be used to stop the aircraft.

Embraer has analysed both events and the root causes are highlighted below:

In the first incident the crew had inadvertently applied a deflection on the RH brake unit during the final stages of the approach and landing (most likely during final drift correction). The RH brake deflection was released just at touchdown. However a residual or 'dummy' pressure was still present in the system when brake pressure became available after touchdown. As there was insufficient time for the system logic in the BCU to reset, the BCU closed the SOV and triggered the BRK FAIL condition.

In very basic terms the BCU saw brake pressure but no pedal application after wheel spin up and WoW sensor it diagnosed uncommanded pressure and showed a BRK FAIL CAS.

Analysis of the event revealed the following:

- Pedals were slightly applied prior and at touchdown (dummy pressure is noted).
- RH pedal released immediately after touchdown but RH pressure remained, (therefore, un-demanded pressure).
- BRK FAIL CAS message is triggered ~1 sec after touchdown.

We have all been taught that the BCU on the Phenom provides touchdown protection, and so if a crew lands with their feet on the brakes the BCU will provide this protection. However the BCU logic does not currently factor for the above combination of events.

Following the analysis of this event Embraer is working on BCU # 10 that will address this issue. The BCU 10 will be a software update that should be on all aircraft by October 17. Currently only one aircraft is left to be fitted with BCU 10.

In the second incident, the BRK FAIL CAS was activated during a very smooth landing and subsequent bounce where the WoW sensor was not activated immediately after touchdown. The aircraft wheels initially touched the ground very smoothly and the wheel spin up sensor detected wheel speed, however the WoW sensor did not detect a landing and therefore the spoiler did not deploy straight away.

The aircraft momentarily got airborne again and wheel stopped spinning due to anti-skid activation. The pedal brake pressure was applied after wheel spin-up but before WoW contact, thus 'confusing' the BCU logic into a BRK FAIL condition.

Analysis of the event revealed the following:

- Pedals were applied only after wheels' spin-up, while WoW remains in AIR mode.
- Aircraft bounces and wheels lose contact to the ground and enter in skid.
- BCU (anti-skid system) removes pressure to avoid wheel lock. Condition remains for longer than 2 seconds.
- BCU monitor saw pedal application without pressure for more than 2 seconds and activates Manual Brake mode, which declares a BRK FAIL.

This issue will also be addressed by the BCU #10 software. A more permanent solution will be with us in mid-2017 with a new BCU (hardware) that will be fitted to the jets.

Note: REFER TO FOL 010/13 and FOL 005/16

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5.2 GPS - TOTAL FAILURE

EXTRACT FROM 'LOSS OF GPS ON THE PHENOM' MEMO DATED
AUG 2016

GPS is the only positioning input to the Garmin 3000 FMS. If both GPS signals are lost, the aircraft does not have a conventional backup for the FMS to have positional information.

The VOR and DME can be used for conventional navigation but cannot be fed into the Garmin FMS.

The AHRS takes a signal feed from the GPS, Magnetometer, and Air Data and if any of these feeds are interrupted, an AHRS FAULT CAS is seen by the crew which means the AHRS is working in a degraded mode of operation.

5.2.1 AHRS FAIL SCENARIO VS. AHRS FAULT SCENARIO

It is important to establish the difference between the AHRS FAIL scenario, which is the loss of the AHRS system due to technical issues, and the above mentioned AHRS FAULT scenario that is related to the lack of the GPS signal degrading the AHRS.

While in degraded mode, high airspeed can increase the chances of the AHRS units exceeding the miscompare limits between themselves.

If the speed advice in the QRH is not followed, there is a greater chance AHRS miscompare. The latest software update (Load 1X++) is on all jets and means that now the ventral rudder would still be available in this event to ensure lateral oscillations (Dutch Roll) should not occur.

It is very important to emphasise that it is not a cause / effect situation, thus the loss of the GPS signal DOES NOT directly lead these systems to become unavailable.

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Recommendations & Procedures

If any of the following Annunciations are seen:

**GPS LOI
GPS NAV LOST
NO FMS POSITION**

Or

If the DR/LOI (Dead Reckoning/Loss of Integrity) indication is seen when flying with GPS source:

If the GPS loss is due to jamming or an 'external' fault the FMS navigation mode (magenta) should still be available in DR mode but the navigation will need to be continued using raw data information (green needles). The AP will still be available, however, the aircraft should be flown in HDG mode while the navigation using 'green' needles is confirmed.

Note: Dead Reckoning Mode (DR) only functions in Enroute (ENR) or Oceanic (OCN) phase of flight. In all other phases, an invalid GPS solution produces a 'NO FMS POSITION' annunciation on the map and system stops using GPS.

1. Perform a manual reversion to raw data information using the CDI softkey and tune the appropriate NAV frequencies.
2. Use VOR / DME to continue navigating the aircraft.
3. Consider the type of approach available and remember to select a 'conventional' approach such as ILS or VOR/DME.

Normally only one CAS message will be shown. In case of both AHRS 1 FAULT and AHRS 2 FAULT after GPS signal loss:

1. Reduce the speed to the values shown in the QRH to avoid an AHRS 1 and 2 miscompare.
2. Follow QRH checklist: **Airspeed.....MAX 240KIAS/0.63M**

Note: No change in altitude is necessary when the AHRS 1 FAULT and AHRS 2 FAULT CAS messages are seen.

3. When the GPS signal is regained, resume GPS Navigation by pressing the CDI softkey until the Magenta GPS indication is shown. If needed, use the HDG mode to help during the transition for re-engagement of the GPS navigation mode.

There is no reason for a GPS failure to lead directly to an AHRS and AP/YD failure and even less of a chance of this happening if the correct maximum speed is flown.

However, if any of the following CAS are seen:

AHRS 1/2 FAIL
AP FAIL
YD FAIL

Carry out the QRH checklist items which include guidance about the possibility of lateral / directional oscillations.

The loss of the AP/YD is not a guarantee to get into Dutch Roll. Follow the advice in the QRH.

5.3 EGPWS WARNINGS / LOAD 1X+

The EGPWS on the Phenom is more 'sensitive' than other aircraft. The Garmin EGPWS looks ahead 45 seconds in a turn and 60 seconds in straight line and at a lower aspect for climb than Honeywell systems. This means that the Phenom is likely to have more terrain warnings than other aircraft going into some of our special airports. Garmin and Embraer were asked to improve the warning criteria for some of the special airports, especially Lugano where we cannot fly the procedures at night or IMC because we would have to react to the TAWS warning. Only four airports (Lugano, Bolzano, Sion, and Innsbruck) have been modified and for all other airports the normal TAWS envelope will remain the same.

The new customised look ahead for the 4 airports is 50%, so 30 seconds straight flight and 23 seconds ion the turn. Since this software was upgraded we have had a massive reduction in the number of warning and are successfully going into Lugano using the reduced minima.

Crews should always follow the OM-A guidance regarding TAWS alerts and make a positive correction regardless of where the alert is seen to be.

5.4 TCAS

We have had 3 events of TCAS traffic alerts (including one with a 'climb climb' alert) that have occurred with the Transponder in AUTO while on the ground. We believe for all events and certainly for the 'climb climb' alert that the radalt was indicating and altitude whilst still on the ground (we have a video showing the aircraft at 1100 ft on the radalt).

Embraer have said that the XPDR can wake up in AUTO which means that TA/RA could occur on the ground.

We have carried out a couple of tests and if ON is selected the transponder will NOT go automatically to ALT once airborne. The transponder needs to be in AUTO for the transfer to work.

We also need to make sure we are ready for ADS-B OUT which exchanges additional information with ATC.

POH 5-55 Page 23 additional information about the radalt:

It is also important to point that in case oscillations reach values greater than 1000 ft, the TCAS II may be automatically commanded to TA/RA mode. In case this occurs it is recommended to set transponder to STBY mode, if needed, in order to avoid interference with ATC.

Therefore, with immediate effect please select STBY at power up and while all of the system tests are carried out and all of the programming and then select AUTO just before taxi.



Aircraft was on the ground but note the Radalt is showing over 1000 ft. TCAS Climb shown (Picture is screenshot from video taken by crew).

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5.5 SPOILERS / SPEEDBRAKES

There have been several incidents where a Spoiler Fail CAS and yellow cross on the spoiler MFD has occurred after attempting to deploy the speedbrakes while airborne.

The cause of this has been due to one of the Spoiler Surface Sensors (LVDT) in each spoiler which has been a bit too sensitive and has thought that an asymmetric situation has occurred and locked the spoiler operation. This means that the spoilers will not be available on the ground for landing.

Each time this has happened, the spoilers have come back on-line after a normal power down and power up. The QRH has guidance for any additional landing factor to be applied.

The cause has been established as water ingress into the sensors after heavy rain. This means that one sensor reacts a little slower than the other and the asymmetric situation is assumed. The sensors are in the process of being changed but on all occasions so far, the sensor has worked on the next trip after it has dried out.

If the aircraft is heading to a performance limited runway as long as the spoilers are not selected in flight (as a precaution) the ground spoilers will deploy on the ground because the same logic is not applied when the WOW switch is on the ground.

5.6 SYNTHETIC VISION SYSTEM

The SVS was a new arrival to our aircraft during 2017 and has greatly enhanced our situational awareness and allows us to use the Garmin 3000 to near its full capacity.

There are no rules promulgated when to have the SVS on the PFD and pilots are free to choose whether they wish to use it or not in most situations. The only time that we want at least one pilot with the SVS displayed is when operating is mountainous terrain because the effect as shown below far exceeds what can be seen on the standard PFD.



The SVS can be turned ON or OFF using the soft keys on each pilots PFD. 'PFD Settings' leads to the 'Attitude Overlays' button, which leads in turn to the Synthetic Terrain button. Pressing the Synthetic Terrain soft key leads to features of the SVT that can be turned ON or OFF within the main 'program'. These are Airport Signs, Horizon Heading, and Pathways (these three in turn can only be activated when the Synthetic Terrain key is activated).

The SVS makes a three-dimensional depiction of terrain, obstacles, and traffic that appear in the primary field of view, so that their proximity is more easily understood during instrument scanning. This provides crews with a greater awareness of the aircraft position relative to surrounding terrain, obstacles, and traffic.

SVS is not intended to be used independently of the traditional instrumentation, it is an operational enhancement. The traditional attitude instrumentation will always be visible in the foreground with SVS features in the background.

The SVS has numerous features which all enhance the visual environment for the crew and provide additional flight critical information, functionality, and safety information. The features include:

Major SVS Features

Terrain Depiction
 Terrain Conflict Colouring
 Obstacle Depiction
 Obstacle Conflict Colouring
 Pathways
 Traffic Depiction

Additional SVS Features

Flight Path Maker (FPM)
 Runway Depiction
 Zero Pitch Line
 Horizon Heading Marks
 Airport Signs
 Water Depiction

As shown in the following pictures, probably the most graphic and useful of the features is the Terrain Avoidance, however, the 3D traffic from the TCAS can also be very valuable in helping where to look for a potential conflict.



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CHAPTER 6 - WINTER OPERATIONS GUIDE

PHENOM Winter Operations Guide (28 Jul 2018)

6.1 INTRODUCTION

This guide has been prepared as an aid to winter operations. It has been created to have some useful information in one place but while there is some sharing of information and techniques with other documents please remember to also check the other publications listed below. The aim is to eventually have all of this information in one place but at the moment this is still a general guide with some fleet specific information and the OM-B and POH still retains the specific checklists to be carried out.

The Phenom Winter Operations Guide will be revised every year. Therefore please do not hesitate to send us additional information and feedback that will improve it as the years go on.

Additional Reading Material

OM-A 8.2.4

POH 2-15

AFM 3-04

OM-B 2.24

A number of useful videos are also available at www.flyembraer.com under Flight Operations / Downloads / Videos.

6.2 AIRCRAFT PREPARATION

'Icing conditions may exist whenever the Static Air Temperature (SAT) on the ground or the Total Air Temperature (TAT) in flight is 10°C or below and visible moisture in any form is present.'

Aircraft performance is calculated for a clean aircraft. Therefore, it is vital to make sure the aircraft's critical surfaces are free of frozen contaminants prior to operating the aircraft.

What are the aircraft critical areas?

- Wings, tail, and control surfaces;
- Pitot tubes (IASP) and SAT/TAS probe;
- Engines;
- Air-conditioning inlets and exits;
- Landing gear associated area and wheel bay;
- Vents (fuel tank, air conditioning);
- Fuselage.

Of course the areas to focus on are not limited to those listed above, but in general these are considered **critical areas**.

Contamination in this guide is understood as all forms of frozen or semi-frozen moisture such as frost, snow, ice, or slush.

Contamination may:

- Reduce maximum lift available at rotation;
- Decrease the climb rate;
- Decrease the stall angle, possibly without triggering the stall warning system;
- Lead to roll or pitch upset;
- Flame out the engine(s);
- Break off and get ingested into the engine(s);
- Cause faulty and confusing airspeed, altitude, AOA, or engine indications;
- Decrease service ceilings;
- Increase fuel consumption and decrease maximum range;
- Decrease missed approach performance.

Tests have shown that the biggest degradation of performance occurs within the first few minutes of ice build-up. Amounts of ice equivalent to medium grit sandpaper covering the leading edges or upper surfaces of wings can increase the stall speed up to 20 knots.

Exposure to clear icing for two minutes can double the drag, reduce the maximum lift by 25-40%, and reduce the critical AOA significantly.

6.2.1 PRE-FLIGHT

The key to winter operations is planning ahead as much as you can. When the temperatures are forecast to be close to or below zero crew should always plan ahead and try to anticipate where problems and delays may occur. Some airports will have very good and well used winter operations, some of the other places we use might not.

Consider some of the following suggestions that could be thought about the day before that might help your day:

- Anticipate additional travel times to and from the hotel.
- Time the airport opens and begins clearing operations.
- What are the de-icing procedures for the airport.
- Where is de-icing carried out (pre or post engine start, on stand or remote).
- How easy is it to book the de-icing truck.
- Do you have gloves and warm clothing.
- Have a torch ready for the morning.
- Will you have to remove all of the liquid stock.
- Will you need a toilet service in the morning to refill.
- Will you require an earlier show time.
- How will the pax be brought out to the aircraft.
- If you are taking over an aircraft make sure you know if the liquid stock was removed and where it was kept.
- Would you require to heat the cabin before a pax flight?
- Can the FBO provide a cabin heater?

Personal Precautions

In severe weather the journey from the hotel to the FBO could take substantially longer than normal, be aware of this when planning your times.

Dress appropriately with weather resistant gloves, wear a warm scarf and a hat, and avoid wearing shoes with smooth soles. Nights are longer in the winter and early starts will mean that tasks in the dark are being performed. Ensure you have a personal torch available for these tasks. The aircraft torches are part of the emergency equipment and are not to be used for pre-flight, etc.

Your skin can stick to very cold metal in conditions below -5°C or when the aircraft has been exposed to extreme temperatures. Be very careful when touching the aircraft skin.

Ramps can be very icy and there is a high risk of slipping and falling. Be careful when walking and avoid icy patches. Often the GA ramp is the last place to be cleared of ice and snow.

- Walk cautiously, your arms help you balance so keep them out of your pockets.
- Be alert to the risk of slipping and falling on an unseen patch of ice.
- Slipping is sometimes unavoidable. If you fall the best techniques are to tuck and roll, try to avoid breaking your fall with your hands as this will often cause a wrist injury.

Be aware of wind chill and its effects. Wind chill is a function of air temperature and wind speed as shown in the table below. It is the apparent temperature felt on exposed skin

		Air Temperature (Celsius)																
		0	-1	-2	-3	-4	-5	-10	-15	-20	-25	-30	-35	-40	-45	-50	-55	-60
Wind Speed (km/hr)	6	-2	-3	-4	-5	-7	-8	-14	-19	-25	-31	-37	-42	-48	-54	-60	-65	-71
	8	-3	-4	-5	-6	-7	-9	-14	-20	-26	-32	-38	-44	-50	-56	-61	-67	-73
	10	-3	-5	-6	-7	-8	-9	-15	-21	-27	-33	-39	-45	-51	-57	-63	-69	-75
	15	-4	-6	-7	-8	-9	-11	-17	-23	-29	-35	-41	-48	-54	-60	-66	-72	-78
	20	-5	-7	-8	-9	-10	-12	-18	-24	-30	-37	-43	-49	-56	-62	-68	-75	-81
	25	-6	-7	-8	-10	-11	-12	-19	-25	-32	-38	-44	-51	-57	-64	-70	-77	-83
	30	-6	-8	-9	-10	-12	-13	-20	-26	-33	-39	-46	-52	-59	-65	-72	-78	-85
	35	-7	-8	-10	-11	-12	-14	-20	-27	-33	-40	-47	-53	-60	-66	-73	-80	-86
	40	-7	-9	-10	-11	-13	-14	-21	-27	-34	-41	-48	-54	-61	-68	-74	-81	-88
	45	-8	-9	-10	-12	-13	-15	-21	-28	-35	-42	-48	-55	-62	-69	-75	-82	-89
	50	-8	-10	-11	-12	-14	-15	-22	-29	-35	-42	-49	-56	-63	-69	-76	-83	-90
	55	-8	-10	-11	-13	-14	-15	-22	-29	-36	-43	-50	-57	-63	-70	-77	-84	-91
	60	-9	-10	-12	-13	-14	-16	-23	-30	-36	-43	-50	-57	-64	-71	-78	-85	-92
	65	-9	-10	-12	-13	-15	-16	-23	-30	-37	-44	-51	-58	-65	-72	-79	-86	-93
	70	-9	-11	-12	-14	-15	-16	-23	-30	-37	-44	-51	-58	-65	-72	-80	-87	-94
	75	-10	-11	-12	-14	-15	-17	-24	-31	-38	-45	-52	-59	-66	-73	-80	-87	-94
	80	-10	-11	-13	-14	-15	-17	-24	-31	-38	-45	-52	-60	-67	-74	-81	-88	-95
	85	-10	-11	-13	-14	-16	-17	-24	-31	-39	-46	-53	-60	-67	-74	-81	-89	-96
	90	-10	-12	-13	-15	-16	-17	-25	-32	-39	-46	-53	-61	-68	-75	-82	-89	-96
	95	-10	-12	-13	-15	-16	-18	-25	-32	-39	-47	-54	-61	-68	-75	-83	-90	-97
	100	-11	-12	-14	-15	-16	-18	-25	-32	-40	-47	-54	-61	-69	-76	-83	-90	-98
	105	-11	-12	-14	-15	-17	-18	-25	-33	-40	-47	-55	-62	-69	-76	-84	-91	-98
	110	-11	-12	-14	-15	-17	-18	-26	-33	-40	-48	-55	-62	-70	-77	-84	-91	-99
		0 to -10 Low			-10 to -25 Moderate			-25 to -45 Cold			-45 to -59 Extreme			-60 Plus very Extreme				

If the aircraft has been parked outside in cold temperatures, be sure you will be given enough time for the pre-flight preparation. OM-A Chapter 7 (paragraph 7.2.5.4) states the requirements for an additional 15 minutes. show time if icing conditions are expected and the aeroplane has not been parked in a hangar overnight. These 15 min. especially in the dark, **will be the minimum time required** to complete a thorough pre-flight and to ascertain whether de/anti-icing is required. If you think you will require the extra 15 minutes or even longer make sure Scheduling is informed and your duty day is updated correctly.

Remember to refill all of the liquid stock and have a toilet service if the toilet was drained the previous night. Simple things such as finding out where the stock was placed can delay crews and pre-planning for normally simple tasks will be required. If the aircraft batteries were removed they will need to be re-installed, and this takes additional planning and time. Request and connect a GPU as soon as possible to start warming up the batteries. Even in temperatures above -10°C the batteries can be very sluggish especially when trying to power up the start. A GPU will also begin to warm the batteries while they are charging and this will help with their operation.

6.2.2 OPERATIONS WITH THRUST REVERSERS INOPERATIVE

Not Applicable.

6.2.3 CONTAMINATION CHECK

Operating regulations clearly state that no take-off is allowed when snow, ice, or frost is adhering to the aeroplane.

Make sure nose wheel chocks are in place. If required, chock main wheels as well. Remove engine air inlet/outlet plugs and covers from pitot/static tubes, static ports and NACA air inlets. If required, leave engine plugs installed until engine start.

The pilot in command has the final responsibility for ensuring that the aeroplane is clear of ice, frost, or snow. The primary method for the pilot to ensure a clean aeroplane is through close visual and physical inspection prior to take-off. Visually check the wing, control surfaces, engines, and fuselage prior to take-off. In addition, as no frozen contamination is allowed on the wing upper surface, carry out a physical (hands-on) inspection to ensure that there is no ice accumulation. Do not touch the surfaces with bare hands, as the skin may stick to a freezing surface.

Even at intermediate stops, an external walk around is necessary due to the possibility of ice forming after landing from either cold soaking frost, conventional frost, or precipitation freezing on the aeroplane.

During the pre-flight walk-around, ensure that the pitot tubes, pressurisation static ports, all inlets, outlets, and vents are clear of ice and unobstructed.

If the aeroplane has become cold soaked as a result of flight at very cold temperatures, fuel might be at a subfreezing temperature. This can cause ice accumulation if the aeroplane is subjected to high humidity, fog, drizzle, or rain even when the outside air temperature is substantially above freezing.

At the completion of the walk-around, if ice, snow, or frost is discovered, de-icing procedure will be required.

A pilot shall not take off in an aircraft that has:

- Frost, snow, slush, or ice on the windshield, engines or pitot tubes, or SAT/TAS probe;
- Snow, slush, or ice obstructing air conditioning inlets and exits;
- Snow, slush, or ice on landing gear;
- Snow, slush, or ice obstructing fuel tank vents;
- Snow, slush, or ice on the wings, stabilisers or control surfaces;
- Snow, slush, or ice in gaps between the airframe and control surfaces;
- Snow, slush, or ice in gaps between control surfaces and control tabs;
- Frost on the upper surfaces of wings, stabilisers, or control surfaces.

Therefore, a contamination check of the aircraft surfaces shall be performed prior to departure. This check is to determine the need for de-ice and shall be performed via a walk-around.

Share the workload. While one pilot obtains weather information, have the other pilot examine the aircraft. A copy of the snowtam format derived from Jeppesen will aid in coping with the weather information.

Whenever the aeroplane is substantially covered by ice or snow, flight control movement could be restricted. In this case, a flight control check shall not be done prior to de-icing.

6.2.3.1 FUSELAGE, WINGS, TAIL, AND CONTROL SURFACES

Fuselage, wings, tail, and control surfaces shall be free of ice, snow, slush, and frost.

No contamination is permitted on the lower or upper surface of the stabilizer or wing. Polished frost is not allowed.

A thin layer of hoarfrost where you can easily see below the aeroplane marking, letters, or paint lines on the fuselage top is permitted.

All snow and ice must also be cleared from nose radome and fuselage nose forward of windshield, as it is likely to blow back into windshields during taxi or take-off.

Look for snow deposits on flying surfaces and in between gaps in control surfaces; they will not always blow off, and will affect the coefficient of lift and/or possibly jam controls.

Run your fingers across the surface to help identify whether the contamination is frozen or not.

Do not use sharp objects to test it, because damage to the aircraft skin may occur.

This will also help to determine if a layer of water has frozen to form a light ice sheet on top, or conceals a layer of ice at the aircraft surface. Clear ice can also exist on aircraft surfaces below a layer of snow or slush.

Significant deposits of clear ice can form in the vicinity of the fuel tanks on wing upper surfaces as well as under-wing.

Aircraft are most vulnerable to this type of build-up when:

- Wing temperatures remain well below 0°C during the turnaround/turn sit;
- Ambient temperatures between -2°C and +15°C are experienced;
- Ambient humidity is high and/or precipitation occurs when the aircraft is on the ground.

If the aircraft surface is contaminated, try to use mechanical means such as a broom to remove the majority of the contamination from the surfaces. Other means are forced air, heat, heated water, or heated fluids with negative buffer freezing point. If this so called pre-step procedure is used, make sure that the subsequent de-icing process removes all frozen contamination including the contamination that may have formed on surfaces and or in cavities due to the pre-step process. If not already carrying on board, ask the FBO or people cleaning the ramp for brooms or other cleaning tools. When removing ice, snow, slush, or frost from aircraft surfaces care shall be taken to prevent it entering and accumulating in auxiliary intakes or control surface hinge areas. Remove snow from wings and stabilizer surfaces forward towards the leading edge and remove from ailerons and elevators back towards the trailing edge.

Remember we are charged de/anti-icing fluids per litre and there is a lot of money involved.

However, if the snow on the wings is wet and loose, do not take off without de-icing the wings. Remember when the aircraft speed increases, the pressure drops on top of the wings and the temperature will drop rapidly as well, which may cause water to freeze on the wing.



6.2.3.2 PITOT TUBES

Make sure that the IASP's and standby probe are clear of ice, frost, snow, and any fluid residues.

6.2.3.3 ENGINES

Engine inlets, exhaust nozzles, cooling intakes, and sensor probes shall be clear of ice and snow. Engine fan blades shall be clear of ice, frost and snow, and shall be free to rotate.

When you check the fan movement, you may have to step on the wing to look inside the engine air intake. Make sure you step on the right areas and remember the wing may be extremely slippery. If the fan is frozen, ask a heat blower from the FBO.

Also check the adjacent areas. Contaminants can break away during flight and may be digested by an engine. This could lead to a flameout or engine failure due to damaged fan and/or compressor blades.



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6.2.3.4 AIR-CONDITIONING INLETS AND EXITS

Air-Conditioning inlets and exits shall be clear of ice, frost, and snow.

6.2.3.5 LANDING GEAR

Landing gear shall be unobstructed and clear of ice, frost and snow.

Accumulations such as blown snow may be removed mechanically or by a heat blower. However, where deposits have bonded to surfaces, they can be removed by the application of hot air. It is very uncommon for most aircraft that you will find brakes to be frozen, but since you might not notice when the wheel is locked during taxi, ask the marshaller to give a signal for observed wheel rotation. Should they be stuck, try the heat blowers.

6.2.3.6 VENTS AND DRAIN MASTS

Fuel tank vents and stall warning vents as well as all drain masts shall be clear of ice, frost, and snow.

6.2.3.7 FUSELAGE

Unless otherwise stated in the aircraft specific AFMs or OM-Bs, frost is allowable on the fuselage provided the layer is thin enough to distinguish the surface features such as paint lines or markings underneath, but all vents, probes, and ports must be clear of frost.

6.2.4 DE-ICING & ANTI-ICING

It is useful to clarify what is meant by the terms 'Anti-icing' and 'De-icing', since their sometimes random use when applied to aircraft systems suggests that some confusion may exist. In strict terms, the following definitions apply:

- De-icing is the **removal** of ice that has accreted, normally by means of cyclic heating or the application of a physical impulse (commonly achieved on leading edges by pneumatic boot inflation) or the use of De-icing Fluids. It also includes the use of mechanical means like brooms or hot air. Even a heated hangar is considered to be a de-icing procedure.
- Anti-icing is the **prevention** of ice formation, generally by means of heating (electrical, ducted engine bleed air, engine oil, etc.) or as well by using Anti-icing Fluids. These fluids provide for a limited period of time (Hold Over Time) protection against the formation of frost or ice and accumulation of snow or slush on clean surfaces of the aircraft.

6.2.4.1 HOLD OVER TIME (HOT)

This is the ESTIMATED time for which an anti-icing fluid will prevent the formation of frost or ice and the accumulation of snow on the protected surfaces of an aircraft. Holdover times for specific approved fluids should be obtained from current tables published in the OM-A. They will be revised regularly.

With a **one-step de-icing / anti-icing** operation the HOT starts at the beginning of the operation.

With a **two-step operation** HOT starts at the beginning of the final (anti-icing) step.

Holdover time depends on:

- Fluid type and concentration;
- Outside Air Temperature (OAT);
- Precipitation type and intensity;
- Wind velocity;
- Jet blast;
- Aircraft skin temperature.

For Type 1 fluid de-icing Embraer recommends using the Composite Hold Over Times (table 1-c in OM-A, Chapter 8 Appendix 1) which are more conservative with regards to holdover times compared to table 1-b for normal aluminium surfaces. The reason for this is that the tail is made mostly from composite material, therefore, it is considered safer to use the more restrictive composite table.

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6.2.4.2 FLUID TYPE AND CONCENTRATION

There are four basic types of de-icing / anti-icing fluids as described below. Be extra vigilant when performing outside checks after the de-icing / anti-icing procedure and especially during post flight. Experience has shown that Type I, II, and III residues can easily be mixed up with hydraulic and / or engine oil.

Type I

- Type I fluids are 'unthickened' and have a relatively low viscosity.
- They have good de-icing properties but provide negligible protection against refreezing.
- They are used predominantly for removing frozen deposits from aircraft surfaces, either in the first step of a two-step operation, or where precipitation has stopped.
- In undiluted form they are not to be used at ambient temperatures below -10°C due to adverse aerodynamic effects.
- Type I fluids are normally colourless in appearance.

Type II

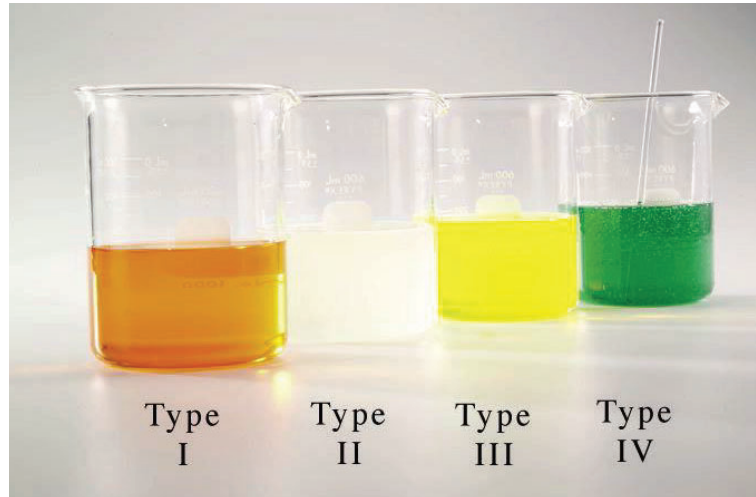
- Type II fluids contain thickening agents which enable the fluid to be deposited as a film and to remain on the aircraft surfaces until the time of take-off.
- This film provides a longer holdover time especially in conditions of freezing precipitation, providing anti-icing protection against re-freezing or further accumulation in precipitation conditions.
- The holdover time can be extended by increasing the concentration of fluid in the fluid/water mix.
- The fluids are designed to flow off the wings when subjected to shear forces at take-off, causing little effect on the aircraft's aerodynamic performance.
- Type II fluids are normally straw coloured in appearance.

Type III

- Type III fluids contain thickening agents which enable the fluid to be deposited as a film and to remain on the aircraft surfaces until the time of take-off.
- Type III fluids reduce in viscosity faster than type II and type IV fluids and thus provide anti-icing protection for a shorter period.
- The holdover time can be extended by increasing the concentration of fluid in the fluid/water mix.
- Type III fluids are normally bright yellow in appearance.

Type IV

- Type IV fluids have been developed in recent years to increase holdover times by the further addition of thickening agents.
- As with Type II, their holdover times can be extended by increasing the fluid concentration in the fluid/water mix.
- Type IV fluids are normally coloured green.



6.2.4.3 OUTSIDE AIR TEMPERATURE (OAT)

Although the freezing point of anti-icing fluid is lower than that of pure water, it can freeze. As the air temperature drops, anti-icing fluids become less able to absorb and melt the freezing precipitation. In general, the colder it is, the shorter the holdover time. In addition, as the temperature decreases, the fluids become more resistant to shearing off the aircraft during the take-off roll.

A Lowest Operational Use Temperature (**LOUT**) is specified for every fluid. If the temperature is below the LOUT, that fluid should not be used.

LOUT is the higher (warmer) of:

- The lowest temperature at which the fluid meets the aerodynamic acceptance test for a given type (high speed or low speed) aircraft; or
- The freezing point of the fluid plus the freezing point buffer of 10°C for type I fluid and 7°C for type II, III, or IV fluids.

6.2.4.4 PRECIPITATION TYPE AND INTENSITY

Anti-icing fluid works by absorbing and melting precipitation. Conversely, precipitation dilutes the anti-icing fluid. The more dilute the anti-icing fluid becomes, the less protection it affords. Ultimately, the anti-icing fluid will fail. In general, the more moisture is present, the shorter the HOT.

The type of precipitation and its intensity are key factors in determining HOT.

Although light freezing rain and freezing drizzle have holdover times, the in-flight ice protection systems may not be able to protect your aircraft in these large droplet conditions. After the anti-icing fluid shears off the wing during take-off, you may encounter icing conditions which exceed the capabilities of your aircraft ice protection systems.

Therefore, take-off is not approved in:

- Moderate and heavy freezing rain.
- Ice pellets.

6.2.4.5 WIND VELOCITY

When there is more than 10 kts of wind, the wind chill factor can affect HOT of anti-icing fluids. As a technique, using the dew point temperature listed in holdover tables gives you an additional safety margin.

6.2.4.6 JET BLAST

Taxiing closely behind another aircraft may degrade the effectiveness of the applied anti-icing fluid and braking. Therefore, use always enough space when taxiing behind another aircraft.

6.2.4.7 AIRCRAFT SKIN TEMPERATURE

COLD SOAK EFFECT

The wings of aircraft are said to be 'cold-soaked' when they contain very cold fuel as a result of having just landed after a flight at high altitude or from having been re-fueled with very cold fuel. Whenever precipitation falls on a cold-soaked aircraft when on the ground, clear icing may occur. Even in ambient temperatures between -2°C and + 15°C, ice or frost can form in the presence of visible moisture or high humidity if the aircraft structure remains at 0°C or below.

In between flights, the aircraft skin temperature may be lower than the OAT. If the OAT is close to the limits of a block OAT of the HOT tables, take the next lower block to determine the HOT. Also a stronger mix (more glycol) may need to be used to ensure a sufficient freeze point buffer.

6.2.5 AREAS TO DE-ICE

6.2.5.1 DECISION AND PREPARATION

If de-icing and in case of (expected) precipitation anti-icing is needed, discuss the specifics of your de/anti-icing request with the handling agent. Make sure what type of fluid is available and have a close look at the holdover tables of the de/anti-icing fluids that will be used. Checking the availability of the de-icing truck in advance will help with your planning.

If active ground icing conditions are forecast, or present during your scheduled departure time, determine whether or not you can de/anti-ice and go, or delay the flight until conditions improve. Do not let the slot or anything else influence your decision making when conflicts with holdover times arise. It is better to arrive late than the alternative.

If there is frost, snow, freezing drizzle, or freezing rain, etc. forecast before departure time, consider using type II, III or IV fluid on the clean aircraft surfaces prior to the start of the potential freezing precipitation. This could minimise the possibility of snow and ice bonding or reduce the accumulation of frozen precipitation on aircraft surfaces and reduce the need for subsequent de-icing.

If de-icing only is required, often a heated but diluted type I is used. Type I fluids form a thin wetting film, which provides only limited holdover time, but can be used just for de-icing an overnight frost for example. In conditions of freezing precipitation with this type of fluid no additional holdover time (HOT) would be provided by increasing the Concentration of the fluid in the fluid/water mix. However, a longer HOT will be obtained by applying the type I fluid in a two-step manner. First spray heated type I fluid to de-ice the aircraft and then apply the anti-icing layer to the clean aircraft. In this case the HOT starts with the beginning of the 'second' anti-icing application.

Anti-icing buys time on the ground. These fluids form a thicker liquid wetting film on external aircraft surfaces, which absorbs and melts the precipitation. However, it does however not remove ice; it must be applied to a clean surface, usually within 3 minutes of de-icing fluid application.

6.2.5.2 PRECAUTIONS BEFORE DE-ICING / ANTI-ICING

Soft Snow can be mechanically removed by blowing cold air across the aeroplane surface, with brooms, soft hand scrappers, or rubber scrappers.

Snow or ice should be removed from the fuselage before the aircraft is heated internally to prevent melting of the snow and subsequent re-freezing, which would make the ice more difficult to remove.

ECS and Air Conditioning shall be selected to OFF.

Engine deposits of snow should be mechanically removed from engine intakes with brooms, soft hand scrapers, or rubber scrappers. Frozen deposits adhering to the lower surface of the intake or to the fan blades should be removed by applying hot air. To ensure the safety of ground personnel and passengers, this should be carefully coordinated with the flight crew.

A post-de-icing/anti-icing check should be performed during or immediately following the ground de-icing and anti-icing process. A check must be performed on all areas where fluid has been applied and to other areas that may be conducive to precipitation accumulation.

Check if all air inlets and pylons are free of snow to avoid ingestion at the moment of turning the fan on or during the fluid application.

If type II, III, or IV fluids are used, any forward area from which fluid can blow back onto windscreens during taxi or take-off shall be free of residues prior to departure. Ensure that all traces of the fluid on flight deck windows are removed prior to departure. De-icing/anti-icing fluid may be removed by rinsing with an approved cleaner and a soft cloth or a wiper obtained from the FBO.

6.2.5.3 DE-ICING APPLICATION

De-icing treatments shall be symmetrical, that is, left-hand and right-hand side of the aircraft shall receive the same treatment, even when only one side of the aircraft is contaminated.

If ice is to be removed, usually a stream of hot fluid is directed at close range onto one spot at an angle of less than 90°, until the aircraft skin is just exposed. The aircraft skin will then transmit the heat laterally in all directions raising the temperature above the freezing point thereby breaking the adhesion of the frozen mass to the aircraft surface. By repeating this procedure a number of times, the adhesion of a large area of frozen snow or glazed ice can be broken. The deposits can then be flushed off with either a low or high flow, depending on the amount of deposit.

Ice, snow, or frost dilutes the fluid. Ensure enough hot de-icing fluid is applied to prevent re-freezing and ensure all contaminated fluid is driven off.

6.2.5.4 ANTI-ICING APPLICATION

Anti-icing treatments shall always cover the entire wing and/or the entire horizontal stabilizer/elevator on both sides of the aircraft. Aerodynamic problems could result if these requirements are not met. The process should be continuous and as short as possible. Anti-icing should be carried out as near to the departure time as operationally possible in order to utilise maximum HOT. The anti-icing fluid shall be distributed uniformly and with sufficient thickness over all surfaces to which it is applied.

6.2.5.5 SPRAY PATTERN

The following surfaces must be protected:

- fuselage;
- wing upper surface and leading edge;
- horizontal stabilizer upper surface and leading edge;
- elevator upper surface;
- vertical stabilizer and rudder.

WING AND STABILIZER

Spray from the tip inboard to the root, sweeping from the leading edge in the aft direction. It is important that the fluid be applied symmetrically to both wings so as to ensure that the aerodynamic effect of the fluid remaining is the same on both sides of the aeroplane.

PITOT/STATIC AND STATIC PORTS

The fluids should be sprayed along the top of the sensors, allowing the fluid to cascade down across the sensor and de-ice it. Do not spray de-icing fluid directly on the probes and static ports.

VERTICAL SURFACES

Start at the top and then work down. The rudder should be in the neutral position.

FUSELAGE

Remove all accumulated snow on the nose to avoid snow blowing back during take-off and restricting pilot visibility. Do not apply de-icing or anti-icing fluid directly to the windshield and to the cockpit windows.

Fluid should be sprayed along the top centreline of the fuselage and then outboard, letting the fluid cascade down and across the windows. Fluid must not be sprayed directly onto windows and window seals, doors and door seals, scoops, or NACA air inlets. Be careful not to cause damage to the antennas installed along the fuselage.

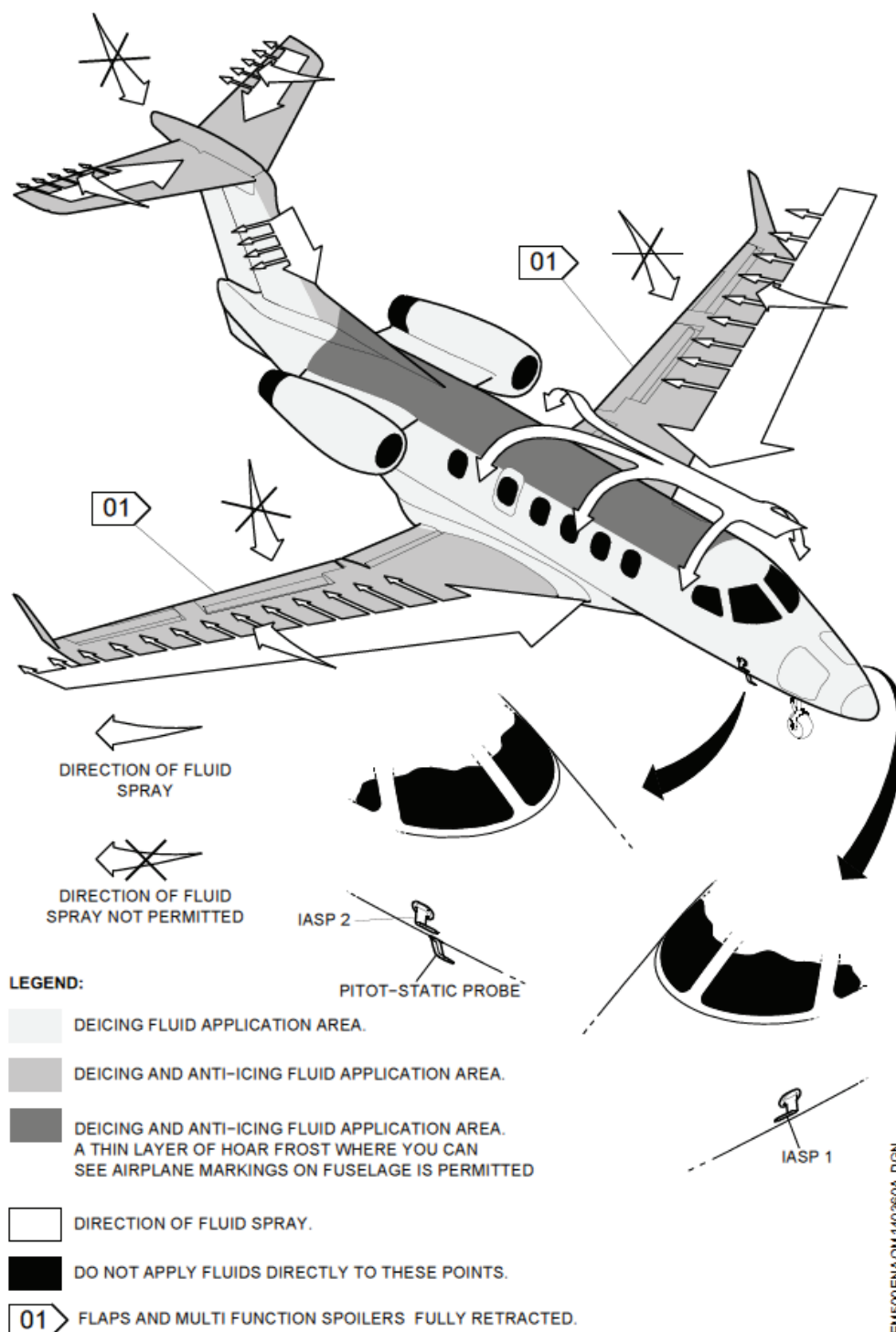
LANDING GEAR AND WHEEL BAYS

Application of fluid in this area must be kept to a minimum. Do not allow water or fluid mixture spray into wheels and brakes to avoid damage to carbon brakes. In these areas of the aeroplane, it is preferable to clear snow or slush using a brush.

ENGINES

Engine deposits of snow should be mechanically removed from engine intakes with brooms, soft hand scrapers, or rubber scrapers. Frozen deposits adhering to the lower surface of the intake or to the fan blades should be removed by applying hot air. To ensure the safety of ground personnel and passengers, this should be carefully coordinated with the flight crew.

CAUTION: DO NOT APPLY FLUID DIRECTLY TO VENTS, DRAINS, OUTLETS OR INLETS, IASP, PITOT / STATIC TUBES AND STATIC PORTS.



FLUID APPLICATION STRATEGY

6.2.5.6 ONE-STEP DE-ICING / ANTI-ICING

This is performed with a heated (60° up to 90°C) anti-icing fluid. The fluid used to de-ice the aircraft remains on aircraft surfaces to provide limited anti-ice capability. The correct fluid concentration shall be chosen with regard to desired holdover time and is dictated by outside air temperature and weather conditions. When using type I fluid, the freezing point of the fluid shall be at least 10°C below the OAT. When using type II, III, or IV fluid, residues may be collected in aerodynamically quiet areas, cavities and gaps. Dried residues may freeze following a period of high humidity, and/or rain conditions. This may impede flight control systems. If removal of contamination is required on the lower side of the wings and the horizontal stabiliser and elevator, de-icing/anti-icing fluid shall be applied sparingly to minimise fluid flow into drain holes. Whenever possible, use type I only.

6.2.5.7 TWO-STEP DE-ICING / ANTI-ICING

The freezing point of the fluid used for the first step shall not be more than 3°C above ambient temperature. If type I is used for the second step, the freezing point of this fluid shall be at least 10°C below the OAT. If type II, III, or IV fluid is used for the second step, application is allowed to -25°C or above. If the OAT is lower, application is allowed provided a 7°C buffer is maintained between the freezing point of the fluid and OAT. In no case shall this temperature be lower than the Lowest Operational Use Temperature (LOUT) for the fluid.

Examples of De-icing / Anti-icing equipment at LSZH vs. LEMD



6.2.5.8 POST DE-ICING / ANTI-ICING CHECKS

The flight crew shall visually check the critical parts of the aircraft after a de-icing / anti-icing operation before take-off. When remote de-icing / anti-icing has been accomplished with the engines running, this check should be limited to the surfaces visible from the cockpit and cabin. Completion of the correct process has to be entered in the Aircraft Flight Log by entering the start time, the type of fluid used, and the mix ratio.

6.2.6 ENGINE START

If the aircraft batteries have been exposed to temperatures of -10°C or less for more than two hours try to use a GPU to assist with starting. If no GPU is available the batteries will need to be warmed up. Battery assisted starts in winter may result in a slower start time and increased ITTs. Even if the aircraft has been left out overnight in just freezing conditions, above the minimum of -10°C, the batteries would still benefit from GPU assistance. Check the correct voltage and operation of all battery powered emergency equipment.

6.2.7 LIMITATIONS

The minimum temperature for starting is -40°C oil temperature, in the unlikely event that crew are operating at this limit ground heating will be required to pre-heat the engines.

The minimum fuel temperature is -37°C.

It is permissible to operate the engine up to 70% N2 in order to warm the oil to the minimum temperature for normal operation (above 10°C).

6.2.8 START-UP

Check for free fan movement before starting the engine. Engine start is likely to be slower in cold weather conditions.

To prevent ice build-up, and thus the possibility of ice ingestion and damage to and/or flame-out of an engine, in icing conditions the engine anti-icing should be on as soon as the engines are started.

Restrict the power to the minimum required until the engine oil has warmed up. Run the engines for a number of minutes before selecting the air conditioning on unless you are sure that no anti-icing / de-icing fluid has been ingested into the engine.

Ensure that the aircraft is not affected by a tail wind. The limit is 10 kts but as a rule of thumb if the tail wind is strong enough to spin the engine blades the 'wrong way' then the aircraft should be turned into wind before start, in order to assist the batteries in the cold temperatures.

The AFM recommends turning the ADS Probes Knob to ON just prior to engine start. This will allow the FADEC to receive reliable information during the start. After the start the ADS knob can return to auto as the system will activate after the first engine is running.

6.2.9 DE-ICING/ANTI-ICING PROCEDURES WITH ENGINE(S) RUNNING

6.2.9.1 REMOTE (OFF-GATE) DE-ICING / ANTI-ICING

In order to improve the speed and efficiency of de-icing/anti-icing operations, to reduce departure delays during adverse weather, and to maximise the use of hold-over times, some authorities allow de-icing/anti-icing with engines running at a remote location at the airport. Mostly these areas are close to the departure runway. When this procedure is allowed each particular aircraft procedures must be followed and experience and engineering tests have proved that any problems can be minimised by taking the following precautions:

- Operate as few engines as possible during the de-icing process.
- Operate at the lowest practicable power settings.
- Select air conditioning 'OFF'.
- Avoid spraying fluid directly into the engines, APU (if installed), and air conditioning system intakes.
- Avoid large run-off of fluid from surfaces adjacent to the engines.
- Minimise the generation of spray in the vicinity of the intakes.

6.2.9.2 PROCEDURES

Communications

During off-gate de-icing / anti-icing a two-way communication between flight crew and de-icing / anti-icing operator/supervisor must be established prior to the de-icing / anti-icing treatment. This is usually done by VHF radio. The 'tail number' (registration) of the aircraft must be used during all communications. Contact with de-icing / anti-icing operator may be closed after anti-icing code and readiness for taxi-out has been announced.

Taxi guidance

Arrange a sufficient taxi and stopping guidance when entering the off-gate de-icing / anti-icing area. Radio contact must be established before entering the area, the signs with clearly marked operation frequency should be visible before entering this area.

General instructions

Information regarding the location of, and standard taxi routing to the de-icing / anti-icing area, means to coordinate the de-icing / anti-icing operation, means to communicate before and during the de-icing / anti-icing operation and information about taxi and stopping guidance must be published by NOTAM or in local AIP.

Responsibilities

After treatment, the result must be checked by a trained and qualified person and the anti-icing code must be given to the Commander, after which the Commander is responsible for the airworthiness of the aircraft.

Terminology (Example)

DIS = De-icing / anti-icing supervisor

COMMANDER = Pilot in command

DIS: 'Set parking-brakes, confirm aircraft is ready for treatment, inform on any special requests.'

After aircraft is configured for treatment:

COMMANDER: 'Brakes are set, you may begin treatment and observe... (any special requests like: ice under wing/flaps, clear ice on top of wing, snow on fuselage, anti-ice with type IV fluid, etc.)'

DIS: 'We begin treatment now and observe... (special request given, like 'ice under wing', etc.). I will call you back when ready.'

ONLY AFTER EQUIPMENT IS CLEARED FROM AIRCRAFT AND ALL CHECKS ARE MADE:

DIS: 'De-icing / anti-icing completed, ANTI-ICING CODE IS:... (plus any additional info needed, e.g. amount in litres). I am disconnecting standby for clear signal at right/left and/or contact ground/tower for taxi clearance.'

COMMANDER: 'De-icing / anti-icing completed, anti-icing code is...'

6.2.10 OPERATIONS ON CONTAMINATED SURFACES

Refer to the OM-A, Chapter 8 and AFM.

Pilots should be aware of the problems imposed by snow, slush, and standing water on runways, i.e. deterioration of braking action.

In addition to the detrimental effects of snow, slush, and standing water on braking action, the possibility of severe damage to the aeroplane from slush impingement shall be taken into account.

Due consideration shall be given to all other relevant factors such as anticipated wind and characteristics of runway concerned, i.e. length, width, slope, type of surface.

Under no circumstances will operations take place on runways where the braking action has been reported as POOR or if there is no reported braking action at all.

Under these circumstances you can still be dispatched, but two suitable alternate airfields will be necessary.

The cleared runway width with no significant snow banks should normally be 45 metres. The commander may accept a width of less than 45 metres for take-off and landing, taking into account the weight, braking conditions, crosswind, and aeroplane status.

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6.3 CROSSWIND TABLE

Contaminated Runway Crosswind Limitation				
Reported Brake Action		Reported Runway Friction Coefficient	Typical Equivalent Runway Condition (see note!)	Max Crosswind
Rep Brake Action (SNOWTAM Index)	MOTNE / OPMET			
Good (5)	95	0.4 and above	Dry, damp and wet runway	Aircraft Limit
Medium to Good (4)	94	0.36 - 0.39	Rwy covered with dry snow	20
Medium (3)	93	0.30 - 0.35	Rwy covered with slush	15
Medium to Poor (2)	92	0.26 - 0.29	Rwy covered with standing water, risk of hydroplaning or with slush	5
Poor (1)	91	0.25 and below	High risk of hydroplaning	Prohibited

Note: The Typical Equivalent Runway Condition stated in the table is used as an example of the runway condition one may expect. They shall not be used instead of the reported braking action or Reported Runway Friction Coefficient. The MOTNE, SNOWTAM Code, or Reported Runway Friction Coefficient will determine the crosswind limitation.

6.3.1 SNOWTAM

How to interpret a SNOWTAM

SNOWTAM and MOTNE RWY reports are only provided for winter conditions (snow, ice, etc.) but not for other hazardous RWY conditions e.g. RWY slippery when wet or, wet combined with dust.

Observations and measurements are made at routine intervals but only when RWY are contaminated and/or when there are significant changes.

There are certain definitions used in SNOWTAM and MOTNE.

DAMP

The surface shows a change of color due to moisture.

WET

The surface is soaked but there is no standing water.

WATER PATCHES

Significant patches of standing water are visible.

STANDING WATER

Standing water of significant depth and area which affects the braking action (aquaplaning).

FLOODED

Extensive standing water is visible.

DRY SNOW

Snow which can be blown if loose or if compacted by hand, will fall apart again upon release.

WET SNOW

Snow which if compacted by hand will stick together and tend to form a snowball.

SLUSH

Water saturated snow which with a heel-and-toe slap-down motion against the ground will be displaced with a splatter.

COMPACTED SNOW

Snow which has been compressed into a solid mass that resists further compression and will hold together or break up into chunks if picked up.

SNOWTAM example

A) LSZH B) 11070620 C) 10 D) 2200 E) 40L
F) 4/5/4 G) 20/10/10 H) 30/35/30MUM J) 30/5L K) YES L
L) TOTAL M) 0900 N) B, D P) YES 12 S) 11070920
T) FIRST 300M RWY 10 COVERED BY 50 MM SNOW, RWY CONTAMI-
NATION 100%

A) LSZH

The ICAO 4-letter location indicator for the aerodrome.

B) 11070620

Day-time group for when the SNOWTAM was issued. The format is month-day-hour-minute UTC.

C) 10

Runway designator.

A SNOWTAM will always tell you the conditions of a runway seen from the end of the lower designator. For example, if the runway orientation is 10/28, then the conditions from rwy 10 will be in the SNOWTAM.

D) 2200

Cleared runway length, if less than published length (in metres).

E) 40L

Cleared runway width, if less than published. (in metres; 'L' or 'R' is added when offset left or right of centre line).

F) 4/5/4

Deposits over total runway length (Observed on each third of the runway starting from threshold having the lower runway designation number).

F) Deposits			
0 or NIL	Clear and dry	6	Slush
1	Damp	7	Ice
2	Wet or water patches	8	Compacted or rolled snow
3	Rime or frost covered (norm. < 1mm)	9	Frozen ruts or ridges
4	Dry snow	/	Not reported (clearance of rwy in progress)
5	Wet snow		

G) 20/10/20

Mean depth (in mm) for each third of total runway length.

H) 30/35/30MUM

Friction measurement on each third of runway and friction measuring device.

H) Friction Measurement	
Measured or Calculated Coefficient	Estimated Surface Friction
0.40 and above	Good - 5
0.39 to 0.36	Medium/Good - 4
0.35 to 0.30	Medium - 3
0.29 to 0.26	Medium/Poor - 2
0.25 or below	Poor - 1
9 - Unreliable*	Unreliable* - 9

* If unreliable is used, it is because the contamination is outside the approved range of whatever equipment is used to measure the braking action. This could either be because the contamination is too deep, or of a type that the equipment is not approved for. For example, measured friction by Grip Tester (GRT) on wet ice is considered to be unreliable and the SNOWTAM will then have the value 9 in field H).

Types of measuring equipment:

- BRD = Brakemeter Dynamometer.
- GRT = Grip Tester.
- MUM = Mu meter.
- RFT = Runway Friction Tester.
- SFH = Surface Friction Tester (high pressure tyre).
- SFL = Surface Friction Tester (low pressure tyre).
- SKH = Skiddometer (high pressure tyre).
- SKL = Skiddometer (low pressure tyre).
- TAP = Tapley meter.

J) 30/5L

Critical snowbanks, if present, insert height (cm) / distance from the edge of runway (m) followed by L or R or LR if applicable.

K) YES L

Runway lights (If obscured insert YES followed by L or R or both LR if applicable).

L) TOTAL

Further clearance (if planned, insert Length (m) / Width (m) to be cleared or insert TOTAL if full dimensions).

M) 0900

Further clearance expected to be completed by (UTC).

N) B, D

Taxiway (if no appropriate taxiway is available, insert NO).

P) YES 12

Taxiway snowbanks (if more than 60 cm, insert 'YES' followed by distance apart in metres).

S) 11070920

Next planned observation/measurement (month/day/hour UTC).

T) First 300M RWY 10 covered by 50 mm

Plain language remarks (including contaminant coverage and other significant information like sanding or de-icing).

Special Codes can be added to the report:

///// = Report not updated.

CLRD = Contamination has gone (e.g. runway has been cleared).

SNOCLO = Airport closed due to snow.

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6.3.2 RUNWAY STATE GROUP

An eight figure Runway State Group may be added to the end of the **METAR (or SPECI)** (following any TREND) when there is laying precipitation or other runway contamination.

- Runway Designator (first two digits).
- Runway Deposits (third digit).
- Extent Runway Contamination (fourth digit).
- Depth of Deposit (fifth and sixth digits).
- Friction Co-efficient or Braking Action (seventh and eighth digits).

D_RD_R = Runway designator			
Parallel runways will be shown i.e.: 27L = 27; 27R = 77 (Right runway always + 50)			
All runways effected = 88			
Repetition of the last message, because no new information received = 99			
E_R = Runway deposits			
0	Clear and dry	6	Slush
1	Damp	7	Ice
2	wet or water patches	8	Compacted or rolled snow
3	Rime or frost covered (norm. < 1mm)	9	Frozen ruts or ridges
4	Dry snow	/	Not reported (clearance of rwy in progress)
5	Wet snow		
CR= Extend of RWY contamination		ERER = Depth of Deposits	
1	Less than 10% covered	00	Less than 1mm
2	11% to 25%	01-90	1... to 90mm
5	26% to 50%	92-98	10cm to 40cm in increments of 5cm
9	51% to 100%	99	Not reported (clearance of rwy in progress)
/	Not reported	//	
MOTNE	Braking coefficients	Braking action	SNOWTAM
95	0,40 and more	Good	-5
94	0,39 - 0,36	Medium to Good	-4
93	0,35 - 0,30	Medium	-3
92	0,29 - 0,26	Medium to Poor	-2
91	0,25 and less	Poor	-1
99	9	Unreliable	-9

NOTE: Runways can only be inspected as frequently as conditions permit, so that a reissue of a previous half hourly report does not necessarily mean that the runway has been inspected again during this period, but might mean that no significant change is apparent.

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6.3.3 RUSSIAN FRICTION MEASUREMENT

Depending on the means of measurement friction coefficient is determined either: directly by the results of measurement or, by converting the results of measurement into a so-called normative friction coefficient with a correlation table. Thus at any Russian airport (no matter what equipment is used to measure braking action) the results of the measurement will always be converted to finally derive a normative friction coefficient. This normative friction coefficient is the only one used officially in Russia and is known as friction coefficient. Flight manuals of any type of Russian aircraft use this normative friction coefficient to determine landing performance depending on RWY conditions.

There are two friction coefficients in Russia:

One is measured by **ATT-2** (a two wheel cart dragged by a vehicle, SFT) and another is the normative friction coefficient (Russian friction coefficient) which is a result of **conversion**.

BRAKING ACTION (RUSSIAN FRICTION COEFFICIENT)	FRICTION COEFFICIENT BY ATT2 (=SFT)	NORMATIVE FRICTION COEFFICIENT
GOOD	0.4 OR ABOVE	0.42 OR ABOVE
GOOD / MEDIUM	0.39-0.36	1.42-0.4
MEDIUM	0.35-0.3	0.39-0.37
MEDIUM / POOR	0.29-0.26	0.36-0.35
POOR	0.25-0.18	0.34-0.31
UNRELIABLE	0.17 OR BELOW	0.3 OR BELOW

6.4 TAXI

Use minimum thrust for breakaway and taxiing, to avoid blowing snow or slush on personnel or aeroplanes nearby.

Maintain ground speed below 10 kt when taxiing in snow covered or icy runways. Lower speeds will also avoid throwing slush on the wheel and brake assembly.

Use firm brake pressure on taxi stops whenever pavement conditions permit in order to warm up the brakes and dry moisture buildup within the disk stack. Anti-skid protection is not provided below this speed, so apply brakes accordingly.

During taxi, 'cold set' (the condition where the tyre retains the flat shape it had while parked) may induce vibration in the airplane. Vibration should disappear as the tyres recover their elasticity during taxi. Do not initiate your take-off run before the 'cold set' disappears.

Turns should be performed at the largest turning radius, preferably at a speed which does not require braking during the turn.

Maintain a greater than normal distance from other aeroplanes while taxiing in snow or slush-covered runways, to avoid contamination by snow blown by jet blasts.

If snow and/or slush are present, you should delay flap extension until shortly before line up. Contaminants may accumulate in the flap mechanism, freeze during take-off run and subsequently cause damage during flap retraction. Look out for CAT II/III holding points.

Short frequent applications of the brakes are recommended to warm up the brakes and help clear slush / ice deposits.

Prolonged running of engines in icing conditions can result in a build-up of ice on the rear side of the fan possibly indicated by unusual vibration. Observe aircraft specific procedures and limitations according to the OM-B and AFM.

6.5 TAKE-OFF

6.5.1 PRE-TAKE-OFF

Prior to take-off the commander shall assess whether the applied holdover time is still appropriate. However, if the commander decides that the aircraft needs de-icing and anti-icing while within the HOT, do so.

The final decision as to the airworthiness of the aircraft rests with the commander, who will have to visually check the critical surfaces for contamination. For this, the wing light may be useful. If the condition of the critical surfaces of the aircraft cannot be effectively assessed by a pre-take-off check, a pre-take-off contamination check is required, which is normally performed from outside the aircraft.

When inspecting the wing, during the pre-take-off contamination check, look at the entire upper surface and not only at the leading edge or wing tip. Although the wing tips can be seen from the cockpit, almost the entire wing is visible from a cabin window. Therefore, it is strongly advised that the visual inspection be done by a crew member from the cabin. Additionally, the crew should ask for the assistance of trained and qualified personnel outside the aeroplane to assist in the pre-take-off and check to make sure that the tail and fuselage, which are not visible from the cockpit or cabin, are free of any ice contamination.

It is the pilot's responsibility to decide whether or not to accept the aeroplane for flight. If contamination is suspected, the aeroplane should return for additional de-icing or anti-icing.

If any doubt exists about the ice-free state of the aircraft, the aircraft must return to the ramp for a complete two-step procedure.

6.5.2 TAKE-OFF

You may want to use the weather radar to check the climb out lane for poor weather before commencing the take-off roll. Although the aircraft is approved for flight in icing conditions, good airmanship dictates that icing conditions should be avoided whenever possible. The aircraft/airframe ice protection systems shall be operative.

Use positive forward pressure on the nose wheel and apply thrust slowly to prevent asymmetry using rudder for directional control as soon as it becomes effective. The rotation should be as smooth as possible to an initial 3° pitch attitude.

After take-off, if performance is not limiting, consider delaying gear retraction for 10 to 15 seconds to allow the airflow to blow off any snow or slush, which might otherwise cause the undercarriage to remain frozen up on recovery. Another option is to cycle the landing gear to help clear any contamination from the gear system, however, aircraft specific procedures according to the AFM have to be observed.

6.6 CLIMB

Operation in moderate to severe icing conditions may induce ice build-up on the fan spinner and/or blades. If ice accumulates, its asymmetrical shedding may result in high fan vibration.

NOTE: Engine vibration indication may peak to the maximum value prior to ice shedding; however, this will not affect the engine.

When flying in icing conditions or after flying in icing conditions, ice accretion on unprotected areas may cause vibration at high speeds. If vibration and/or buffeting occur, a change in the current airspeed will eliminate these effects. At high speeds, reduce the airspeed as required.

If encountering icing in a later stage of the climb and climbing out of the icing quickly is not likely, descending may be the best option. If the freezing level is known and it is safely above the MEA, a descent to warmer air may be your best option.

6.7 CRUISE

The build-up of airframe ice in flight may be very rapid, even relatively small amounts of ice can have a significant and dangerous effect on an aircraft's performance and control.

Icing is most frequent when the SAT is between +2° and -20°C. In severe icing conditions, the rate of ice accretion may be as high as five centimetres per minute. Keep well clear of cumulonimbus clouds, which can give hail, icing, freezing rain, and turbulence. Regular visual checks of the leading edges of the wings shall be done when cruising in icing conditions. This can be done at night by using the wing leading edge light.

The WASAIS is inhibited above 30000 ft and if left on the CAS will alert crew with an INHB message. All of our Phenoms are post MOD SB 505-00-0001 which allows operations in icing conditions up to 45000 ft with All Engines Operating. With One Engine Inoperative the maximum limit for flight in icing conditions is 15000 ft and the WASIS is automatically inhibited above this level.

6.7.1 USE OF RADAR

As a rule of thumb when using the weather radar to avoid storm cells, choose a heading which avoids by 5 NM when the OAT is above 0°C, 10 NM when the OAT is below 0°C, and 20 NM when flying above FL200.

Flight Altitude FL	Echo Characteristics			
	Shape	Intensity	Gradient of Intensity	Rate of Change
	Avoid by 10 miles Echoes with hooks, finger, scalloped edges or other protrusions	Avoid by 5 miles Echoes with sharp edges or strong intensities	Avoid by 5 miles Echoes with strong gradients of intensity	Avoid by 10 miles Echoes showing rapid change of shape, height or intensity
20 - 25	Avoid all Echoes by 10 miles			
25 - 30	Avoid all Echoes by 15 miles			
>30	Avoid all Echoes by 20 miles			

6.7.2 SIX DIGIT TURBULENCE AND ICING CODE

The Turbulence and Icing Reports will be delivered in a six-digit format usually placed behind the QNH reading in the METAR/TAF. This can give you already some idea where you can expect icing and their intensity.

TURBULENCE '5' GROUP	
Code Figures	Type of Turbulence and Intensity
0	None
1	Light Turbulence
2	Moderate Clear Air Turbulence Occasional
3	Moderate Clear Air Turbulence Frequent
4	Moderate in Cloud Turbulence Occasional
5	Moderate in Cloud Turbulence Frequent
6	Severe Clear Air Turbulence Occasional
7	Severe Clear Air Turbulence Frequent
8	Severe in Cloud Turbulence Occasional
9	Severe in Cloud Turbulence Frequent
X	Extreme Turbulence
Occasional	Occurring less than 1/3 of the time

ICING '6' GROUP		
Code Figures	Type of Icing	
0	No Icing	
1	Light Icing	MIXED
2	Light Icing in Cloud	RIME
3	Light Icing in Precipitation	CLEAR
4	Moderate Icing	MIXED
5	Moderate Icing in Cloud	RIME
6	Moderate Icing in Precipitation	CLEAR
7	Severe Icing	MIXED
8	Severe Icing in Cloud	RIME
9	Severe Icing in Precipitation	CLEAR

For example:

510403 (Turbulence Group)

- 5 = Turbulence Indicator.
- 1 = Type of Turbulence and Intensity, e. g. Light Turbulence (see table above).
- 040 = Height of the Turbulence layer base in hundreds of feet AGL, e.g. 4000 feet.
- 3 = Thickness of the Turbulence layer in thousands of feet, e. g. 3000 feet.

650057 (Icing Group)

- 6 = Icing Indicator.
- 5 = Type of Icing, e. g. Moderate Rime Icing in Cloud (see table above).
- 005 = Height of the Icing layer base in hundreds of feet AGL, e. g. 500 feet.
- 7 = Thickness of the Icing layer in thousands of feet, e. g. 7000 feet.

6.7.3 PILOT ACTIONS ENCOUNTERING ICING IN CLOUDS

In general, when the edges of clouds appear sharp and well defined, the clouds are likely to contain mostly liquid water.

If the edges appear diffuse or fuzzy, the clouds are likely to be composed largely of frozen ice crystals. Ice crystals will not accrete on the aircraft.

When flying above a cloud deck and the sun is shining, look at the cloud deck about where the aircraft's shadow would be. If you see coloured rings or a rainbow-like pattern, the cloud is likely to have liquid water.

Icing in cumulus clouds is usually limited laterally, but can extend vertically for thousands of ft. Icing in stratiform clouds may be quite extensive laterally, but is usually less than 3000 ft thick vertically. Icing often is most intense in frontal conditions.

In 'cold' clouds, where the temperature is below 0°C, ice crystals and supercooled droplets co-exist. In these 'mixed' clouds, the air is close to being saturated with respect to liquid water, but is super saturated (an unstable phase) with respect to ice. Consequently, in mixed clouds, ice crystals grow from the vapour phase much more rapidly than do the nearby droplets. This is usually known as the Bergeron - Findeisen process.

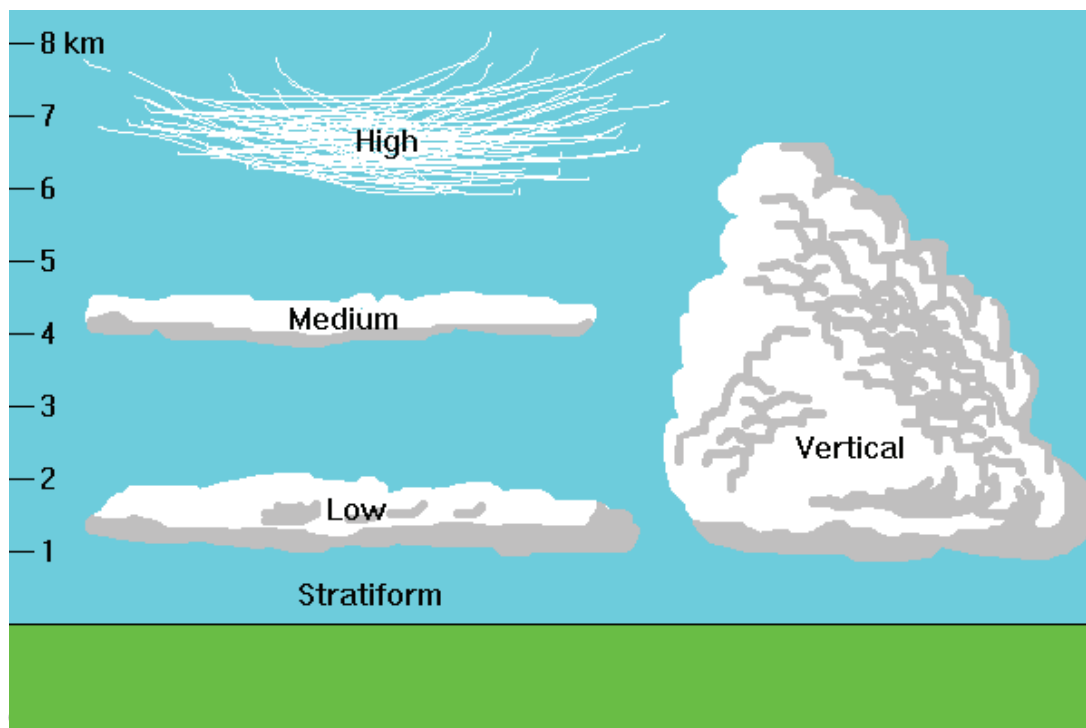
Below -40°C almost all clouds consist entirely of ice crystals, which are not harmful anymore to an aircraft skin or leading edge.

Therefore, if using the engine anti-ice below -40°C at high altitudes might lead to unexpected engine behavior like compressor stalls or even damage.

As a reminder, by using the SIGMET Charts you will already get a good idea where to expect icing conditions and your avoidance routing.

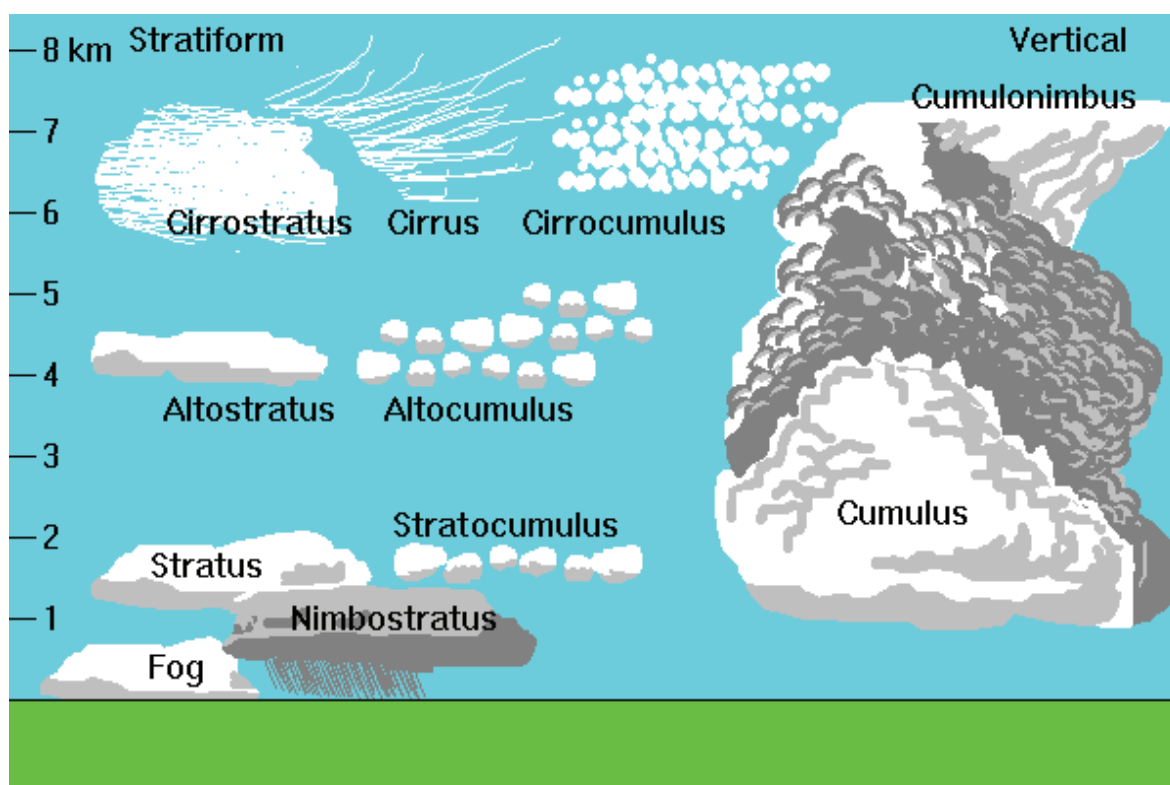
STRATIFORM CLOUDS

Activate the ice protection systems and change the altitude by 3,000 ft, which will take you out of the icing environment, even if the aircraft remains in the clouds.



CUMULUS CLOUDS

Navigate laterally around cumulus clouds.



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ALONG WINDWARD SIDE OF MOUNTAIN RANGE

Consider flying along the leeward side to reduce the risk of ice build-up, but watch out for mountain wave turbulence.

ALONG A WARM FRONT

Fly behind the warm front. A change of 3,000 ft will take you out of the icing environment.

ALONG A COLD FRONT

Fly behind it and do not rely solely on weather radar. Cells that are not yet visible on weather radar can produce substantial icing.

ALONG A SHALLOW COLD FRONT

Consider flying ahead of a shallow cold front, and be prepared to change altitude by 3,000 ft if encountering icing.

6.7.4 PILOT ACTIONS WHEN IN SEVERE ICING CONDITIONS**WARNING**

SEVERE ICING MAY RESULT FROM ENVIRONMENTAL CONDITIONS OUTSIDE OF THOSE FOR WHICH THE AEROPLANE IS CERTIFICATED. FLIGHT IN FREEZING RAIN, FREEZING DRIZZLE, OR MIXED ICING CONDITIONS (SUPERCOOLED LIQUID WATER AND ICE CRYSTALS) MAY RESULT IN ICE BUILD-UP ON PROTECTED SURFACES EXCEEDING THE CAPABILITY OF THE ICE PROTECTION SYSTEM, OR MAY RESULT IN ICE FORMING AFT OF THE PROTECTED SURFACES.

THIS ICE MAY NOT BE SHED USING THE ICE PROTECTION SYSTEMS, AND MAY SERIOUSLY DEGRADE THE PERFORMANCE AND CONTROLLABILITY OF THE AEROPLANE.

During flight, severe icing conditions that exceed those for which the aEROPlane is certificated shall be determined by the following visual cues. If one or more of these visual cues exists, immediately request priority handling from Air Traffic Control to facilitate a route or an altitude change to exit the icing conditions:

- Extensive ice accumulation on the airframe in areas not normally observed to collect ice.
- Accumulation of ice on the wing aft of the protected area.

Since the autopilot may mask tactile cues that indicate adverse changes in handling characteristics, use of the autopilot is prohibited when any of the visual cues specified above exist, or when unusual lateral trim requirements or autopilot trim warnings are encountered while the aEROPlane is in icing conditions.

PROCEDURES FOR EXITING SEVERE ICING CONDITIONS

These procedures are applicable to all flight phases from take-off to landing. Monitor the ambient air temperature.

While severe icing may form at temperatures as cold as -18°C , increased vigilance is warranted at temperatures near freezing with visible moisture present. If the visual cues for identifying severe icing conditions as specified above are observed, accomplish the following:

- Immediately request priority handling from Air Traffic Control to facilitate a route or an altitude change to exit the severe icing conditions in order to avoid extended exposure to flight conditions more severe than those for which the aEROPlane has been certified.
- Avoid abrupt and excessive manoeuvring that may exacerbate control difficulties.
- Do not engage the autopilot.
- If the autopilot is engaged, hold the yoke / control column firmly and disengage the autopilot.
- If an unusual roll response or uncommanded roll control movement is observed, reduce the angle of attack.
- Do not extend flaps during prolonged operations in icing conditions. Operation with flaps extended can result in a reduced wing angle-of-attack, with the possibility of ice forming on the upper surface further aft on the wing than normal, possibly aft of the protected area.
- If the flaps are extended, do not retract them until the airframe is clear of ice.
- Report these weather conditions to Air Traffic Control.

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6.8 DESCENT

Before starting the descent into icing conditions the aircraft anti-icing systems shall be switched on and checked to be operational.

If icing is expected during the descent, remain above the icing layer as long as possible and try to transit through the ice as quickly as possible. However, be careful of 'slam dunk' approaches.

6.9 APPROACH

Flying a missed approach with a contaminated air foil can be extremely dangerous.

Do not start an approach if the local weather conditions are such that a missed approach may be necessary and the accretion of ice suggests that you may not be able to successfully complete the missed approach.

In this case, a diversion to the alternate may be the best option.

6.9.1 RECOMMENDED PROCEDURES

- Maintain a minimum power required to ensure adequate protection for the engines and intakes by the engine anti-ice system.
- Keeping the speed up on the approach will minimise the risk of wing stall.
- If icing conditions exists, or if ice has formed on the unprotected parts of the airframe prior to approach, add the AFM recommended speed to the normal approach and landing speeds and observe the increased landing distance.
- Consider limiting or delaying flap extension.
- If the runway is reported to be contaminated, check the AFM for adjusted distances, speeds, and crosswind limitations.

6.9.2 ALTIMETER TEMPERATURE ERROR CORRECTION

This correction is made easier in the Phenom for us by going into the minimums page on the GTC you can select TEMP COMP and set the temperature on the ground for auto compensation. The TEMP COMP is shown on the GTC pages by COMP MIN showing on the PFD.



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Apply the altimeter temperature error correction for minima and terrain clearance. Add 4% for ea

Aerodrome Temp. (°C)	Height above the elevation of the altimeter setting source (FEET)										
	200	300	400	500	600	700	800	900	1000	1500	2000
0	20	20	30	30	40	40	50	50	60	90	120
-10	20	30	40	50	60	70	80	90	100	150	200
-20	30	50	60	70	90	100	120	130	140	210	280
-30	40	60	80	100	120	140	150	170	190	280	380
-40	50	80	100	120	150	170	190	220	240	360	480
-50	60	90	120	150	180	210	240	270	300	450	590

Aerodrome Temp. (°C)	Height above the elevation of the altimeter setting source (METERS)										
	60	90	120	150	180	210	240	270	300	450	600
0	5	5	10	10	10	15	15	15	20	25	35
-10	10	10	15	15	25	20	25	30	30	45	60
-20	10	15	20	25	25	30	35	40	45	65	85
-30	15	20	25	30	35	40	45	55	60	85	115
-40	15	25	30	40	45	50	60	65	75	110	145
-50	20	30	40	45	55	65	75	80	90	135	180

Pressure altimeters are calibrated to ISA conditions. Any deviation from ISA will result in error proportional to ISA deviation and the height of the aircraft above the aerodrome pressure datum. The error is approximately 4 ft/1000 ft per °C difference. When temperature is LESS than ISA an aircraft will be LOWER than the altimeter reading. Table values should be ADDED to published/calculated altitudes or heights.

The error corrections in the tables are properly a function of deviation from ISA, but for simplicity the aerodrome temperature may safely be used for aerodromes up to 1000 ft above sea level. At higher aerodromes the ISA deviation should be used. The temperature at ISA is + 15°C minus 2°C per 1000 ft above sea level. The ISA deviation is the ambient temperature minus the temperature at ISA. (e.g. An airfield 2500 ft above sea level at -30° C has an ISA deviation of $-30 - (+10) = -40$).

When to apply corrections

When the aerodrome temperature is -10°C or colder, temperature error correction MUST be added to:

- DH/DA or MDH/MDA and step fixes inside the FAF.
- ALL low altitude approach procedure altitudes in mountainous regions (defined as 3000ft amsl or higher).

When pilots intend to apply corrections to the FAF crossing altitude, procedure turn, or missed approach altitude, they MUST advise ATC of their intention and the correction to be applied.

Pilots may refuse IFR assigned altitudes if altimeter temperature error will reduce obstacle clearance below acceptable minima. However, once an assigned altitude has been accepted, it must not subsequently be adjusted to compensate for temperature error.

Minimum sector altitude (MSA)

When the temperature is -30°C or colder, add 1000 ft to the MSA to ensure obstacle clearance.

6.10 LANDING

Again, check your crosswind limits for all runway conditions.

If the runway is covered with snow or slush, ensure to slow down as soon as possible, even with a long runway ahead. The last half of the runway may be slippery due to preceding aircraft's braking.

Conduct a positive landing to ensure initial wheel spin-up and initiate firm ground contact upon touchdown, achieving wheel load as quickly as possible. Such technique avoids hydroplaning on wet runways and reduces the strength of any ice bond that might have been eventually formed on brake and wheel assemblies during flight.

The factors that influence the occurrence of hydroplaning are high speed, standing water, and poor runway macrotexture. When hydroplaning occurs, it causes a substantial loss of tyre friction and wheel spin-up may not occur.

Icy runways can be very slippery at all speeds depending on temperature.

Stopping the aeroplane with the least landing run must be emphasised when landing on wet or slippery runways.

- Anticipate the approach procedures and speeds: a well-planned and executed approach, flare, and touchdown minimise the landing distance.
- Lower nose wheel immediately to the runway. It will decrease lift and will increase main gear loading.
- Apply brakes with moderate-to-firm pressure, smoothly and symmetrically, and let the anti-skid do its job.
- If no braking action is felt, hydroplaning is probably occurring. Do not apply Emergency / Parking Brake, as it will remove anti-skid protection.

Maintain runway centreline and keep braking until aeroplane is decelerated.

Keep the minimum aquaplaning speed in mind (8.6 times the square root of tyre pressure in psi) and go easy on the brakes until well below it.

For reference:

180 psi equals aquaplaning speed of 115 kt. 147 psi equals aquaplaning speed of 104 kt.

If the tyre is not rotating, is locked, or stops spinning for any reason, full aquaplaning speed can be reduced to as low as 7 times the square root of the tyre pressure in psi.

IF THE NO CAS MESSAGE IS PRESENT THE BRAKES ARE WORKING LET THEM DO THEIR JOB.

DO NOT USE THE EMERGENCY BRAKE AS YOU WILL LOOSE ANTI-SKID PROTECTION AND MAKE THE SITUATION WORSE.

6.11 VACATING THE RUNWAY AND TAXI IN

Slow down to a low speed before exiting the runway. It is a good habit to bring the aircraft to almost a complete stop before turning.

Turn the nose wheel carefully and make a note where the aircraft starts to slide. When sliding, turn the nose wheel neutral again to enable grip and turn it to enable a turn when grip is recovered. Consider differential braking to aid in turning the aircraft.

If icing conditions were encountered during the approach, or the landing was on a contaminated runway, consider leaving the flaps extended for taxiing in. Subsequently, ensure the flaps and flap mechanisms are inspected, and if necessary, de-iced before retraction.

Once safely on the taxiway, taxi half the normal taxi speed. Bring the aircraft to almost a complete stop before turning.

Turning off from the taxiway towards the apron, be careful with any snow ridges originating from cleaning the apron. The snow ridges may have frozen and may cause damage to the undercarriage.

The taxiways might not have been as well cleared as the main runway, in addition the runway may well be grooved which will assist braking action and feel of the aircraft.

Check the clearance of the taxiway, how much snow has been cleared, is the snow piled up next to the taxiway, are all taxiways available. The route you have taken many times before may not be available.

6.12 PARKING AND POST-FLIGHT

Put on the chocks as soon as possible and release the parking brake. Check the brakes after parking and brush the snow and/or slush off to prevent the brakes from freezing in applied position.

The pilot opening the main cabin door should go out first to assist the passengers exiting the aircraft and remain near to the aircraft steps to ensure the passengers will not fall on the slippery surface. Be careful when disembarking the passengers because they may not have suitable winter footwear or may have come from somewhere much warmer and are not prepared for being exited into the cold. Try to make sure the FBO has suitable transport available to ease the process and consider the use of a mat.

Expedite putting on all covers when it snows, to avoid blowing snow to get into the pitot tubes and engine intakes.

If ground icing conditions are forecast during your ground stay, arrange to protect your aircraft to spare the de-icing expense.

IF YOU CAN, HANGAR THE AIRCRAFT

Even an unheated hangar gives good protection against the snow and especially the wind. Check the cost but it is often cheaper than having to de-ice the aircraft the next day. The request should be addressed to LIS-GroundServices@netjets.com. Do check that conditions the following day would not require anti-icing anyway otherwise the cost for the hangar may not be justified.

As guidance, if the temperature is forecast to be near -10°C or below and you have to leave the aeroplane out for the night, make sure you take out all the items that may freeze, like water bottles, soda cans, coffee water container, wine, and champagne bottles. Do not forget the extra stock in the baggage bay.

Try to find out where the FBO will store all of the stock so that you can make sure you will have easy access to it again the following morning.

Do not forget that if you are leaving the aircraft at the end of a tour and you do not know when it might be crewed again just because the temperature is ok now does not mean that in 3 days' time it will still be. Most stock exploding events have occurred after the aircraft was left for a few days by an off-going crew and the temperature dropped lower than when they left it.

If you have any doubt please remove the liquid stock.

The lavatory should be emptied but not refilled until just before the next flight. If this is not possible consider adding glycol or vodka to the toilet fluid as a last resort to prevent it from freezing.

Embraer recommend removal of the aircraft batteries when the aircraft is left outside for more than 12 hours at a temperature of -10°C, or in excess of just 3.5 hours at ambient temperatures below -15°C. However, given the fact that this is a maintenance action and would be very difficult to carry out, Embraer also state that this temperature limitation does not apply when using a GPU for the next start. (FOL 505 -013/11). Therefore, if it is going to be extremely cold before the next flight try to ensure a GPU will be available.

When closing the main cabin door, make sure that the steps are clean of snow and slush and the door seals as well. This will prevent from snow on the door seals to melt due to the warm cabin and freeze during the night.

6.13 CONCLUSION

Winter Operations are one of the most challenging tasks we are facing every year in our flying environment. As ever, **safe** execution will be our highest priority. If there is any doubt, then you as a crew shall take responsibility to inform the office as soon as possible. We are all here trying to give you the best support on the line.

CHAPTER 7 - MISCELLANEOUS SYSTEM ADVICE

7.1 GARMIN DEFAULT PROFILE

The following profile has been saved into the Phenom Garmin 3000 as a NETJETS DEFAULT for the LOAD 1X to allow for all avionics and map settings to be standard across the fleet. This setting is designed to make it easier to begin pre-flight because all settings will be the same for the beginning of each flight. Individuals are free to use their own preferences and settings but must not alter or change the NETJETS DEFAULT profile that is saved in the Utilities / Crew Profile page.

We will also programme the L1X aircraft to have this as the power up default setting (you will know it has worked if you have split screens on power-up)

This latest amendment has considered the activation of the Connex Weather

Garmin 3000 Profile Setup

This section describes the standard setup for PFD1, PFD2, and MFD. Operational requirements (e.g. mountain airports, uncontrolled airspace) may dictate modification of this setup. Any modification must be briefed.

PFD SETTINGS (Using the Soft Keys along the bottom of each PFD)

- PFD Mode - Split - Split with Chart and PFD.
- Bearing 1 - OFF - to be set as required during pre-flight. ILS is normally set on Nav1.
- Bearing 2 - OFF - to be set by PF but normally set to raw VOR for departure.

Other PFD settings

- Wind - Option 3.
- Alt Units - HPA.

MFD SETUP

Select - Home>Map>Map Settings:

MAP SETTINGS

- Orientation - Track Up.
- Map Sync - OFF.
- Map Detail - Most.

SENSOR TAB

- Traffic - ON
 - Traffic Settings Tab
 - XPDR/TCAS - Auto
 - Altitude Range - Normal
 - Altitude Display - Relative
 - Map Settings Tab
 - Labels - On
 - Symbols - 75 NM
 - Labels - 75 NM
- Terrain - Absolute
 - Settings Tab
 - Map Setting Tab
 - Terrain 250 nm
 - Absolute Terrain Scale - ON
- Weather Radar - OFF
- Graphical Metars - OFF
- Connex Radar - OFF
- Connex Lightning - OFF
- TFR - OFF
- WX Source - Connex

INSET WINDOW TAB

- VERT Display- OFF
 - Settings
 - Mode - Auto
 - Winds - OFF
- Flight Plan Text - ON
- Flight Plan Progress - OFF

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AVIATION TAB

- Low Altitude Airways - Off
- High Altitude Airways - Off
- Airspaces - On
 - Settings
 - Smart Airspace - ON
 - Class B/TMA - 50 nm
 - Restricted - 50 nm
 - Class C/CTA - 50 nm
 - MOA - 50 nm
 - Class D - 10 nm
 - Other/ADIZ - 50 nm
- Airports - ON
 - Large - 100
 - Medium - 50
 - Small -25
- VOR - ON
 - Settings
 - VOR 50 nm
 - Compass Rose OFF
- INT - Off (25 nm)
- NDB-On (40 nm)
- User Waypoint - On (50 nm)
- Safe Taxi - On (1.5 nm) IF FITTED
- Runway Extension - On (25 nm)

LAND TAB

- Obstacle Data - ON
 - Obstacle Data Settings - 2.5 NM
- Roads - On (Default Ranges 50/15/4/2.5 nm)
- Railroads - On
 - 7.5 nm
- Cities - On (Default Ranges 100/50/25 nm)
- State/Province - On
 - 400 nm
- River/Lake - On
 - 75 nm

OTHER TAB

- Auto Zoom - OFF
- Track Vector - On
 - 60 Secs
- Wind Vector - On
- Fuel Ring Rsv - ON
 - Set 45 minutes
- Selected Altitude Range Arc - On
- Field of View - OFF
- Lat/Long Lines - OFF

Home>Utilities>Setup>Avionics Settings

AVIONICS SETTINGS

System Tab

- Time Format - UTC
- Flight Director Active Format - Dual Cue
- GPS CDI - AUTO
- COM Channel Spacing - 8.33kHz
- Nearest Airport Runway Surface - Hard Only
- Nearest Airport Minimum Runway Length - 3000 ft
- Sync CDI - OFF
- Sync Alt Baro - OFF

Units Tab

- Nav Angle - Magnetic
- Altitude/Vertical Speed - Feet
- Cockpit Cabin Temp - Celsius
- Position Format - HDDD°MM.MM

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Alerts

- Baro Transition Alert - 1000 FT
- Enable - ON
- Surface Watch - Inhibit OFF (Only shown if fitted)
- Airspace Alert Altitude Buffer - 1000 FT
- Arrival Alert - Off
- Class B/TMA - Off
- Class C/TCA - Off
- Class D - Off
- Restricted Airspace - Enable
- MOA (Military) - Off
- Other/ADIZ - Off

MFD Fields

- Field 1. - ESA
- Field 2. - MSA
- Field 3. - GS
- Field 4. - TAS
- Field 5. - DIS
- Field 6. - ETE
- Field 7. - ETA
- Field 8. - FOD

Audio Tab

- COM Frequency Change Tone - OFF

Home > Utilities > Trip Stats

Trip Statistics

- Flight Time - In-Air
- Departure Time - In-Air

All Resets to be 'Manually'

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Utilities > Weight and Fuel

Operating Weights

- BEW - 11704 lbs (PHA as an example)
- Pilots & Stores - 656 lbs
- BOW - 12360 lbs (PHA as an example)

Landing

- Fuel Reserves - 0 lbs (to be set as per OFP each flight)
- Holding Time - 30 Minutes

If, for any reason, the NETJETS DEFAULT profile is not on the jet (normally after maintenance), make sure the above settings are in the system and then:

Select - **Utilities - Crew Profile**

Press - **Add Profile**

Input the name to assign to the crew profile in this case **NETJETS DEFAULT**

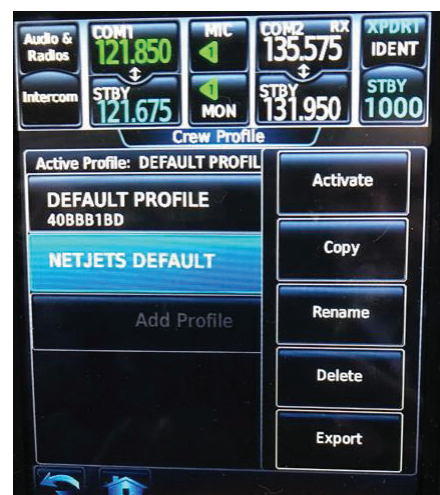
Press - **Enter**

7.1.1 SETTING UP THE DEFAULT PROFILE

On all jets the NETJETS DEFAULT PROFILE should come up after power on with no requirement to have to select it from the menu.

Normally this is shown by the split screens on the PFDs coming up straight away. HOWEVER, after any maintenance, the default profile may have dropped out. These instructions will allow you to reload it.

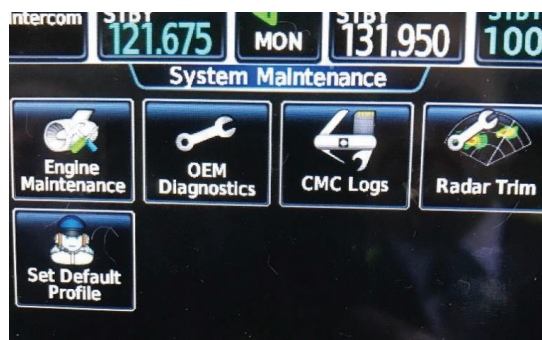
1. Put the NAV SD card in the top MFD slot and turn the power ON.
2. First you need to EXPORT the profile to the SD card.
3. Select Crew Profile as usual but in the normal ACTIVATE window the EXPORT button will now be lit white because the SD card is fitted.
4. Select EXPORT.



5. Next, from the Aircraft system page go into the Maintenance section and add the secret code EMBRG3K (same as for a CMC download).



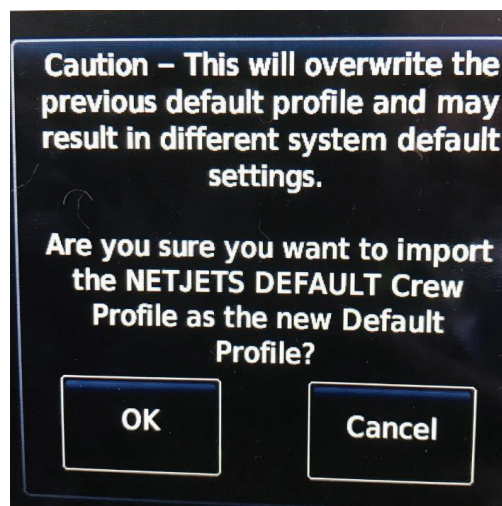
6. Select the Set Default Profile button



NETJETS DEFAULT will be shown because it is on the SD card.



Select NETJETS DEFAULT and OK the caution.



Go back to the System Maintenance home page and select END Maintenance.

Note: By default 'DEFAULT PROFILE' is the active crew profile. This profile cannot be deleted or renamed. The NetJets default will show as active by split screens coming up after power-on.

7.2 CMC DOWNLOAD

The following instructions are the ones required to carry out a CMC download for maintenance.

A CMC download is normally required or requested after any maintenance event or whenever you think the additional information would be useful for Embraer. A CMC download contains a detailed recording that can be analysed by the engineers much more than is seen on the AHEAD system and has proved invaluable so far for Embraer and ourselves.

With crew help doing the download, we can save many hours of the aircraft being off the line because the first thing Embraer will do when the aircraft comes into Maintenance is a CMC download and then analyse the results, and then order the bits. If the fault allows that to be done while the jet carries on flying, the advantages to us are obvious.

Place a blank or spare SD card into the TOP slot of the MFD (the CMC or NAV data card from the silver box).

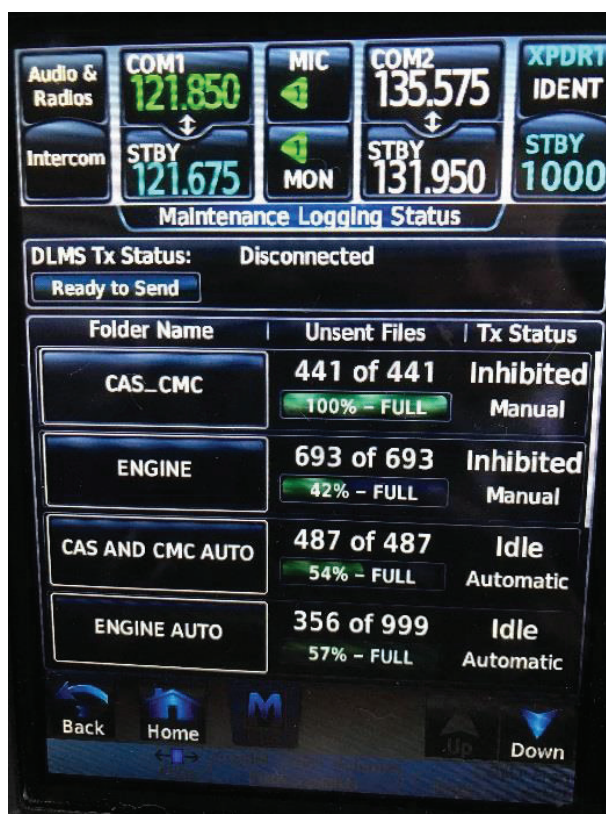
Turn on the power ON and when the jet is up, select 'Aircraft Systems' and then 'Maintenance'.



Type in the Maintenance access password - EMBRG3K.



Select 'CMC Logs' on the System Maintenance page.



Select 'CAS CMC'

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In the instructions on C365 it says to 'Save All' but you do not need to do this (these instructions will be removed from C365 after this Handbook has been published) the file would be huge and most would be deleted. Follow the instructions below and you can just download the files from the flight you want to send the details off.

Select the folder you need from the list (e.g. CAS-CMC).



The individual flights are now shown on the left-hand side of the page.

Select the flight you need (usually the top one).

Select 'Save CMC Log File to Card'.



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The CMC files will be loaded on to the SD card.



Select BACK and repeat for as many files as you need. If it is an engine fault then you will need to go back out to the folder page and select the Engine Folder and repeat. If you are only downloading individual files, then this process is very quick.

These files can then be emailed to Maintenance lis-mxembraer@netjets.com.

Note: You should power the jet down before removing the SD card from the upper MFD slot, but if you forget, you will just get a message on the GTC telling you the card has been removed before power off.

7.3 CONNEXT WEATHER

Connex Weather is available on the fleet. It provides weather data not only for the Connex page but also for the airport information pages if you want to see the latest actual without bringing up the Connex page again.

We pay for the data and whilst there is no limit to data usage, please make sure Auto Request is turned OFF, otherwise the fleet will be sending out weather requests every 15 minutes. With the AUTO Request off you can send an Immediate Request whenever you need to update the weather and save money and data usage.

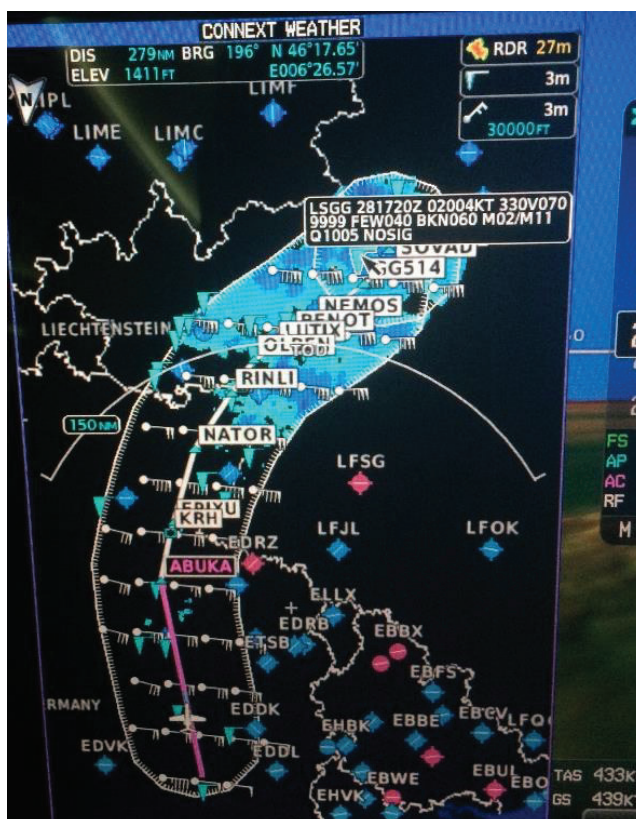
Select the CONNEXT page and then define the coverage you want and make sure you select METAR as well as any other items you want.

Select the parameters on the Connex Weather Coverage page. Probably the simplest is to select the Flight Plan and a sensible width (the example below is 100 nm) then press the 'Send Immediate Request' and it should show you the time to receive the weather.



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Another important reason to define the coverage is to speed up the time it takes to receive the data. If you define all of Europe then it will take time to download.



The weather can be seen at the airports of your choice by moving the cursor over the flag next to the airport. This will display the TAF and METAR.



It will also show the weather in the airport waypoint options and again the small flag will be shown along with the observed weather and the time stamp.

The Connex is has specific codes and these have been known to drop out on occasion of after maintenance. Here is a list of the aircraft specific codes that can be tried if the Connex is not playing. If after entering the code the weather is still down, it will more than likely be a SatPhone/Comm fault, so please have a go at making a call on the phone and see if their whole system is down before calling it in.

Connex Access Codes

A/C TYPE	TAIL	S/N	Access Code
Phenom 300	CS-PHA	50500203	12330772
	CS-PHB	50500209	13088381
	CS-PHC	50500214	13438519
	CS-PHD	50500225	14416248
	CS-PHE	50500252	17324181
	CS-PHF	50500260	17999977
	CS-PHG	50500264	18060976
	CS-PHH	50500270	18475981
	CS-PHI	50500332	24350572
	CS-PHJ	50500336	25147636
	CS-PHK	50500433	45388298
	CS-PHL	50500460	128350340
	CS-PHM	50500471	128350642

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7.4 FLEET OPERATIONAL LETTERS (FOLs)

FOLs are intended to be manufacturer advice or best practice for problems or issues that have been reported to Embraer. It is only advice from the manufacturer but the fact that these FOLs come out very quickly after a problem allows us to troubleshoot before grounding the aircraft. Following an FOL advice will not compromise in any way the MEL or maintenance procedures.

For example - carrying out a reset of the system after a BRAKE FAIL CAS will only help prevention of grounding the aircraft. If the BRAKES actually have a fault, then the CAS will not be cleared after a reset of the aircraft systems.

From Embraer:

FOLs are meant to help Operators with the best practices and procedures based on field reports, aiming to disseminate the information as quick as possible in a constant quest of better support the customer and to avoid unnecessary AOG. The data contained in the FOL is intended to assist the operator and is not a substitute for the Operator's Manuals. Fault correction, maintenance, and the aeroplane's return-to-service tests shall be accomplished in full accordance with proper operator's manuals.

It is not the intention to repeat all of the FOLs currently active with Embraer, however, some of the more frequently used ones are highlighted below along with the more common events that have occurred.

Fuel Pump Fail CAS on Power-Up FOL 001/14 and FOL 012/14

Fuel system CAS messages being triggered when the aircraft is powered up due to navigation database being updated using the lower MFD slot.

Only load the Navigation database via a spare SD card into the top slot of the MFD.

Battery Best Practices FOL 001/13 and FOL 013/11

These FOLs have been issued to complement and reinforce some aspects of battery operation in order to minimise the risk of main battery premature problem. They also give recommendations for ground operations with GPU connected.

FCE Related CAS Messages FOL 015/11

This FOL summarises the recommended procedures in case of CAS messages, related to the Flight Control Electronics (FCE), being displayed during the aircraft power up. These CAS include Flap fail, Spoiler fault, and Pitch Trim faults.

Engine Does Not Start (S/Gen Does Not Start to Run) - FOL 006/13

Embraer has been receiving reports of engine not starting, associated with the starter generator (S/GEN) not running. During the analysis of these events, two root causes have been identified.

ECS - Transitory Condition During Climb and Initial Cruise -FOL 010/14

FOL for ECS 1(2) VALVE FAIL message momentarily shown in flight during climb phase or at initial cruise phase after level off.

ECS Recommendations During Cold Weather Operations - FOL 011/13

During cold weather operations and under external temperatures below 8°C, the Environmental Control System (ECS) will optimise the cockpit temperature utilising the high flow mode (AUX HEAT MODE).

Recommendations Before Turning On the Anti-Ice System Below 13000 Ft - FOL 002/14

Simultaneous BLEED 1 FAIL and BLEED 2 FAIL Messages triggered with associated 'X' indication in the synoptic page, after the A/I WINGSTAB system is switched on.

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7.5 PHENOM CREW COMMS LIST ON COMPLY365

The same as the FOLs it is not the intention to repeat everything that is found on Comply365, however it may be useful for some to have a list of relevant Memos and Alerts that can be found on Comply365.

As this manual matures some communications and briefs will be placed in here, and removed from Comply365. However, at the moment while we receive feedback on the handbook a list of the more important ones can be found below.

2016-07-05 Phenom Wi-Fi Memo - Wi-Fi on the Phenom

Roll out of Wi-Fi on the fleet.

2016-07-04 Loss of GPS Signal Phenom 300 Rev 01

Guidance in the event of loss of GPS signal.

2016-05-23 Safety Alert - BRK FAIL Conditions - BCU

Brief released after a couple of brake fail incidents in the USA.

2016-03-23-EMB 505S-EH-OXYMASKS

Briefing on the differences between the oxygen masks on the aircraft. All aircraft from PHI onwards will have a new type of mask. Due to both masks being very similar and having the same immediate actions there is no intention to change the older style masks.

2015-02-08 EMB-505S EH Performance Guidance for Phenom_300

Briefing on the performance figures and data used on the aircraft.

2015-10-06-EMB 505S -NCN-Phenom 300S GURU Performance Guide

Copy of the official Flyprestanda performance data and limitations used in GURU.

2016-01-18 Guru Pilots Guide

Unofficial TSO written guide of the GURU performance information received pre-flight.

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CHAPTER 8 - UPDATING THE GARMIN DATABASES

8.1 AIRCRAFT LAPTOP

Each aircraft has a laptop to assist with Garmin Updates, CMC downloads, and for doing some trouble shooting on the Go-Go as shown in the paragraphs above.

The laptop is standalone so that we can download programs and data onto it without needing to go through the office. To be allowed to do this the laptops are free from all NJE restraints and, therefore, must be looked after in terms of possible file corruption and viruses. A downside of having the laptop remote from the office is that IT cannot solve any problems and if they get locked out they have to come back to Lisbon.

The laptop bag should have a cable and, for some of the newer laptops, an adapter for an USB to SD card because the newer laptops only have mini sd card slot or none at all.

The laptops have codes in the bag on A4 sheet of paper but they are unlocked with the following codes:

Normal Account Details:

Account: NJPilot

Password: Phenom300

Power Use Account Details

Account: NJPoweruser

Password: NJE-phenom300

The Poweruser password will be required whenever you want to download anything to the laptop (other than when using the Jeppesen Application) and for updating any of the websites or Jepp App.

If, for any reason, the computer is locked out or not responding to the normal password, then try the poweruser account details and this can often unlock the normal one.

For day to day use of the Garmin site and the Jepp App, the NJPilot account should be used.

There is no MS Office on the laptop and so if CMC files need to be sent then access via your own email account or the OM-A will need to be done to attach anything onto the emails.

8.2 GUIDANCE FOR UPDATING THE GARMIN G3000

The notes and screenshots that follow are a guide of how to update the Garmin 3000 databases on the Phenom 300S. Each aircraft has a laptop on board to use for these updates.

Additional information is in Appendix B of the *Embraer Prodigy Flight Deck 300 Pilots Guide*.

The Garmin 3000 has two SD card slots in both PFDs and the MFD. The top slot is empty and only used to update the Navigation Database (or CMC downloading), the lower slots house the Garmin SD card and this has all the other Garmin Databases and Jeppesen Charts on it. This card remains in the lower slot at all times.

WEF 08 NOV 2018 ALL DATABASES ARE NOW ON THE JEPPESEN APPLICATION ONLY

NO DATABASES ARE AVAILABLE ON THE FLYGARMIN SITE

ALL DATABASES ARE NOW ONLY ON THE JEPPESEN DISTRIBUTION MANAGER (JDM)

Navigation Database

The Navigation Database should ONLY be loaded using an SD card into the top MFD slot.

It is then either loaded directly onto the system if the current database has already expired, or automatically transferred into the Standby Navigation folder.

ONLY USE THE TOP SD SLOT OF THE MFD TO UPDATE THE NAVIGATION DATABASE

The Navigation Database, which expires every 28 days, can be downloaded either at home onto an SD card or using the aircraft laptop and onto the 'spare' SD card kept on the aircraft in the SD card box in the laptop bag.

The new Navigation Database is available on the website 7 days ahead of the expiry date of the current one and is the only database that can be loaded onto the Garmin before the expiry of the old one.

The Garmin 3000 has a Standby Navigation folder which means the next Navigation Database can be loaded as soon as it is available. When the current navigation database expires the system will automatically confirm if you wish to upload the new one during power up.

If the Navigation Database is loaded onto the lower Garmin SD card there is a possibility of spurious Fuel CAS messages on power up, see **FOL 01/14**.

The advantage of using the JDM App, which is also loaded onto the aircraft laptop, is that the Navigation database can be loaded onto the laptop harddrive first and then put onto an SD card later without the need for an internet connection.

Other Databases:

All other databases are saved directly onto the Garmin MFD SD card. This card **MUST** always remain in the lower MFD SD slot except when being prepared for an update.

Crew should normally only need to update the Garmin SD card in the MFD (lower slot), because during the subsequent power-up the Garmin will recognise the new data and transfer this across the entire system (PFD 1/2, MFD & GTCs). Therefore, normally only one Garmin SD card needs to be removed and updated.

NOTE: The Terrain Database and the Jeppesen Charts do not get synchronised from the MFD to the PFDs (file size is too large). Therefore, for these updates only it is necessary to upload the charts onto all 3 aircraft SD cards (MFD and both PFD).

The following is a list of databases used by the Garmin 3000 system. All of these can be checked for currency and updated using the JDM Application on the laptop.

G3000 Database	Description	Valid for	Lead Time	SD Card
Obstacle Data		56 days	7 Days	Lower MFD SD Card Only Will auto sync across the system
Safe Taxi	Europe safe Taxi	56 days	7 Days	Lower MFD SD Card Only Will auto sync across the system
Airport Directory	International by AC-U-KWIK	56 days	7 Days	Lower MFD SD Card Only Will auto sync across the system
Base Map	Worldwide Base Map	As required (normally 12 months+)	7 Days	Lower MFD SD Card Only Will auto sync across the system
Terrain	Worldwide Terrain	As required (normally 12 months+)	7 Days	Update required onto all 3 lower SD cards.
Navigation	Navigation Database is the ONLY one that legally MUST be in date	28 Days	7 Days Can be pre-loaded onto the Garmin using the Stby Nav Folder	Top MFD SD Card Only Will auto sync across the system.
E Charts	Jepp E Charts	28 Days	3 Days Can be pre-loaded onto the Garmin as soon as available on the Jepp App. See instructions further down for details.	Update required onto all 3 lower SD cards.

Note: All databases may be uploaded to the laptop hard drive / desktop up to either 3 or 7 days prior to the current expiry date. This allows crews to pre-load the laptop anytime and then just update the SD card on the aircraft on the day required without the need for an internet connection. This should prevent the need to rush around on the day the databases expire looking for an internet connection before the first flight of the day.

If you have the chance, please think of your colleagues and update the laptop whenever you have time.

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8.2.1 JEPPESEN DISTRIBUTION MANAGER (JDM)

The JDM NOW HAS ALL DATABASES REQUIRED FOR THE G3000 ON IT.

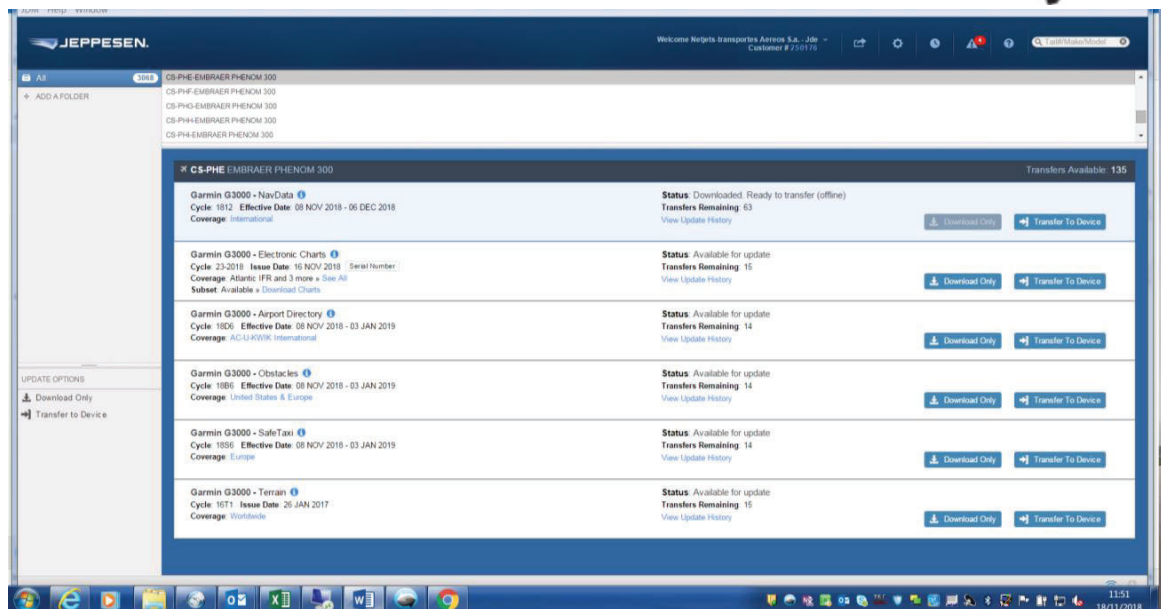
Note: JDM is an updated version of the Jeppesen Services Update Manager App (JSUM).

Any updates to the JDM App itself can be made by using the PowerUser log on when requested by the computer to authorise any update.

If the User Account Control screen appears, type in the PowerUser account and password.

Account: NJPoweruser

Password: NJE-phenom300



The ECharts (and Terrain) do not get 'spread' across the system automatically, so you need to update all 3 lower SD cards (MFD and 2 x PFD) using the laptop.

However, if you use the download described below, you only need to load the database once from the website using the internet and then it will allow you to load all 3 cards from the one download stored onto the laptop. This is the preferred method of updating the jet to reduce the risk of losing the SD cards.

However, if the aircraft laptop is not available, the crew may use an FBO or a personal PC / Mac. The JDM application is available for both systems.

Caution: Some FBO computers will not allow downloading of programs onto them.

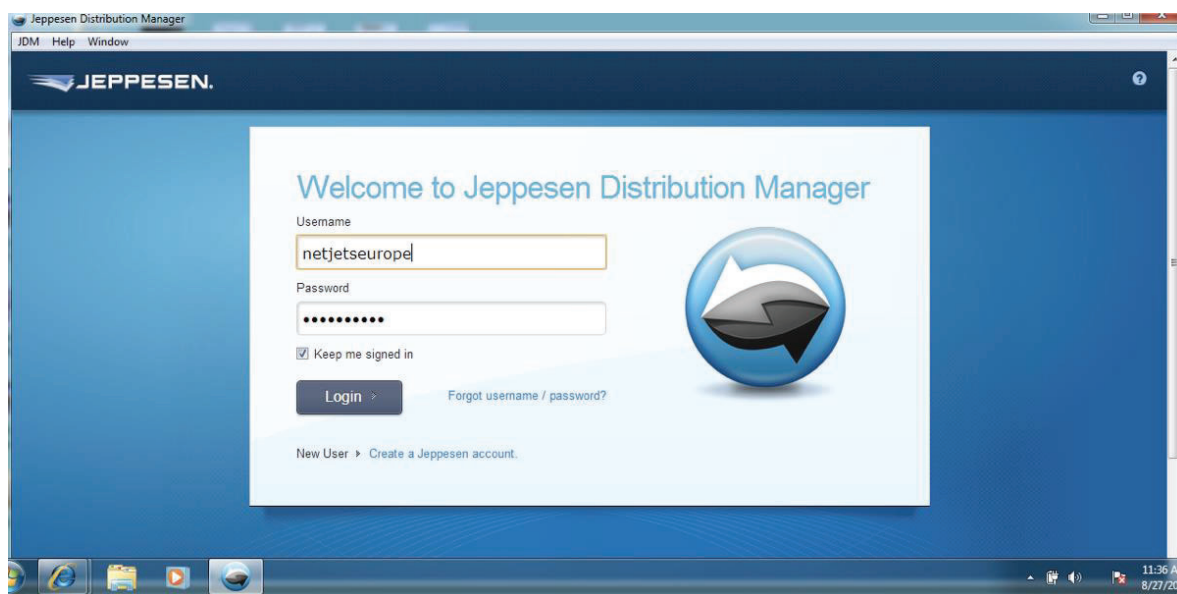
The JDM application is available for download on the Jeppesen website using the link below:

[http://ww1.jeppesen.com/support/technical_support_details.jsp?prodNameTxt2=Jeppesen%20Distribution%20Manager%20\(JDM\)](http://ww1.jeppesen.com/support/technical_support_details.jsp?prodNameTxt2=Jeppesen%20Distribution%20Manager%20(JDM))

Note: There are separate downloads for Apple and Windows products. If using your own laptop, make sure you download the correct version. The applications differ slightly between Apple and Windows. Examples of both systems with screenshots are shown below.

8.2.2 WINDOWS APPLICATION SCREENSHOTS

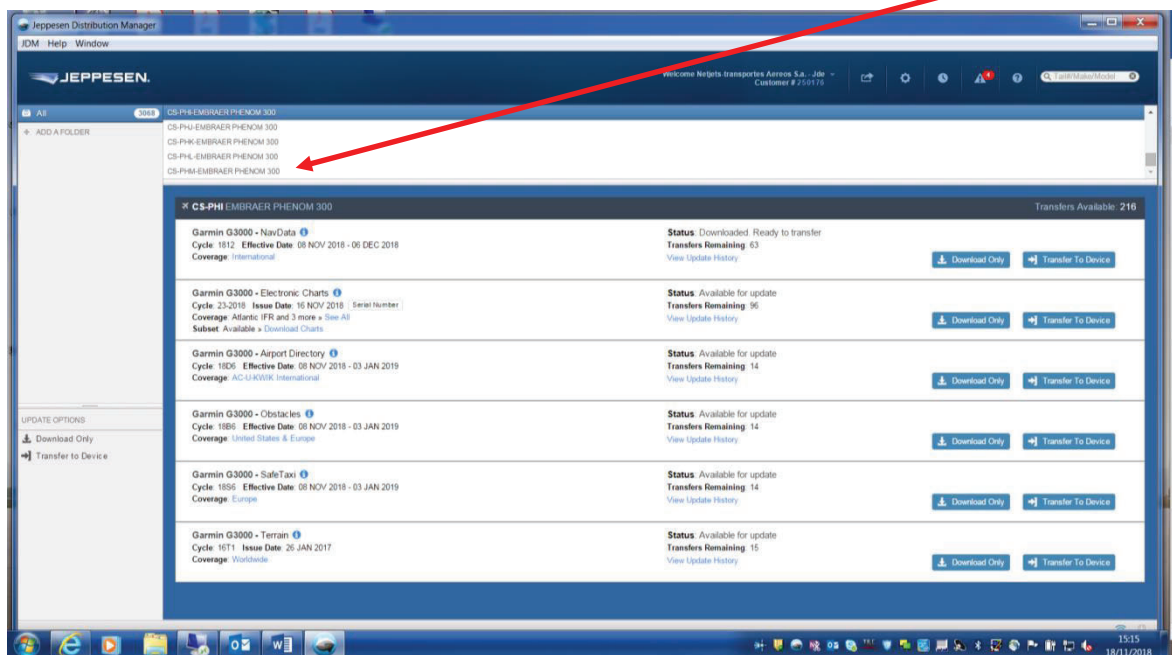
1. Open the JDM application and login using the following credentials:



Username: netjetseurope Password: Hawker800

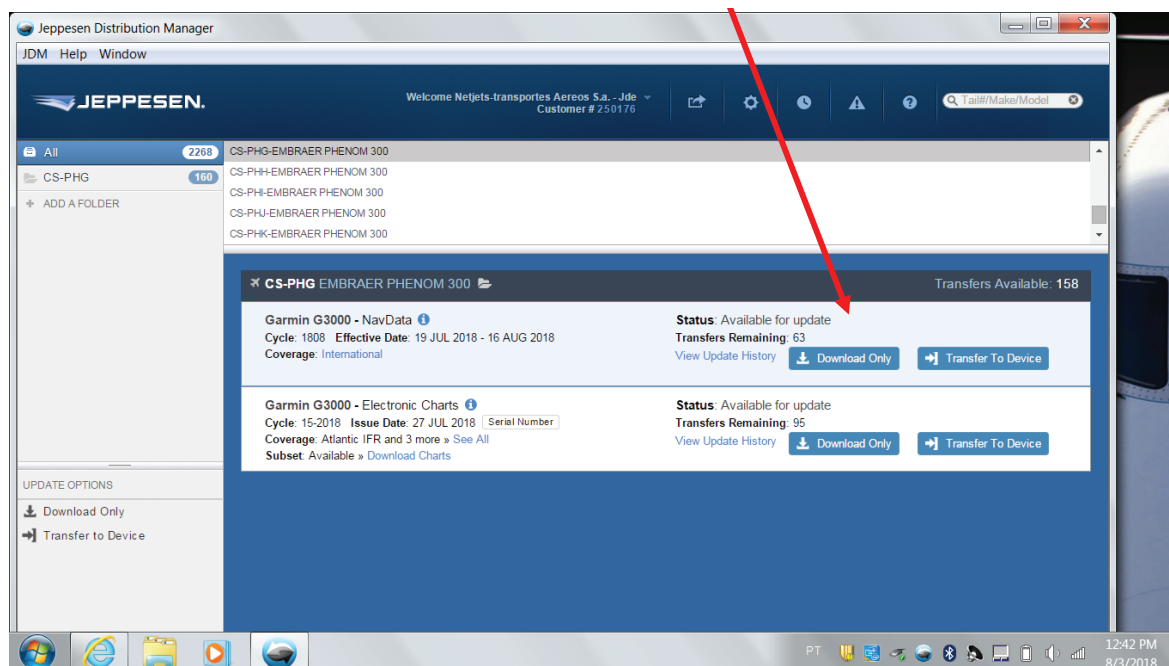
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- Select your aircraft from the list at the top. Scroll down to find your tail.

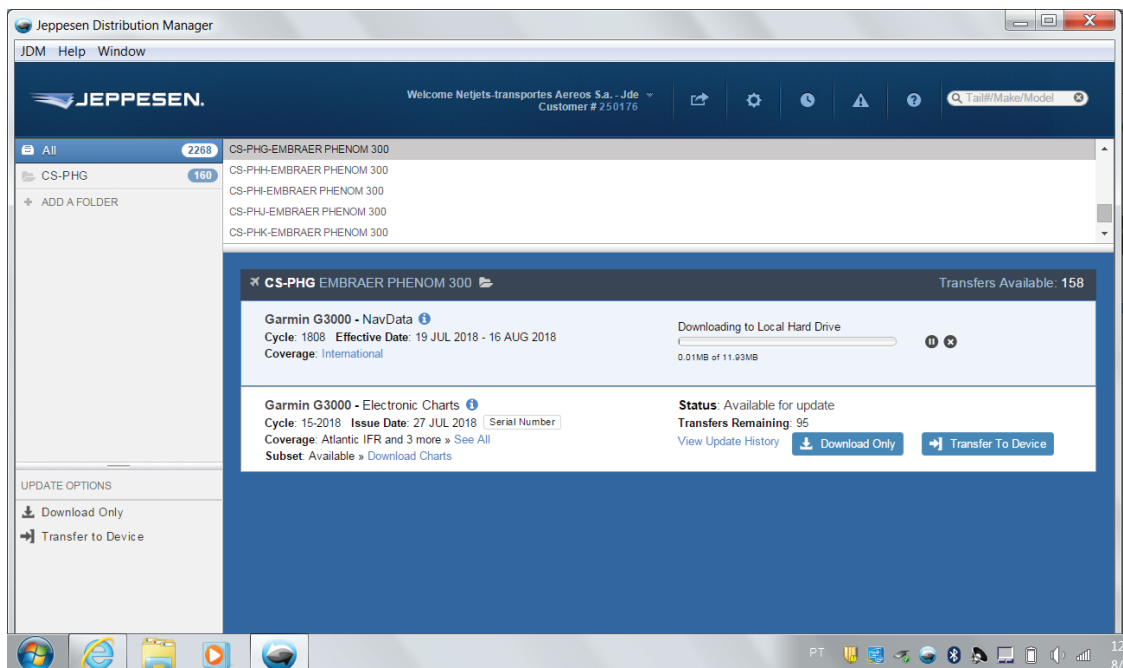


*Note: Because we need to update all 3 SD cards from the aircraft, there is an option to download the Jeppesen file to the computer using the **'Download Only'** option on the left hand side. This will then allow you to download the file to the computer just once using an internet connection and then take the computer to the aircraft and upload each card from the saved file. This saves a great deal of time and means you do not need an internet connection once the initial download is complete.*

- To download the database, simply click on the square 'Download Only' for download onto the hard-drive or 'Transfer to Device' to load an SD card directly.

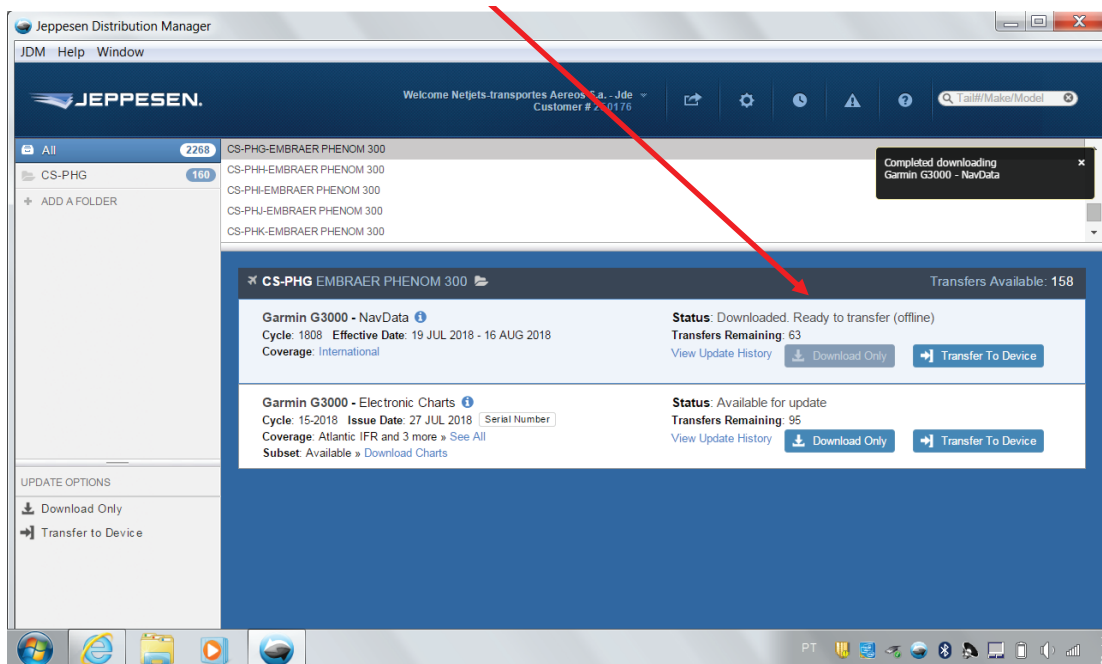


- You will then see this screen with the downloading progress.

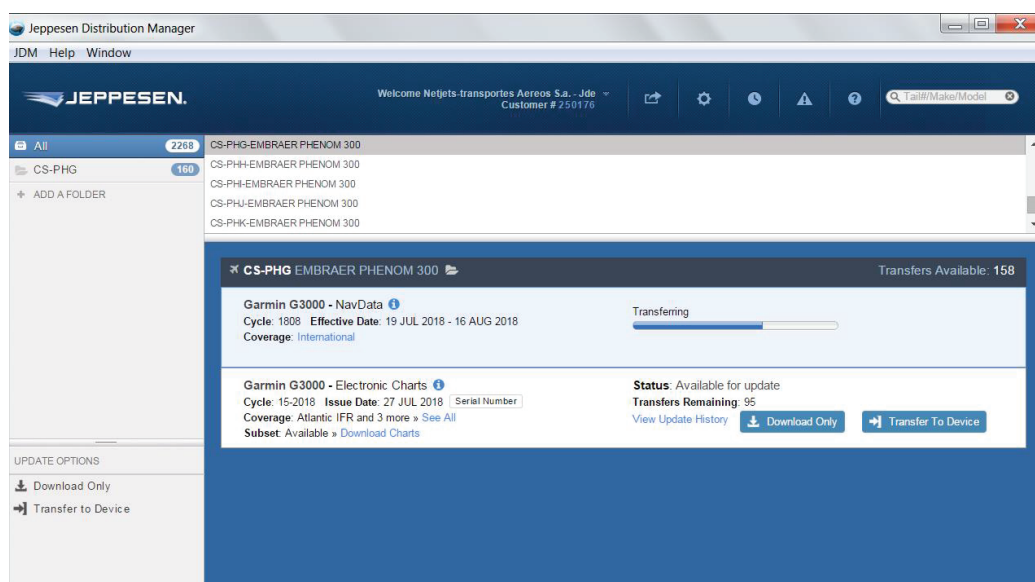
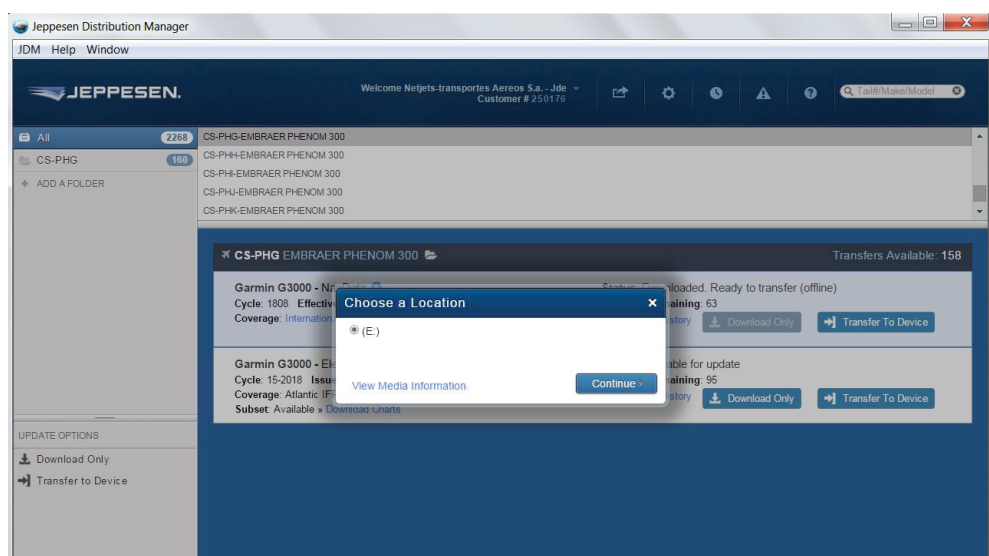


You will get another message when the upload is complete, and you will see that the databases are downloaded and 'Ready to Transfer (offline)'

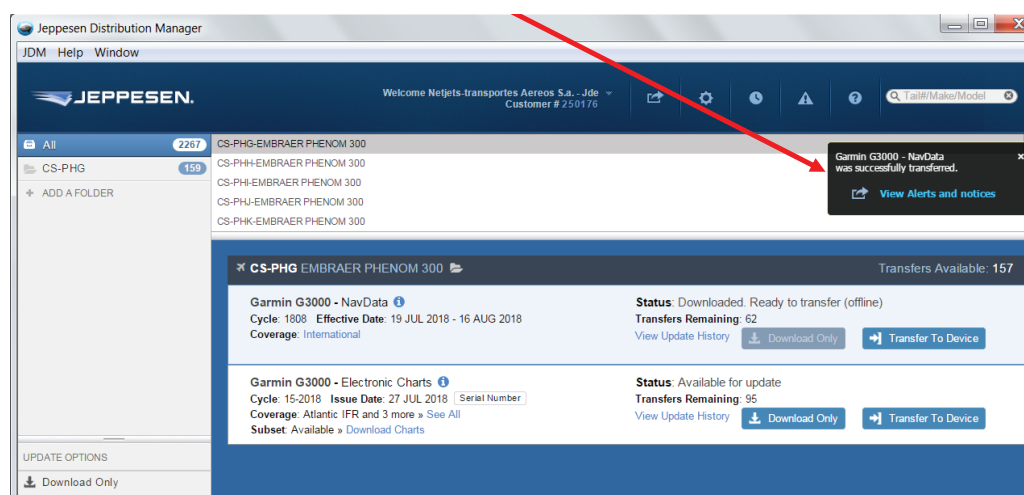
- The Download button will be greyed out if the database is on the hard-drive.



6. To upload to the SD card, drag and drop the 'box' again (or click on) the 'Transfer to Device' option and then choose the SD card to load.



7. The program will inform you when the card is updated.

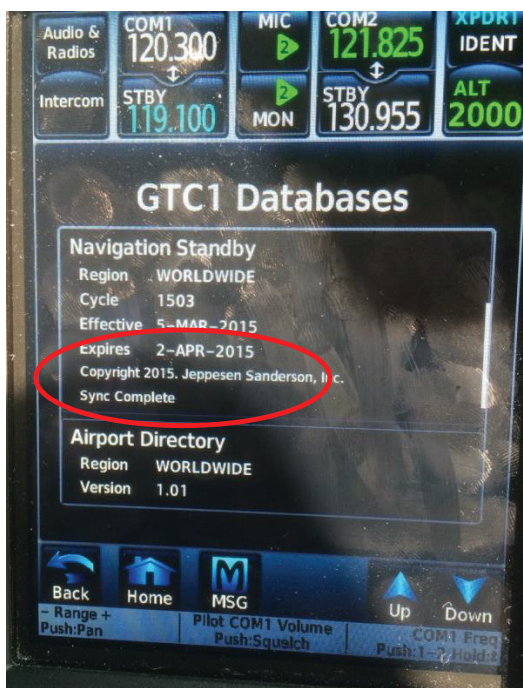


8.2.3 JEPPESEN UPLOAD ONTO GARMIN 3000

You will need to load each of the 3 aircraft SD cards (MFD, PFD 1, and PFD 2) because the Jeppesen data does not get spread across the system like the Garmin databases do.

With the aircraft powered down replace the SD cards on the PFD 1/2 and MFD lower slots making sure that each card is placed in their respective unit. The cards are labelled PFD 1, PFD 2, and MFD.

Power up the aircraft and when the sequence is complete, go to the GTCs and confirm that the database has been updated and that all three units are in date. It can take some time for the units to synchronise especially the Jeppesen charts. Synchronisation status can be verified / monitored in the Avionics Status screen, accessed in the GTC by selecting from Home: Utilities -> Setup -> Avionics Status and the Database tab. The database status for each LRU (MFD, PFD1, PFD2, GTC1, and GTC2) can be selected individually. After the status 'Sync Complete' is displayed for all LRUs, the system shall be power cycled to complete the update process.



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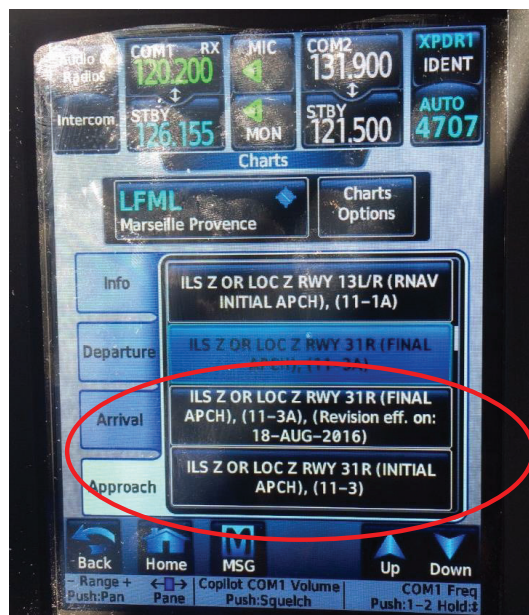


When you do the Jeppesen Charts, sometimes you get a yellow date of 01 Jan 1990, which usually means a transfer error but please wait a while and see if the sync starts.

If there is an error with the synchronisation, then a power down and back up again will be required to reboot the system and hopefully, start the synchronisation.

8.2.4 LOADING NEW JEPPESEN CHART BEFORE OLD EXPIRES

If a new chart is valid after the upload date, both the current chart and the new chart will be seen on the GTC.



In the picture above there are 2 charts for ILS Z, the current and the new chart.

(Eff on: 18-Aug-2016)

When the new chart becomes valid, the old one will automatically be removed from the GTC.

LFML-ILS Z OR LOC Z RWY 31R (FINAL APCH). (11-3A)				
LFML/MRS MARSEILLE/PROVENCE		MARSEILLE/PROVENCE, FRANCE		
20 MAY 16		11-3A Eff 26 May		
D-ATIS	PROVENCE Approach		PROVENCE Tower	Ground
125.350	North 131.225	South 120.2	133.1	121.9

LFML-ILS Z OR LOC Z RWY 31R (FINAL APCH). (11-3A)				
LFML/MRS MARSEILLE/PROVENCE		MARSEILLE/PROVENCE, FRANCE		
12 AUG 16		11-3A Eff 18 Aug		
D-ATIS	PROVENCE Approach		PROVENCE Tower	Ground
125.350	North 131.225	South 120.2	133.1	121.9

With the charts selected on the PFD it is important to check the Effective date. In the above example, the date of the flight was 15th August and both charts were available.

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8.3 DATABASE UPDATE SCHEDULE

Aeronautical Information Publications (AIP) are issued regularly and to a structured 28-day cycle. The structure and contents of AIPs are standardised by international agreement through ICAO. AIPs normally have three parts - GEN (general), ENR (en route), and AD (aerodromes).

AIPs are kept up-to-date by regular revision on a fixed cycle. For operationally significant changes in information, the cycle known as the AIRAC (Aeronautical Information Regulation and Control) cycle is used: revisions are produced every 56 days (double AIRAC cycle) or every 28 days (single AIRAC cycle).

Jeppesen Charts

Jeppesen and Garmin have stated that we can use the charts from whenever they appear on the website which is normally 4 days before the effective date. Garmin have stated that for all G3000 version 4.0 and better (ours) the available date and issue date have been removed allowing the system to take the charts as soon as they are on the website.

Garmin customers utilising the Jeppesen ChartView Database have requested to use the data upon receipt but are hindered by the effective date being shown on the GTC. Garmin has removed the effective date for the Jeppesen ChartView Database in GTC v4.00 and later. This memo is intended to formalize Garmin and Jeppesen's position on using the ChartView database upon receipt.

Therefore, instead of waiting until the expiry date you can upload the next cycle anytime within 4 days which gives much better chance of having time to do this. In addition Jeppesen have released a Service Advisory statement which says that the Chartview database is valid 7 days after the expiry shown on the GTC. What this means in effect is that you have almost 11 days to update the Jeppesen and that any data in the current charts is valid until 7 days after the shown expiry date which co-insides with the normal AIRAC cycle.

For example:

Jepp Cycle 05/2017 Issue Date 10 Mar 17 shows expiry of 24 Mar 17 (same as next issue date).

Jepp Cycle 06/2017 is available on 21 Mar 17 with an Issue Date of 24 Mar 17.

Garmin state that we have 7 days to update the 05/2017 charts because the data in then is in line with AIRAC cycle 1704 below:

AIRAC Cycle for Navigation Database 1704 Effective Date 30 Mar 17.

Service Advisory No. 1651 Rev A dated August 26, 2016

TO: Owners and Operators of Garmin G2000, G3000, G5000, and Embraer Prodigy® Touch Systems.

DATE: August 26, 2016

SUBJECT: ChartView Database Expiration Date

DESCRIPTION

The ChartView Database Tab will indicate Chartview has expired before the actual expiration date.

PILOT ACTION

Pilots should be aware that the ChartView database is valid until seven days after the expiration date shown on the GTC Database Tab.

RESOLUTION

A future software release will correct this issue.

8.3.1 NOTIFICATION OF DATABASE UPDATE

On completion of a database update, an email should be sent to lismxembraer@ntasa.pt with the following information:

- Aeroplane Registration.
- Date completed.
- Name of FCM carrying out the update.
- Date range of new database.
- SD Card number.
- Location where update completed.

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8.4 GARMIN TROUBLESHOOTING - IF YOU ACTIVATE THE NAV DATABASE TOO EARLY

The Garmin will ask if you want to activate the new Navigation Database when it sees a newer date.

Make sure you do not load the next database into the ACTIVE system until the actual day. It will load if you accidentally press YES.

If you do press YES see below.

We have had incidents where the next database has been loaded by accident into the active folder. Because the system always checks for the most up-to-date data it can be difficult to reload the old data. If you reload to the MFD, then after power up the Garmin will see the newer database on the PFDs and reload the MFD again.

The Embraer procedure for reloading an older database (or the current one again) is to pull the c/bs for the MFD and PFD 2 and turn the power ON. Load the current database into PFD 1 and when the upload is complete, power off and then pull the c/bs PFD 1 and only then reset the c/bs for the MFD and repeat. Finally, pull the c/bs for the MFD again and reset them for the PFD 2.

This way when the power is then turned fully back on, all 3 will have the same database and the 'system' will not be able to change it.

MFD PWR1 (Left Hand CB panel - position A9);
MFD PWR2 (Right Hand CB panel - position A24);
PFD1 (Left Hand CB panel - position A7);
PFD2 (Right Hand CB panel - position B24).

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CHAPTER 9 - PERFORMANCE TIPS

The aim of this chapter is to give general guidance and education on some Phenom performance issues. The information comes from within the POH, however, it has been tabulated into what is hoped to be an easier to use guide. The aim is to provide you with some quick information that is factually correct to assist you in managing your sectors.

9.1 PHENOM MIN FUEL CONSUMPTION GUIDANCE

The table below shows you what is quite simply a good guide for the max altitude that is possible to utilise the least fuel. This is really for those little hops of less than 200 nm. This should help to provide guidance on what levels to ask for when the OFP may be suggesting something a little lower.

The basic assumptions are a climb at 225/M0.6, a minimum of 5 mins in the cruise, and a descent planned at 3000 ft/min (M0.75/250).

Opt Altitude Min Fuel Consumption - Max Speed Cruise

	Dist and FL			
TOW	50nm	100nm	150nm	200nm
17600	>MLW	>MLW	240	390
17200	170	220	320	400
16800	170	220	320	400
16400	170	230	330	410
16000	180	230	330	420
15600	180	230	340	420
15200	180	240	340	420
14800	190	240	350	430
14400	190	240	360	440
14000	200	240	360	440

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Sector	Fuel cost < 150	
Length	Cost difference	
< 1.5	≥ 5	tanker
1.5 - 3.0	≥ 8	tanker
3.0 - 5.0	≥ 15	tanker
> 5.0	n/a	n/a
fuel index 151-200		
< 1.5	≥ 7	TANKER
1.5 - 3.0	≥ 15	TANKER
fuel index > 200		
< 1.5	≥ 10	TANKER
1.5 - 3.0	≥ 20	TANKER

9.2 PHENOM MAX SPEED CRUISE GUIDANCE

The following information is provided to highlight altitudes you can ‘make’ performance wise and also give you a clue as to what Max Speed the jet will settle at in the cruise. As can be seen from the tables below, if you are heavy and it is ISA+, then maybe FL390 is a more sensible altitude to ask for rather than trying 410 or above. A stepped climb may be a thought. The table below has a colour coding with respect to your G protection above the stall should you encounter turbulence of some sort.

Max Speed Cruise - Fuel Burn						
ISA - 10 Ice Protection OFF						
G Protection above Stall	TOW		FL370	FL390	FL410	FL430
	18387	Mach	0.75	0.74	0.71	0.69
		Range/100 lbs	35.8	38.7	41	43.2
	17200	Mach	0.76	0.75	0.73	0.71
		Range/100 lbs	36.1	39.2	41.8	44.1
	16000	Mach	0.76	0.75	0.75	0.73
		Range/100 lbs	36.4	39.5	42.6	45.1
	15200	Mach	0.77	0.76	0.75	0.75
		Range/100 lbs	36.5	39.7	42.9	45.9

2.0 - 1.75 G
1.74 - 1.6 G
< 1.6 G

Max Speed Cruise - Fuel Burn					
ISA Ice Protection OFF					
TOW		FL370	FL390	FL410	FL430
18387	Mach	0.75	0.74	0.71	0.69
	Range/100 lbs	35.8	38.6	40.9	43.1
17200	Mach	0.76	0.75	0.73	0.71
	Range/100 lbs	36.1	39.1	41.8	44.1
16000	Mach	0.76	0.75	0.75	0.73
	Range/100 lbs	36.4	39.5	42.6	45
15200	Mach	0.77	0.76	0.75	0.75
	Range/100 lbs	36.5	39.7	42.9	45.8

Max Speed Cruise - Fuel Burn					
ISA + 10 Ice Protection OFF					
TOW		FL370	FL390	FL410	FL430
18387	Mach	0.72	0.69	0.67	0.63
	Range/100 lbs	37.2	39.7	42	42.8
17200	Mach	0.73	0.71	0.69	0.67
	Range/100 lbs	37.7	40.5	42.9	45.2
16000	Mach	0.73	0.72	0.7	0.69
	Range/100 lbs	38	41.2	43.9	46.4
15200	Mach	0.74	0.73	0.72	0.7
	Range/100 lbs	38.3	41.5	44.5	47

G Protection above Stall

2.0 - 1.75 G

1.74 - 1.6 G

< 1.6 G

As can be seen, the performance has a marked drop off at above FL400 when it gets warm. Not so relevant in the winter months but certainly worth consideration during the warmer months. Further below will be some guidance on the climb. This will also give you the best chance of having a decent speed at TOC.

9.3 OEI CLIMB GRADIENT GUIDANCE

The following tables give you the climb gradient achievable for a given altitude and a given temperature. We provide this information to give you some confirmation of what gradient you can achieve in the unlikely event of an engine failure after V1, it also clearly allows you to know what climb you can achieve if the OEI event happens a couple of thousand feet above the airfield. Something that the current Flygprestanda guidance does not cover on the Guru. This may help for those Take Off Briefs where you might wish to consider an alternative to the Guru OEI procedure or indeed if there is not one.

At this point, it is important to be aware of the framework that surrounds building an SID within Standard Pans Ops. An SID will be designed to provide obstacle clearance for a 2.5 deg climb gradient. As you will all remember from your ATPL performance exams, there is a gross and net engine performance. Therefore, you must consider the net engine performance degradation which is 0.8%. Therefore, to 'guarantee' 2.5% climb you will need a minimum of 3.3%. The tables below are highlighted with anything that falls below 3.3%.

On ANY Jepp plate, the SID will be annotated with minimum climb gradients. These MUST have an explanation attached if it is for anything other than obstacle clearance. The majority are just for Air Traffic Management purposes. For example, in EGWU, off 07 (CPT5X) it states that if you can get to 230 ft DER, then you can use a minimum of 3.6% to 750 ft, and then you require 13.2% until D18.0 OCK. They then clarify this is for ATM purposes. Thus, should you have an OEI event and you can only manage 6.5%, you can continue to follow the SID as you are SAFE from obstacles. You WILL have to tell ATC because they may need to reroute other traffic, however, your Navigation is sorted, the FMS and AP can remain engaged, and ATC importantly know exactly where you are going. Keeps it all pretty simple really. The figures are for Flap 1 and V2 to V2+10, so if you lose a donk after you have already cleaned up, you will outperform these numbers below.

	ISA -10		Climb Gradient OEI Ice Protection OFF				Flap 1
OAT	AMSL	14400	15200	16000	16800	17600	18386
5	0	12.5	11.2	10	8.9	8.1	7.3
3	1000	12.4	11.2	10	8.9	8.1	7.3
1	2000	12.3	11	9.8	8.8	7.9	7.1
-1	3000	11.9	10.6	9.4	8.5	7.6	6.8
-3	4000	11.4	10.1	9	8.1	7.3	6.5
-5	5000	10.9	9.6	8.6	7.7	6.9	6.1
-7	6000	10.2	9	8	7.2	6.4	5.7
-9	7000	9.4	8.3	7.3	6.5	5.8	5.1
-11	8000	8.8	7.7	6.8	6	5.3	4.6
-13	9000	8.3	7.3	6.4	5.6	4.9	4.3
-15	10000	7.9	6.9	6	5.3	4.6	4
-17	11000	7.5	6.5	5.7	4.9	4.3	3.6
-19	12000	7	6.1	5.3	4.6	3.9	3.3
-21	13000	6.6	5.7	4.9	4.2	3.6	3
-23	14000	6.1	5.2	4.5	3.8	3.2	2.6
Avg Climb to 15K		9.5	8.4	7.4	6.6	5.9	5.2
Avg Climb to 10K		10.8	9.6	8.5	7.6	6.8	6.1

	ISA		Climb Gradient OEI Ice Protection OFF				Flap 1
OAT	AMSL	14400	15200	16000	16800	17600	18386
15	0	12.4	11.1	10	8.9	8	7.2
13	1000	12.4	11.1	9.9	8.9	8	7.2
11	2000	12.2	10.9	9.8	8.8	7.9	7.1
9	3000	11.8	10.6	9.4	8.4	7.6	6.8
7	4000	11.4	10.1	9	8.1	7.2	6.5
5	5000	10.9	9.6	8.6	7.7	6.9	6.1
3	6000	10.2	9	8	7.1	6.4	5.7
1	7000	9.3	8.2	7.3	6.4	5.7	5
-1	8000	8.6	7.6	6.7	5.9	5.2	4.5
-3	9000	8.3	7.2	6.4	5.6	4.9	4.2
-5	10000	7.9	6.9	6	5.3	4.6	3.9
-7	11000	7.4	6.4	5.6	4.9	4.2	3.6
-9	12000	7	6	5.2	4.5	3.9	3.3
-11	13000	6.6	5.6	4.9	4.2	3.5	2.9
-13	14000	6	5.1	4.4	3.7	3.1	2.5
Avg Climb to 15K		9.5	8.4	7.4	6.6	5.8	5.1
Avg Climb to 10K		10.8	9.5	8.5	7.6	6.8	6.0

	ISA +10		Climb Gradient OEI Ice Protection OFF				Flap 1
OAT	AMSL	14400	15200	16000	16800	17600	18386
25	0	11.9	10.6	9.5	8.5	7.6	6.8
23	1000	11.3	10.1	8.9	8	7.2	6.4
21	2000	10.8	9.5	8.5	7.6	6.8	6
19	3000	10.2	9	8	7.1	6.4	5.6
17	4000	9.7	8.5	7.6	6.7	6	5.3
15	5000	9.1	8	7.1	6.3	5.5	4.9
13	6000	8.4	7.4	6.5	5.7	5	4.4
11	7000	7.7	6.7	5.9	5.1	4.4	3.8
9	8000	7.1	6.1	5.3	4.6	3.9	3.3
7	9000	6.8	5.8	5	4.3	3.7	3.1
5	10000	6.4	5.5	4.7	4	3.4	2.8
3	11000	5.9	5	4.3	3.6	3	2.4
1	12000	5.5	4.6	3.9	3.3	2.7	2.1
-1	13000	5.1	4.3	3.5	2.9	2.3	1.8
-3	14000	4.6	3.8	3.1	2.5	1.9	1.4
Avg Climb to 15K		8.0	7.0	6.1	5.3	4.7	4.0
Avg Climb to 10K		9.3	8.2	7.2	6.4	5.7	5.0

	ISA +20		Climb Gradient OEI Ice Protection OFF				Flap 1
OAT	AMSL	14400	15200	16000	16800	17600	18386
35	0	9.2	8.1	7.2	6.4	5.6	4.9
33	1000	8.8	7.7	6.8	6	5.3	4.6
31	2000	8.4	7.4	6.5	5.7	5	4.3
29	3000	8	7	6.2	5.4	4.7	4
27	4000	7.7	6.7	5.8	5.1	4.4	3.8
25	5000	7.3	6.3	5.5	4.7	4.1	3.5
23	6000	6.8	5.8	5	4.3	3.6	3
21	7000	6.2	5.3	4.5	3.8	3.2	2.6
19	8000	5.8	4.9	4.2	3.5	2.9	2.3
17	9000	5.4	4.6	3.9	3.2	2.6	2.1
15	10000	5	4.2	3.5	2.8	2.3	1.7
13	11000	4.5	3.7	3	2.4	1.8	1.3
11	12000	4.1	3.3	2.6	2	1.5	
9	13000	3.7	2.9	2.3	1.7		
7	14000	3.2	2.5	1.8	1.3		
Avg Climb to 15K		6.3	5.4	4.6	3.9	3.6	3.2
Avg Climb to 10K		7.4	6.4	5.6	4.8	4.1	3.5

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 Aircraft still climbing but unable to meet PAN Ops SID climb gradient for obstacle clearance (3.3%).

 Aircraft not climbing.

All figures are Flap 1. Climb at V2 to V2 +10.

9.4 CLIMB TECHNIQUE GUIDANCE

The POH provides climb data in 3-40, this particular schedule is, as we all know, 225 KIAS up to the Mach change over at 29,600 ft whereby you then climb at M0.6. This is up to FL450.

That sounds simple enough. However, as most of you know, M0.6 at FL450 is NOT where you want to be on the mythical Drag Curve. Mainly because you will be behind it. Which is a rubbish place to be.

This is not an article on advance aerodynamics and we do not have the luxury of our SMART probes displaying AOA on any of the screens, so we are only able to feel and experience the response of the aircraft at altitude and make decisions based upon that.

The aircraft's performance markedly falls away above FL400 when ISA is anywhere from +5 and above, clearly, this is exacerbated with a heavy jet. In [9.2 Phenom Max Speed Cruise Guidance](#), the coloured areas of the tables are where your g-stall margin is degraded. As you will see, high, warm, heavy = not good.

Sweptwing performance essentially means do not climb at M0.6 above FL300 as you will not be doing yourself any favours. As you are aware, we do not have a display of AOA. If we did, we would climb at a single AOA unit. IF we had this, you would experience a gradual increase in the Mach as you increased altitude whilst maintaining the wings most efficient climb angle.

A technique to mimic this is to climb to FL300 using 225 KIAS (or preferable 250 above FL200 for the same reasons highlighted in the last paragraph), select VS and depending on temp and weight, use either 1500 ft/min, above FL400 use 500 ft/min. You will notice that you get to TOC with some actual airspeed and then after using up to a maximum of 5 mins at climb thrust (IF NEEDED) before transitioning to cruise power you will have a splendid transit Mach. There is not a significant difference in fuel burn.

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